Kernel Affine Hull Features for Time-Series Forecasting: Supplementary Studies

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Abstract

This document presents supporting studies of Kernel Affine Hull Machine representations for forecasting. It defines the evaluated encodings and decoders, establishes the supporting information and residual-risk identities, and specifies the controlled, natural-data, scaling, additive, metric-learning, reconstruction-training, and online-adaptation experiments. Complete comparison families, source-level results, resource measurements, and unsuccessful or unavailable configurations accompany their protocols. The studies use different procedures and evaluation populations. Their results are interpreted separately from the bounded reconstruction-departure experiment in the accompanying main paper.

S1 Scope, evidence, and notation

The supporting experiments examine different ways in which reconstruction geometry can affect fitted forecasts.[1] They test information retention, finite-decoder access, source transfer, residual correction, supervised geometry, and online adaptation. None is a replication of the main paper's bounded reconstruction-departure procedure. Their candidate sets, tuning rules, data populations, cost scopes, and loss normalizations remain specific to each study.

Sections S2 and S3 define the common interpretive framework and controlled representations. Sections S4–S6 describe the controlled, natural-data, and scaling comparisons. Sections S7–S10 examine residual correction, learned metrics, direct reconstruction training, and online updates. The appendices give complete proofs, source rosters, numerical specifications, comparison tables, and reproducibility details. All crossreferences in this document are local; the prefix S distinguishes its sections, equations, propositions, tables, and figures from those of the main paper.

Vectors are columns, and reference matrices store one transposed observation per row. For a vector v, $v^\top$ denotes transpose, $\|v\|_2$ denotes the Euclidean norm, and $\|v\|_1$ is the sum of its absolute coordinates. The identity matrix I and all-ones vector $\mathbf{1}$ have the dimensions specified by their surrounding products. Expectations, variances, and covariances refer to the probability law stated in each result. A history contains only observations available at its forecast origin; a horizon-h target occurs h steps later. Fitted transformations use only the data permitted by each protocol.

S2 Predictive usefulness and computational cost

Let $X \in \mathbb{R}^d$ be an observed history, $Y \in \mathbb{R}^{d_y}$ its future response, and $\Gamma_{\mathcal{D}}$ a representation fitted on development data $\mathcal{D}$. Expectations below are conditional on the fitted state. Each experiment specifies its own evaluation distribution and loss aggregation. A deterministic representation can discard predictive information; appending it to X preserves the raw history and hence its Bayes risk (Proposition S8). This does not ensure equal performance of finite, regularized fitted decoders.

[1] AI tools assisted with protocol development, implementation, numerical and statistical verification, analysis, and manuscript preparation. The authors are responsible for the scientific claims, evidence, and final text.

Residual information relative to a baseline. Fix $f_0 : \mathbb{R}^d \to \mathbb{R}^{d_y}$, put $e = Y - f_0(X)$ and $Z = \Gamma_{\mathcal{D}}(X)$, and assume $\mathbb{E}\|e\|_2^2 < \infty$. Write $R(f) = \mathbb{E}\|Y - f(X)\|_2^2$ and let $L^2(P_Z; \mathbb{R}^{d_y})$ contain the measurable corrections with finite second moment under the distribution P_Z of Z. Conditional expectation gives the best correction $q^*(Z) = \mathbb{E}[e \mid Z]$ and

$$R(f_0) - \inf_{q \in L^2(P_Z; \mathbb{R}^{d_y})} R(f_0 + q \circ \Gamma_{\mathcal{D}}) = \mathbb{E}\|\mathbb{E}[e \mid Z]\|_2^2. \tag{S1}$$

The circle denotes composition. Appendix SJ.1 proves the identity. Improvement is possible in this unrestricted correction class precisely when the residual conditional mean is nonzero with positive probability. The identity does not determine the performance of a correction estimated in a finite class.

Fitted procedures and computation. For fitted forecasts $\widehat{f}_0$ and $\widehat{f}_s$, evaluation unit $\mathcal{E}$, and specified squared loss ℓ, define

$$\Delta_s = \mathbb{E}[\ell(\widehat{f}_0; \mathcal{E}) - \ell(\widehat{f}_s; \mathcal{E}) \mid \mathcal{D}]. \tag{S2}$$

Positive Δ_s favors the richer procedure. A comparison may require a minimum gain $\delta > 0$ and cost budget $B > 0$, with measured complete procedure cost $C_s \le B$. Feature construction, fitting, and selection are charged even when the selected prediction equals the baseline. Evaluation outcomes cannot enter fitting or selection. Independently sampled repetitions, overlapping windows, and selected source families supply different units of evidence. Missing predictions without a prespecified fallback make the affected full-roster aggregate undefined; restriction to successful fits changes the evaluated population. No experiment here estimates unrestricted natural-data Bayes risk or certifies the minimum complexity necessary to achieve a target accuracy.

A correction that retains the fitted baseline. For geometric family g, fitted correction $\widehat{q}_g$, and scalar mixing weight γ, define

$$\widehat{f}_{g,\gamma}(x) = \widehat{f}_0(x) + \gamma \widehat{q}_g(x), \qquad 0 \le \gamma \le 1. \tag{S3}$$

Zero correction recovers the same fitted baseline. Residual fitting is established (Bühlmann & Yu, 2003); these studies compare the geometric features supplied to the correction. With square-integrable residual and correction, expansion of squared loss yields

$$R(\widehat{f}_{g,\gamma}) - R(\widehat{f}_0) = \gamma^2 \mathbb{E}\|\widehat{q}_g(X)\|_2^2 - 2\gamma \mathbb{E}\langle Y - \widehat{f}_0(X), \widehat{q}_g(X)\rangle. \tag{S4}$$

Here $\langle \cdot, \cdot \rangle$ is the Euclidean inner product. For a correction nonzero with positive probability, positive expected alignment is necessary and sufficient for improvement at some sufficiently small positive γ. A finite candidate grid may miss that interval, and validation can select a worse forecast. Baseline availability is therefore a candidate-class property, not a performance guarantee; Appendix SJ.2 gives the derivation.

S3 Geometry, predictive encoding, and decoder access

Matched reconstruction rules. Let $z \in \mathbb{R}^d$ be a column vector obtained by standardizing an input with training-set means and standard deviations. The controlled study fits three independent k-means partitions, with 2, 4, and 8 local models, respectively. Index the resulting 14 models by c. Model c contains $n_c \le 24$ reference observations, stored as the rows of $V_c \in \mathbb{R}^{n_c \times d}$. Let $\mu_c \in \mathbb{R}^d$ be their mean and let $P_c \in \mathbb{R}^{d \times p_c}$ contain orthonormal principal directions, where $p_c \le \min(2, d-1, n_c - 1)$ is the effective local rank. Input scaling, partitions, references, principal directions, and the score construction are shared by the two reconstruction rules:

$$r_c^{\mathrm{A}}(z) = \mu_c + P_c P_c^\top (z - \mu_c), \qquad r_c^{\mathrm{K}}(z) = \frac{V_c^\top B_c k_c(z)}{\mathbf{1}_{n_c}^\top B_c k_c(z)}. \tag{S5}$$

Here $k_c(z) \in \mathbb{R}^{n_c}$ contains the Gaussian similarities between the query and the reference observations in the local projected covariance metric; $B_c \in \mathbb{R}^{n_c \times n_c}$ is the symmetric regularized inverse of the reference kernel matrix; and $\mathbf{1}_{n_c}$ is the vector of n_c ones. Appendix SB specifies the kernel, regularizer, and treatment of degenerate banks. The normalized KAHM weights can be negative. In the controlled implementation, reference values are stored in 32-bit floating-point precision, and an exactly zero reconstruction denominator is replaced by one. A separate numerical sensitivity analysis examines Gaussian underflow. Sharing the bank makes this a comparison of reconstruction operators within one specified procedure.

The features supplied to the predictor. For either reconstruction r_c, define a radial discrepancy q_c and an angular discrepancy b_c by

$$q_c(z) = 1 - \exp\{-\|z - r_c(z)\|_2\}, \qquad b_c(z) = \pi^{-1} \arccos\left(\mathrm{clip}\left(\frac{\langle z, r_c(z)\rangle}{\|z\|_2 \|r_c(z)\|_2}, -1, 1\right)\right),$$
$$T_c(z) = \sqrt{\{q_c(z)^2 + b_c(z)^2\}/2}, \qquad w_c(z) = (1 - T_c(z) + \tau)^\omega, \qquad \phi_c(z) = \frac{w_c(z)}{Z_{b(c)}(z)}, \tag{S6}$$

where $\mathrm{clip}(u, a, b) = \min\{b, \max\{a, u\}\}$, $\tau = 10^{-6}$, and $\omega = 16$. If the denominator in the cosine is zero, its value before clipping is defined to be zero. The index $b(c) \in \{1, 2, 3\}$ identifies the partition containing model c, and $Z_b(z) = \sum_{j:b(j)=b} w_j(z)$ is that partition's normalizing constant. Thus the affinities sum to one within each partition. Let $\phi(z) \in \mathbb{R}^{14}$ and $\mathbf{T}(z) \in \mathbb{R}^{14}$ be the concatenated affinity and score vectors, respectively. The feature comparisons provide the decoder with affinities, full scores, one $\log Z_b$ per partition, raw inputs, or gated local coordinates. The last construction is

$$\Gamma_{\mathrm{local}}(z) = \mathrm{col}\big(\phi(z), \{\phi_c(z) P_c^\top (z - \mu_c)\}_{c=1}^{14}\big), \tag{S7}$$

where col concatenates its column-vector arguments. This is a local-tangent feature construction (Yu & Zhang, 2010); the study examines how the geometric gates affect prediction.

Two information properties follow directly from these definitions. First, the score normalization has the inverse

$$T_c(z) = 1 + \tau - \exp\{(\log \phi_c(z) + \log Z_{b(c)}(z))/\omega\}. \tag{S8}$$

Consequently, $\mathbf{T}$ and the pair consisting of ϕ and all three log normalizers retain the same information in exact arithmetic. Their finite fitted decoders can nevertheless have different risks because the inverse is nonlinear. Second, suppose all gates are positive and the matrix formed by stacking $P_c^\top$ has full column rank d. Dividing the local-coordinate blocks by their gates recovers $P_c^\top (z - \mu_c)$. Adding $P_c^\top \mu_c$ then yields a full-rank linear system for z. Proposition S3 makes this argument precise. The 144 inspected controlled forecasting banks have full numerical rank and positive gates on their evaluated inputs. These checks do not establish uniformly positive gates or a stable inverse on the entire input domain (Appendix SC.5). Conversely, a rank-deficient stack does not by itself establish information loss, because the gates may encode additional distinctions between inputs.

A controlled target intervention. Let S take the values -1 and $+1$ with equal probability, let $T \sim \mathrm{Unif}[-1, 1]$, and let $\epsilon \sim \mathrm{Unif}[-0.1, 0.1]$. Assume these three variables are independent. For curvature $c \in \{0, 1\}$ and target-mixture weight $\lambda \in \{0, \frac{1}{2}, 1\}$, define

$$X = \begin{pmatrix} T \\ c\cos(\pi T) + 0.5S \end{pmatrix}, \qquad Y_\lambda = \sqrt{1 - \lambda}\, S + \sqrt{3\lambda}\, T + \epsilon. \tag{S9}$$

Here S identifies the branch and T gives position along it. Within each repetition and curvature, changing λ leaves the input observations, noise, and fitted bank unchanged. The predictable signal has variance one, and the Bayes mean squared error is $\mathrm{Var}(\epsilon) = 1/300$. As λ increases, the target changes from branch identity to signed position. Its accessibility to a particular decoder can change even though the signal and noise variances remain fixed.

Define the reflection $Q(x_1, x_2) = (-x_1, x_2)$. A representation satisfying $\Gamma(QX) = \Gamma(X)$ cannot retain the sign of the position component $\sqrt{3\lambda}T$. Its Bayes mean squared error is therefore at least $1/300 + \lambda$. There is also a quantitative statement for an approximately invariant representation. Let d_U be a pseudometric on its feature space, let g be a decoder satisfying $|g(u) - g(v)| \le \kappa d_U(u, v)$ for all feature values u, v, and define

$$R(g \circ \Gamma) - R_X^{\mathrm{Bayes}} \ge \left(\sqrt{\lambda} - \kappa\delta_\Gamma/2\right)_+^2, \qquad \delta_\Gamma^2 = \mathbb{E}[d_U(\Gamma(X), \Gamma(QX))^2]. \tag{S10}$$

In this expression $R(f) = \mathbb{E}[(Y_\lambda - f(X))^2]$, $R_X^{\mathrm{Bayes}} = 1/300$, and $(a)_+ = \max\{a, 0\}$; the claim assumes the quantities are square-integrable. A pseudometric satisfies the usual metric inequalities but may assign distance

zero to distinct points. Appendix SA proves both the exact and approximate statements. The measured reflection differences are descriptive estimates: they do not supply a population upper bound for δ_Γ, and the fitted experimental decoders have no imposed common Lipschitz budget.

The complete branch results and reconstruction and scoring diagnostics are reported in Appendix SD. Affine local coordinates recover accurate position prediction in the straight-branch example; the information advantage there is not exclusive to nonlinear KAHM reconstruction.

S4 Conditional gains in controlled forecasting

The controlled forecasting laws are nonlinear autoregression (AR), Van der Pol, and Duffing. Each repetition independently generates 128 training, 128 validation, and 512 test episodes of length 72. Histories have length 12 and predict offsets 1, 4, and 12 at 13 origins per episode. Entire episodes stay within a split; MSE averages origins and horizons within episodes and then episodes. Appendix SB specifies the numerical generators and fitting rules.

The first study has 24 full data-and-fit repetitions in each of nine primary settings: six target functions on the branch construction and three forecasting laws. Adding local coordinates to KAHM affinities improves ridge prediction beyond the fixed practical margin in 8 settings. On Van der Pol, mean MSE falls from 0.111108 to 0.014475, although affine local coordinates perform better with the same ridge head. The first study uses 95% Bonferroni-adjusted paired Student intervals for a family of 29 contrasts. This supports retaining local position, while leaving the complete representation–decoder choice unresolved.

A follow-up specifies its candidate models, practical margins, and analysis before generating 24 additional independent repetitions per forecasting law. It compares raw histories, affine local features, KAHM local features, and full KAHM scores with 64 or 256 Gaussian Nyström landmarks. The head appends linear coordinates to the finite nonlinear basis, standardizes basis columns, and uses an ordinary Euclidean ridge penalty. Validation selects among three bandwidth multipliers and seven penalties; the fitted model uses training observations only. Equal landmark counts do not imply equal realized ranks, function classes, or computational costs.

Table S18 reports the complete comparison family. KAHM local features outperform raw and affine local features on Van der Pol at both widths in every repetition. At 256 landmarks, mean MSEs are 0.000871969, 0.001654150, and 0.001078003, respectively: descriptive reductions of 47.3% and 19.1% relative to the controls. The two simultaneous lower endpoints, 0.00075496 and 0.00018100, exceed the prespecified practical margin 0.00014891; both 64-landmark comparisons also clear it. Raw inputs win on AR at both widths, and both controls win on Duffing. KAHM–affine differences on AR remain unresolved. The separate 29- and 12-contrast families do not imply a joint 95% coverage statement. Their inferential units are complete repetitions, not windows or horizons (Appendix SC).

Numerical sensitivity and an adverse natural-data check. The original Gaussian query evaluation can produce an all-zero row before signed normalization. Writing ℓ_j for a query log-kernel entry, replacing $\exp \ell_j$ by $\exp(\ell_j - \max_i \ell_i)$ preserves exact normalized weights whenever the denominator is nonzero (Proposition S7). In a post-hoc check reusing the 24 Van der Pol follow-up samples and banks, with affected heads refitted and validation-tuned, exactly zero normalizers fall from 99,762 among 2,236,416 query–local-model evaluations to zero. KAHM-local mean MSE changes by less than 4×10^{-13} at either width, whereas full-score losses improve by roughly 0.37–0.39%. Row scaling addresses this underflow pathway; it does not prevent signed cancellation or establish stability on independently generated data.

On a fixed grouped split of Airfoil Self-Noise (Brooks et al., 1989), raw RBF obtains mean group-weighted MSE 15.6898 dB^2, compared with 20.9313 for KAHM-local RBF. The validation selector chooses raw RBF in all 24 fits. These seeds characterize fitting variability on one static-regression dataset and split, not independent natural forecasting evidence. Together, the controlled results motivate testing local features as conditional candidates. The branch symmetry result does not explain the Van der Pol gain, and these simulations do not establish transfer to new source families; Section S5 examines that question separately.

S5 Natural forecasting: transfer, cost, and predicting benefit

The study fixes four development families (Electricity, NN5, Tourism, and FRED-MD) and four confirmation families (Traffic, Hospital, Saugeen River flow, and US births). Training-prefix eligibility and identifier hashes select 26 series. Two history–horizon regimes, two capacities, and two seeds yield 208 runs, including 80 confirmation runs. Every method uses the same available windows and chronological 60/20/20 splits. All fitted transformations use permitted training observations; validation selects hyperparameters, and a final fit uses eligible pretest windows. Appendix SE defines all twelve procedures; Appendix SF supplies source versions, chronology, capacity grids, loss scales, selection rules, and cost scopes. The four confirmation families remain the units of transfer despite multiple configurations and overlapping windows within them.

The primary matched comparison is raw finite-RBF prediction against raw history augmented by KAHM gates and gated principal coordinates. The augmentation preserves history information, but changes Gaussian distances, the finite basis, and regularization. All 208 runs complete with every candidate present, yielding 2,496 task–method losses. Raw plus KAHM has 4.26% higher source-balanced loss than raw RBF at 2.30 times its measured warm-library pipeline cost; raw ridge is more accurate than both (Table S21). KAHM raises loss against raw RBF by 0.50% on Traffic, 12.27% on Hospital, and 3.53% on US births, and lowers it by 1.53% on Saugeen. Omitting any one confirmation family leaves the aggregate unfavorable. These are descriptive selected-source comparisons, without population-wide confidence intervals.

The structural controls and development results further limit attribution. Raw plus KAHM improves aggregate loss over raw plus affine and raw plus locality by only 0.46% and 0.50%, respectively; each comparison changes sign across source means. FRED-MD contributes 96.62% of the development gain over raw RBF, while raw plus affine has lower mean loss and cost than KAHM in every development family. Table S22 separates substantial selection cleanup from search and refitting. The reported 2.30-fold ratio describes the supplied warm workflow, not cold-start deployment or intrinsic complexity.

Four ridge diagnostics predict log relative benefit from metadata, geometry, a reduced-capacity validation probe, or their combinations. They are trained on development outcomes with whole-source exclusions and frozen before confirmation (Section SF.4). Adding geometry to metadata plus the paid probe changes none of the 80 confirmation decisions. Both variants attain scaled MSE 0.807085 at about 0.3309 seconds per task; log-benefit RMSE increases from 0.184549 to 0.184680. Their aggregate gain over always raw is only 0.054%, at 1.96 times its pipeline cost. Metadata plus geometry also makes the same decisions as metadata alone. Complete methods, policies, source outcomes, and diagnostic predictions appear in Appendix SG. These data do not validate the tested geometric adoption rule; they do not rule out a different diagnostic.

S6 Post-hoc sensitivity to geometry scaling

A specified sensitivity analysis replaces training z-score geometry scaling by training min–max scaling to $[-1, 1]$, without clipping future values. Raw-history components and comparators remain standardized. Targets, splits, seeds, head grids, capacities, and query conventions are fixed. The protocol precedes these fits but follows inspection of the original outcomes. These paired comparisons reuse sources and draws and supply no fresh confirmation.

All 208 natural configurations are rerun for the three raw-augmented geometries under both scalers. On the four formerly reserved families, KAHM loss changes from 0.8419 to 0.8422, an increase of 0.03%. Min–max KAHM remains 4.29% above raw RBF and 8.00% above ridge. Its measured warm method time is 0.5561 seconds, compared with 0.3804 for standardized KAHM and 0.1762 for the remeasured raw RBF. The timings include a retained Saugeen search/cleanup outlier and do not identify an intrinsic cost of scaling.

All 24 original controlled draws per law and both landmark counts are also reused. The principal ordering persists: raw RBF is better on autoregression and Duffing, while KAHM local features are better than raw and both local controls on Van der Pol. Individual changes can be substantial: 256-landmark Duffing KAHM MSE falls by 7.35%, yet remains above the raw, affine, and locality controls. Appendix SI retains the complete paired cells, source heterogeneity, and resource accounting. The original benefit diagnostic is not refitted or reevaluated under min–max geometry.

S7 Preserving raw forecasts while testing geometric utility

An additive experiment separates the contribution of KAHM features from changes to the raw predictor. It tests whether a local geometric correction improves a fixed raw forecast, how this improvement depends on training size and raw-decoder capacity, and whether a validation rule identifies useful corrections on four additional archive collections. The protocol, candidate grid, source roster, and implementation were archived before full evaluation. This was a prospective execution checkpoint informed by earlier findings, without external preregistration.

S7.1 Forecasts, controls, and selection

The raw baseline is a finite Gaussian Nyström predictor with appended linear history coordinates, column standardization, and a ridge head. Its requested M landmarks and regularization are fitted using the raw history only. For each $g \in \{\text{KAHM}, \text{affine}, \text{locality}\}$, we fit a separate ridge head on the local-coordinate features of Equation (S7), using the raw model's training residuals as targets. The additive forecast is Equation (S3). The correction has its own standardization and ridge penalty; it does not change raw distances, landmarks, coefficients, or regularization, and it introduces no additional Nyström basis. Residuals are in-sample training residuals, not cross-fitted residuals. This choice is part of the evaluated procedure and can limit the structure left for the correction to learn.

All geometries use the same partitions, principal-direction configuration, reference-selection rules, and fitting seeds. The geometry bank uses three blocks of 2, 4, and 8 local models, requested rank two, at most 24 references per local model, and at most 512 fitting histories. KAHM uses the stable Gaussian query arithmetic described in Section S4; affine and locality controls do not fit KAHM kernels. The concatenated raw-plus-KAHM architecture from the original natural study is also fitted under this extension's limits. It preserves raw information but changes the finite kernel geometry, so it is a different complete procedure, not a restricted version of the additive model. Raw ridge, persistence/seasonal-naive prediction, and a fixed histogram-gradient-boosting control complete the comparison.

Validation first chooses the raw RBF penalty from $\{0.01, 1, 100\}$. Holding this raw forecast fixed, each correction searches the same three penalties and weights $\gamma \in \{0, 0.25, 0.5, 1\}$. There are ten distinct correction candidates: one zero correction and nine positive-weight choices. Ties favor smaller γ, followed by the fixed penalty order. These extra validation opportunities are part of the additive procedure; equal landmark counts do not equalize search size, effective function classes, or cost. The main KAHM policy uses its selected correction only if validation MSE is at least 5% below that of the selected raw baseline. A second declared policy applies the same threshold to the validation-best choice across all three geometries. The 5% threshold is a research sensitivity, not a test-set guarantee or an application-derived minimum. Every policy pays for the corrections it examines, including corrections it declines.

S7.2 Training size, raw-decoder capacity, and archive sources

The synthetic factorial crosses the three forecasting laws in Appendix SB, $N \in \{32, 128\}$ independent training episodes, and $M \in \{16, 64\}$ raw RBF landmarks. Twelve fresh repetitions use seeds 9100–9111, for 144 configurations. Within a law and repetition, the 32 training episodes are a prefix of the same 128-episode collection, and all four $N \times M$ settings share 64 validation and 128 test episodes. Episodes retain length 72, history 12, offsets 1, 4, and 12, and 13 origins per episode. All training origins can fit the decoder; the 512-history limit applies only to the geometry bank. Selected forecasts use training-only fitting, without adding validation episodes, so N remains an interpretable factor. This changes the raw predictor's capacity while holding the correction's declared architecture fixed; it does not vary every source of model complexity.

Synthetic MSE averages target coordinates and origins within episodes, then episodes. The primary loss differences compare additive KAHM with raw RBF and concatenated KAHM in each of the 12 law–N–M cells. All 24 contrasts use a separate family of 95% Bonferroni paired Student intervals over 12 complete repetitions. Their coverage relies on a normal approximation for repetition-level differences; origins and horizons do not increase the replication count. Affine, locality, and practical-control means are reported without promoting their descriptive comparisons into that primary interval family.

The archive-source evaluation uses Weather, Solar Weekly, COVID-19 Deaths, and Car Parts with missing values from the Monash archive (Godahewa et al., 2021; 2020l;j;i;h). They were absent from the original eight-source study. Solar Weekly is an aggregation of simulated NREL solar-production series; it is not an independent observational solar deployment. Three training-prefix-eligible series per collection are selected by a fixed hash ordering. Eligibility and selection cannot use validation/test forecasting performance, and selected series are not replaced because later evaluation is unfavorable or unavailable. The roster is listed in Appendix SJ.4.

Each series has chronological 60/20/20 training, validation, and test segments. Weather and COVID use a 28-step history and all seven next steps as direct targets; Solar Weekly and Car Parts use a six-step history and two-step target. Fits use at most 512 available training windows and evaluations at most 64 origins per segment, sampled evenly. After validation selection, selected configurations are refitted on admissible pretest windows under the same cap. Inputs use only past observations, including already observed test values at later origins; test targets cannot retune the fitted model. Missing values are not interpolated. A training-only seasonal-difference MSE, with a declared first-difference fallback, supplies the fixed scale. The two requested landmark counts give 24 configurations. Aggregation weights capacities equally within series, series equally within source, and the four sources equally. These are descriptive comparisons on four selected collections, not 24 independent domains or a random sample of applications.

The main model-pipeline cost covers raw search, geometry and correction search, validation forecasts, selected refits, final prediction, and model serialization. Shared raw work is charged once for an additive procedure or policy. Data preparation is recorded separately, as are model size and prediction cost. Imports, downloads, prediction-artifact writes, and research reporting are outside model timing. Policy costs are unions of measured required components, not separately timed full-policy executions. One task and one numerical-library thread are active during timing. Wall time and available CPU components are recorded separately. The accuracy–cost comparison therefore describes this implementation and hardware, without claiming a minimum necessary representation complexity. Additional fitting and audit details appear in Appendix SJ.4.

S7.3 Complete outcomes and fit coverage

Tables S1, S43, and S44 report every synthetic cell and primary contrast. Five of twelve comparisons with raw RBF favor additive KAHM, none favors raw RBF, and seven remain unresolved. Against concatenated KAHM, six favor the additive construction, four favor concatenation, and two remain unresolved. Simpler additive geometries have lower mean error in every Van der Pol and Duffing cell; the raw-baseline contrasts do not establish an advantage specific to KAHM.

Table S1: New synthetic factorial: mean test MSE multiplied by 10^3, over 12 fresh complete repetitions per cell. N counts training episodes; M counts raw RBF landmarks. All correction heads are ridge heads. These are cell means; simultaneous primary intervals appear in Table S44.

Law	N	M	Raw RBF	Concat.	+KAHM	+Affine	+Locality
Nonlinear AR	32	16	4.8715	4.9851	4.7525	4.7586	4.7584
Nonlinear AR	32	64	5.0133	5.3388	5.0157	5.0190	5.0173
Nonlinear AR	128	16	4.6648	4.6592	4.5701	4.5805	4.5753
Nonlinear AR	128	64	4.4806	4.7006	4.4806	4.4840	4.4832
Van der Pol	32	16	22.0913	7.0925	15.6100	15.1578	12.8848
Van der Pol	32	64	9.2066	2.6518	8.2892	8.3880	8.0522
Van der Pol	128	16	20.5463	7.5690	14.4955	13.8986	11.5027
Van der Pol	128	64	8.4031	2.4087	7.4507	7.5359	7.0687
Duffing	32	16	7.3080	8.5619	6.6154	5.3215	5.6464
Duffing	32	64	5.3159	5.0348	5.2959	4.7693	5.0240
Duffing	128	16	6.6937	7.6252	5.8873	4.7225	4.8299
Duffing	128	64	4.3215	4.0529	4.2667	3.6788	3.8379

All 144 synthetic configurations complete. On the archive roster, 22 of 24 configurations complete every procedure. Both capacities of Car Parts T1324 have 23 training histories but only seven distinct ones;

the declared eight-group bank fails. Concatenated KAHM and all three additive geometries fail on both configurations, giving eight failed fits and four unavailable dependent policy outputs. These are shared bank-fitting failures, not an isolated KAHM-kernel failure. Raw ridge, raw RBF, naive, and boosting complete all 24 configurations.

Table S2: New archive-source coverage and losses. Full-roster scaled MSE uses all 24 configurations and is undefined (dash) for procedures with failed fits. Matched loss and wall seconds use only the post-hoc 22-configuration common-support diagnostic (11 series, including two Car Parts series), with equal source weights. Pipeline costs include all required searches, validation, selected refit, prediction, and serialization. Policy costs are unions of measured required components, including rejected corrections; they are not separately timed full policy executions. Conditioning on success does not complete the prespecified full-roster evaluation.

Procedure	Outputs	Full-roster MSE	Matched MSE	Matched seconds
Raw ridge	24/24	1.2848	0.8466	0.0015
Raw RBF	24/24	9.6633	9.1730	0.0037
Concatenated KAHM	22/24	—	1.2111	0.0691
Additive KAHM	22/24	—	9.1443	0.0498
Additive affine	22/24	—	9.3312	0.0369
Additive locality	22/24	—	9.5624	0.0379
Persistence / seasonal naive	24/24	1.5720	1.2062	0.0012
Histogram gradient boosting	24/24	10.7552	10.3174	0.6380
KAHM 5% rule	22/24	—	9.2460	0.0467
All-geometry 5% rule	22/24	—	9.2624	0.0869

The full-roster geometric and policy losses are undefined. The separate post-hoc matched-success analysis conditions on 22 configurations, with eleven series and equal source weights. Additive KAHM gains only 0.31% over raw RBF at 13.29 times its measured model-pipeline cost; ridge is more accurate and cheaper. Neither policy improves aggregate loss on this subset. COVID is the only source mean improved by additive KAHM, Solar is unchanged, and Weather and Car Parts worsen; excluding COVID reverses the aggregate direction. Appendix SJ.5 retains failures, source means, interval families, chosen corrections, and decoder sizes; Appendices SJ.6 and SJ.7 give complete costs and the saved-forecast residual-alignment identity.

S8 Shared proxy-metric geometry and donor residuals

The additive experiment leaves two methodological questions open: whether residuals from models that exclude their query observations improve correction fitting, and whether a supervised input metric improves KAHM forecasts. We study these factors with a raw ridge baseline and equally trained geometric controls. The protocol, input roster, implementation, and numerical dependencies were archived before full evaluation. The design uses earlier findings but evaluates additional simulation draws and four additional source collections.

S8.1 Baseline, residual targets, and matched procedures

The base f_0 is a raw-history ridge predictor, with its penalty selected on a separate validation segment. Four gates use the same partition and reference-selection rules: KAHM reconstruction, affine reconstruction, locality-aware affine, and a center-distance RBF gate. All retain the affinity-gated principal coordinates of Equation (S7). The RBF-local control therefore describes local affine experts; it is distinct from the raw Nyström predictor. Requested banks contain $2, 4, 8$ groups, rank two, at most 24 references per group and 512 fitting histories. KAHM queries use the log-shifted Gaussian evaluation in Section S4.

For each gate we fit three procedures: an identity metric with in-sample residuals, an identity metric with out-of-fold (OOF) donor residuals, and a learned metric with those same OOF residuals. The geometry is fitted on the training histories, subject to the 512-history cap. All three correction heads use exactly the same eligible rows, so the natural-data OOF procedure does not gain a different correction sample simply by excluding its initial donor prefix. For gate family g, let $\Gamma_g(x)$ denote its local feature vector and let q_g be the fitted residual correction, including column standardization and a ridge readout. Its output has H

coordinates, one for each forecast offset. The forecast is

$$\widehat{f}_g(x) = f_0(x) + \gamma q_g\big(\Gamma_g(x)\big), \qquad \gamma \in \{0, \tfrac{1}{4}, \tfrac{1}{2}, 1\}.$$

The baseline and correction penalties use the same seven-value grid $\{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 100\}$. The base is selected first and held fixed. The candidate set contains zero correction and 21 positive penalty–weight combinations. Validation chooses a nonzero correction only when its error is strictly smaller than the baseline error; subsequent ties retain the earlier candidate in the fixed grid order. The synthetic and archive evaluations retain their selected training-only fits, without a pretest refit. The zero correction makes the baseline available during selection, but validation selection does not guarantee improved test error.

Synthetic donors exclude complete independent episodes in four folds. Natural-series donors use three forward blocks following an initial 40% of training origins; every donor target must have been available at the first forecast origin of its query block. A donor selects its own ridge penalty using an inner split of its own available data. The final base's outer-validation choice never feeds backward into donor fitting. Appendix SK.2 gives the boundary and small-sample rules. Episode exclusion exploits independent synthetic trajectories, whereas the forward-block construction respects target availability within a natural series. The appropriate cross-validation design depends on the forecasting assumptions (Bergmeir et al., 2018).

Residuals from different baselines. Condition on the fitted representation Γ, final baseline f_0, and donor baseline f_d. Let (X, Y) be a fresh evaluation pair with $Y \in \mathbb{R}^H$, and write $Z = \Gamma(X)$. All expectations in this paragraph refer to that evaluation distribution. Define

$$m_0(Z) = \mathbb{E}[Y - f_0(X) \mid Z], \qquad b_d(Z) = \mathbb{E}[f_0(X) - f_d(X) \mid Z].$$

The correction minimizing the donor's population residual risk is $q_d(Z) = m_0(Z) + b_d(Z)$. Assuming Y, $f_0(X)$, and $f_d(X)$ have finite second moments, define $R_H(f) = H^{-1}\mathbb{E}\|Y - f(X)\|_2^2$. Then

$$R_H(f_0) - R_H(f_0 + q_d \circ \Gamma) = \frac{1}{H}\mathbb{E}\|m_0(Z)\|_2^2 - \frac{1}{H}\mathbb{E}\|b_d(Z)\|_2^2. \tag{S11}$$

This elementary projection identity, proved with its fitted-correction extension in Appendix SK.1, explains why donor exclusion alone does not establish improvement of a different final predictor. Chronological folds need not share the same distribution. Our recorded donor/final disagreement on correction-training rows is a descriptive diagnostic, not an estimate of either oracle term in Equation (S11).

S8.2 A residual-supervised metric shared across geometries

We learn a positive diagonal input metric through a Gaussian kernel-regression proxy, following the supervised metric-learning principle of Weinberger & Tesauro (2007). Let $z_i \in \mathbb{R}^d$ denote a training-standardized history, with $d = L$ equal to the history length, and let $s_i \in \mathbb{R}^H$ denote its donor residual vector, centered by output and divided by one pooled residual standard deviation. The shared scalar division preserves relative original-unit horizon weights. On at most 256 training correction rows, set

$$d_\theta^2(z_i, z_j) = \sum_{k=1}^{d} e^{\theta_k}(z_{ik} - z_{jk})^2, \qquad p_{ij}(\theta) = \frac{\exp\{-d_\theta^2(z_i, z_j)/(2\sigma^2)\}}{\sum_{l \in \mathcal{A}_i} \exp\{-d_\theta^2(z_i, z_l)/(2\sigma^2)\}}, \quad j \in \mathcal{A}_i, \tag{S12}$$

$$\mathcal{J}(\eta) = \frac{1}{|\mathcal{I}|H} \sum_{i \in \mathcal{I}} \left\| s_i - \sum_{j \in \mathcal{A}_i} p_{ij}(\theta) s_j \right\|_2^2 + \frac{0.05}{d}\|\theta\|_2^2, \qquad \theta = \eta - \overline{\eta}\mathbf{1}. \tag{S13}$$

The admissible donor-index set $\mathcal{A}_i$ excludes the query's episode for synthetic data, or origins separated from the query by at most $L + H$ steps for natural data. The set $\mathcal{I}$ contains queries with at least one admissible donor. The vector $\eta \in \mathbb{R}^d$ contains free optimization parameters, $\overline{\eta} = d^{-1}\sum_k \eta_k$, and θ is its centered version. Thus e^{θ_k} is the metric weight of input coordinate k. The bandwidth σ^2 is the identity-metric median positive

admissible squared distance and remains fixed during optimization. The box $|\eta_k| \leq \log(16)/2$ bounds the ratio between the largest and smallest metric weights by 16; their geometric mean is one. Indeed, $\theta_k - \theta_l = \eta_k - \eta_l$ is at most $\log 16$, and $\sum_k \theta_k = 0$. The box constraint applies to η; the centered coordinates θ_k need not lie in the same interval.

The limited-memory bound-constrained quasi-Newton optimizer L-BFGS-B uses the analytic gradient and at most 40 iterations. Identity is the declared fallback for an unavailable or nonfinite objective, constant residuals, or a worse final iterate. Finite improved iterates at the iteration limit are retained with their convergence status. No validation or test outcomes enter this optimization. Natural metric donors may be earlier or later within the training period: this is offline supervised fitting, separate from the forward construction of baseline residuals.

The same learned weights are applied *after* input standardization for all four geometric gates; subsequent partitioning and local models are refitted under that metric. This is a supervised kernel proxy transferred to KAHM and its controls, not end-to-end minimization of KAHM forecasting loss. Its training objective may decrease without improving the final decoder. The comparison with equally trained local controls tests whether any benefit is specific to the KAHM reconstruction.

S8.3 Declared evaluation and complete accounting

Twelve independent repetitions of each of the nonlinear AR, Van der Pol, and Duffing laws give 36 synthetic tasks. Each uses 128 training, 64 validation and 128 test episodes, history 12, direct offsets $1, 4, 12$ and the observation models in Appendix SB. Six prespecified comparisons per law isolate learned-OOF KAHM versus ridge, learned versus identity metric for OOF KAHM, OOF versus in-sample targets at the identity metric, and learned-OOF KAHM versus each of the three equally trained geometric controls. All 18 comparisons receive two-sided Bonferroni-adjusted paired Student intervals over the 12 complete repetition differences. Their coverage uses a normal approximation; origins and horizons do not increase the replication count. These fresh simulations supply new stochastic evidence on known laws, not new-domain confirmation.

Four additional Monash collections (Godahewa et al., 2021) provide Australian state electricity demand, Bitcoin activity indicators, Melbourne pedestrian counts, and KDD2018 air quality. Three series per collection are selected using training-prefix eligibility and fixed hash ordering. The 12-series roster and split rules are specified in Appendix SK.2. These collections were not used in the earlier evaluations, although some application domains overlap. They are not a random sample of future forecasting applications. Natural results use equal weights for series within source and for the four sources; overlapping windows do not create independent source replications.

The 16 procedures comprise the 12 geometric corrections, ridge, raw Nyström-plus-linear regression, naive prediction, and a jointly fitted raw-plus-KAHM ridge readout. The last is a secondary architecture control and differs from the earlier concatenated Nyström procedure. The protocol reduces group counts when local training histories are insufficiently distinct and uses predefined ridge fallbacks for unavailable correction fits. Fewer than eight eligible correction rows or a failed secondary joint head invokes a fallback for the affected arms while retaining the available baselines. These fallback rules belong to this experiment; the additive study in Section S7 used a different failure policy. Full-roster results remain unavailable if a primary baseline or required task output is missing.

Warm pipeline cost charges each method for its complete required baseline, donor, metric, bank, head-selection and serialization work, including rejected candidates and fallbacks. Shared components are charged once per standalone procedure. These are measured component sums on one host, not repeated end-to-end deployment timings or minimum possible complexity. Tables S3 and S4 report the declared comparisons; Appendix SK.4 retains all contrasts, source-level results, fitting diagnostics, and resource records.

S8.4 Results: predictive gains and their attribution

All 48 tasks and 768 method–task forecast outputs completed, without fallbacks. Learned-OOF KAHM reduces mean synthetic MSE relative to ridge by 11.18% on nonlinear AR, 53.26% on Van der Pol, and 19.44% on Duffing. All three adjusted paired intervals favor the correction. However, the three learned-versus-identity

metric intervals and the three OOF-versus-in-sample target intervals all include zero. These comparisons do not establish equivalence; they leave the incremental benefits of the two proposed changes unresolved at the declared precision. Among nine comparisons with equally trained geometric controls, none favors KAHM with an interval excluding zero. Five favor the control: locality and RBF-local on Van der Pol, and all three on Duffing. The other four remain unresolved. Thus the 18-contrast family contains three favorable, five unfavorable, and ten unresolved results for its designated candidates (Appendix SK.4). The secondary raw RBF predictor also has lower mean MSE than learned-OOF KAHM on every law; this mean ranking has no additional confirmatory interval.

On the four archive collections, learned-OOF KAHM has source-balanced scaled MSE 1.272128 versus ridge's 1.300458, a descriptive 2.18% reduction. Its charged mean pipeline time is 0.1622 s versus 0.001299 s for ridge, or 124.9× on this host. The identity-metric in-sample KAHM procedure achieves a lower loss, 1.264589 (2.76% below ridge), at 39.3× ridge's time. The lowest archive mean among the specified procedures is 1.257888 for identity-metric, in-sample RBF-local correction. Learned-OOF affine and locality also have lower means than learned-OOF KAHM, at 1.268679 and 1.266922, respectively. Table S4 reports the archive accuracy changes alongside their measured computational cost; the added metric stage is not supported as a default improvement by these results.

Learned-OOF KAHM improves the mean in each of the four collections, but improves only seven of the 12 individual series, worsens four, and exactly retains ridge on one through zero correction. Omitting each collection in turn leaves descriptive aggregate gains of 1.49–3.35%. This sensitivity check shows that the sign does not depend on retaining any one collection; it does not expand the population represented by four selected collections or supply an independent replication. These results establish improvement over the selected ridge reference in the stated settings while leaving a KAHM-specific advantage unsupported. The gains over ridge can also be obtained by simpler geometric controls under these settings.

Table S3: Fresh synthetic repetitions: mean test MSE $\times 10^3$ over all 12 independent repetitions per law, with equal episode, origin and horizon weights. Every declared procedure is shown. A dash denotes an incomplete full roster; baseline fallbacks retain their actual forecasts. Differences are assessed by the 18 prespecified paired intervals in Table S54, not by ranking these means.

Procedure	Nonlinear AR	Van der Pol	Duffing
Raw ridge	5.057	44.140	6.916
Raw RBF	4.442	8.316	4.172
Persistence / seasonal naive	8.549	293.407	37.977
Joint raw + KAHM	4.465	17.009	5.277
Fixed, IS KAHM	4.478	21.380	5.468
Fixed, IS Affine	4.470	18.986	4.706
Fixed, IS Locality	4.503	13.699	4.508
Fixed, IS RBF local	4.427	15.987	5.318
Fixed, OOF KAHM	4.481	21.383	5.469
Fixed, OOF Affine	4.473	18.983	4.721
Fixed, OOF Locality	4.506	13.700	4.512
Fixed, OOF RBF local	4.431	15.988	5.321
Learned, OOF KAHM	4.492	20.629	5.572
Learned, OOF Affine	4.460	18.879	4.736
Learned, OOF Locality	4.530	13.968	4.536
Learned, OOF RBF local	4.408	15.460	5.328

Table S4: Natural-data forecasting results: equal-collection means of equal-series scaled MSE and charged standalone wall seconds, covering four collections and three series per collection. Scaling uses training-only seasonal-difference MSE (first-difference fallback). IS denotes in-sample residuals and OOF denotes out-of-fold residuals. Change is relative to raw ridge; negative is favorable. Fallbacks predict raw ridge and retain attempted computation. Missing tasks make the affected full-roster means undefined. These are descriptive results, with no population confidence intervals or independence claims for adjacent origins.

Procedure	Outputs	Scaled MSE	Change (%)	Seconds	Fallbacks
Raw ridge	12/12	1.3005	0.00	0.001	0
Raw RBF	12/12	1.9483	49.82	0.005	0
Persistence / seasonal naive	12/12	1.8476	42.07	0.004	0
Joint raw + KAHM	12/12	1.4530	11.73	0.058	0
Fixed, IS KAHM	12/12	1.2646	-2.76	0.051	0
Fixed, IS Affine	12/12	1.2584	-3.24	0.029	0
Fixed, IS Locality	12/12	1.2635	-2.84	0.028	0
Fixed, IS RBF local	12/12	1.2579	-3.27	0.027	0
Fixed, OOF KAHM	12/12	1.2767	-1.83	0.056	0
Fixed, OOF Affine	12/12	1.2731	-2.11	0.034	0
Fixed, OOF Locality	12/12	1.2735	-2.07	0.032	0
Fixed, OOF RBF local	12/12	1.2746	-1.99	0.031	0
Learned, OOF KAHM	12/12	1.2721	-2.18	0.162	0
Learned, OOF Affine	12/12	1.2687	-2.44	0.142	0
Learned, OOF Locality	12/12	1.2669	-2.58	0.141	0
Learned, OOF RBF local	12/12	1.2848	-1.20	0.139	0

S9 Training local reconstruction for forecasting

This experiment asks whether forecasting error can usefully train the KAHM reconstruction and its gates. We compare this intervention with gate training alone and with equally parameterized interventions in three non-KAHM local models. The implementation, task roster, prepared inputs, and analysis were publicly archived before evaluation on split D. The design was informed by earlier experiments; the archive is a prospective execution checkpoint, rather than an external preregistration.

S9.1 Horizon-specific reconstruction and residual prediction

For a scalar series observed through time t, let $x_t = (y_{t-L+1}, \ldots, y_t)^\top \in \mathbb{R}^L$ be the history and y_{t+h} the target at forecast offset h. Training standardization gives $z_t \in \mathbb{R}^L$. Each local model c has m_c reference histories stored as the rows of $Q_c \in \mathbb{R}^{m_c \times L}$, their mean $\mu_c = m_c^{-1} Q_c^\top \mathbf{1}$, and an orthonormal principal basis $P_c \in \mathbb{R}^{L \times r_c}$. Three partition blocks request 2, 4, and 8 local models, respectively, with at most 24 references and effective rank $r_c \le 2$ per model. The reference sets, principal directions, and positive kernel regularizers λ_c are fitted on split A.

Let $f_{0,h}$ be a raw-history ridge forecast fitted on A. For gate family g, the corrected forecast is

$$\widehat{y}_{t+h} = f_{0,h}(x_t) + \gamma_h \left\{ a_h + \beta_h^\top \widetilde{\Gamma}_{g,h}(z_t; \theta_h) \right\}, \qquad \gamma_h \in \{0, 1/4, 1/2, 1\}. \tag{S14}$$

Here $\Gamma_{g,h}$ concatenates the gates and gated principal coordinates of Equation (S7); the tilde denotes column standardization fitted on A. The scalar a_h is an unpenalized intercept, and β_h is a coefficient vector of the same dimension as the feature map. The baseline penalty is selected by an inner split of A, after which the baseline is refitted on A and held fixed. Each local model therefore learns residuals of the same baseline.

For block b and horizon h, the parameter $\xi_{b,h} \in [-\log 4, \log 4]$ changes the kernel bandwidth. With Σ_c the positive-definite covariance of the projected reference coordinates, define

$$D_{c,ij} = \frac{\|\Sigma_c^{-1/2} P_c^\top (q_{ci} - q_{cj})\|_2^2}{2r_c}, \qquad d_{c,i}(z) = \frac{\|\Sigma_c^{-1/2} P_c^\top (z - q_{ci})\|_2^2}{2r_c},$$

where $q_{ci} \in \mathbb{R}^L$ is reference i as a column vector. For $r_c > 0$,

$$K_c(\xi)_{ij} = \exp\{-D_{c,ij}e^{-2\xi}\}, \qquad k_c(z;\xi)_i = \exp\{-d_{c,i}(z)e^{-2\xi}\},$$
$$w_c(z;\xi) = (K_c(\xi) + \lambda_c I_{m_c})^{-1} k_c(z;\xi), \qquad r_c^{\mathrm{K}}(z;\xi) = \frac{Q_c^\top w_c(z;\xi)}{\mathbf{1}^\top w_c(z;\xi)}. \tag{S15}$$

A rank-zero local model returns its center. A second parameter $\eta_{b,h} \in [-\log 4, \log 4]$ multiplies the block's gate logits by $e^{\eta_{b,h}}$ before softmax normalization. Thus θ_h contains three bandwidth parameters and three gate-temperature parameters. Zero parameters recover the fixed model.

Bandwidth changes affect both the reference Gram matrix and the query kernel vector. To see their effects explicitly, suppress local indices, write $M = K + \lambda I$, $w = M^{-1}k$, $s = \mathbf{1}^\top w$, and $r = Q^\top w/s$. Differentiating $Mw = k$ and then the quotient gives

$$dw = M^{-1}(dk - dM\, w), \qquad dr = \frac{(Q^\top - r\mathbf{1}^\top)\, dw}{s}. \tag{S16}$$

These identities require invertible M and $s \neq 0$. They also show why a small signed normalizer can amplify changes in the reconstruction. The implementation shifts query log kernels to avoid common underflow and rejects candidates for which $|\sum_i w_i| / \sum_i |w_i| \leq 10^{-10}$ on A, B, or C. This ratio is defined as zero when all weights vanish. If a selected model fails this check on D, the baseline supplies forecasts for the entire affected horizon.

S9.2 Selection, controls, and evaluation

For each geometry candidate and $\alpha \in \{0.01, 0.1, 1\}$, a ridge head minimizes the sum of squared A residuals plus $\alpha\|\beta_h\|_2^2$. Geometry training minimizes B forecast MSE at $\gamma_h = 1$, normalized by an A-only target-variance scale. Split C chooses the penalty and correction weight separately for each horizon; the zero correction is retained unless a candidate has strictly smaller C error. Split D evaluates the selected forecasts. Its labels are unavailable to every fitting and selection step.

Affine, locality, and RBF-local controls each train three shape parameters and three gate temperatures. The shape changes are, respectively, a principal-projection gain, a balance between projection residual and center distance, and tangent/normal anisotropy. Appendix SL defines these maps. Fixed, gate-trained, and shape-trained versions of the four families give twelve geometric procedures. Raw ridge, raw Nyström RBF, histogram boosting, seasonal naive, and a C-selected non-KAHM policy complete the 17-method comparison. The policy chooses one complete method from its C error and is charged for all candidate computations. Each geometry optimization has the same 256-evaluation ceiling per horizon and penalty. Equal parameter counts and evaluation limits do not imply equal function classes or wall time.

The synthetic evaluation uses twelve independent repetitions of each of the nonlinear AR, Van der Pol, and Duffing laws. A/B/C/D contain 64/32/32/128 independent episodes of length 72. Histories have length 12 and offsets are 1, 4, and 12; A/B/C are capped at 512/256/256 rows, and D contains 1,664 origins. The 18 primary contrasts compare reconstruction-trained KAHM with raw RBF, gate-trained KAHM, each of the three shape-trained local controls, and the non-KAHM policy on each law. Two-sided Bonferroni-adjusted Student intervals use twelve paired repetition differences and 11 degrees of freedom. Their simultaneous coverage relies on the stated normal approximation and applies only to this family.

The natural evaluation selects ten series from Dominick retail, rideshare prices, Wikipedia traffic, and Sunspot daily data (Godahewa et al., 2021). Selection uses training-prefix eligibility and fixed identifier hashes. Chronological A/B/C/D segments contain 50/20/15/15 percent of each series; missing windows are omitted without interpolation. The three selected rideshare series have no complete B windows. Consequently, the planned four-collection aggregate is undefined. The available seven series support a separate descriptive comparison that weights series equally within each of three collections and then weights the collections equally. Full source details, selection chronology, scaling, and the preceding nine-task gate-training pilot appear in Appendix SL.

S9.3 Predictive results and computational cost

All 43 feasible tasks completed, producing 731 of 782 planned method–task records. The 51 unavailable records belong to the three rideshare tasks. No additional task failures, invalid objective evaluations, or numerical fallbacks occurred. Replay of the saved selected states and a separate calculation of the losses and intervals verified the reported results; these checks establish computational consistency, not independent experimental replication.

Reconstruction training lowers KAHM's mean synthetic MSE relative to its fixed version by 0.42%, 61.81%, and 12.76% on AR, Van der Pol, and Duffing, respectively. Relative to gate training alone, the reductions are 0.09%, 41.43%, and 5.51%. Of the 18 primary intervals, 2 favor KAHM, 6 favor its comparator, and 10 include zero (Table S60). The favorable contrasts are against raw RBF on AR, with a 2.87% reduction and absolute MSE-difference interval $[6.65\times10^{-6}, 2.776\times10^{-4}]$, and against gate-trained KAHM on Van der Pol, with a 41.43% reduction and interval $[0.004349, 0.007709]$. Differences are comparator MSE minus KAHM MSE.

Shape-trained affine and the selected policy are more accurate than KAHM on Van der Pol; the policy selects boosting in all twelve repetitions. Raw RBF, shape-trained affine, shape-trained locality, and the policy outperform KAHM on Duffing. No favorable primary interval compares KAHM with an equally shape-trained non-KAHM local control. Direct training therefore improves KAHM in some settings, particularly Van der Pol, without establishing an advantage specific to its reconstruction rule. Intervals containing zero leave their comparisons unresolved and do not establish equivalence.

On the available natural collections, KAHM has scaled MSE 0.371935, compared with 0.385625 for ridge, a 3.55% reduction. Raw RBF (0.363007), boosting (0.352201), and shape-trained RBF-local (0.363835) are more accurate. The ridge gain is concentrated in Dominick; excluding that collection leaves a 0.00229% improvement across the other two collections. Twelve of the 21 available natural horizons select zero KAHM correction, including every Sunspot horizon. Table S61 reports all collection-specific results.

The 3096 geometry fits use 406832 objective evaluations. Of these fits, 2,491 report convergence, 604 reach their evaluation budget, and one has an abnormal optimizer stop; each retains its best finite candidate. Mean synthetic pipeline time is 10.725 s for reconstruction-trained KAHM, 0.914 s for gate-trained KAHM, 0.0774 s for fixed KAHM, 0.0172 s for raw RBF, and 0.653 s for boosting. On the seven available natural tasks, KAHM averages 7.929 s, compared with 0.0130 s for raw RBF and 0.655 s for boosting. Table S62 and Figure S1 report the complete accuracy and resource comparison. These costs include model fitting, selection, querying, and state serialization; they do not measure online update latency.

S10 Budgeted online adaptation

A chronological development experiment evaluates whether updating local KAHM geometry improves forecasts under finite reference and replay budgets. Twelve streams, six methods, and two capacity settings yield 144 runs and 387,900 scored forecasts. Four stationary nonlinear AR streams are paired with four recurring-regime streams through their innovations. The recurring streams alternate between regimes A and B every 256 observations, with both regimes represented in the initial fitting period. Regime labels are hidden from the forecasting methods. Three Dominick weekly series and one Sunspot daily series supply natural-data comparisons. These series had been examined previously, so the results characterize development on selected inputs. Appendix SM specifies the streams, update schedule, and evaluation criteria.

Forecasts and temporal ordering. All geometric methods combine a sliding-window raw ridge baseline with ridge residual corrections. Fixed KAHM updates its correction heads while retaining the initial geometry. Adaptive KAHM, affine, and RBF-local methods share a reference-refresh policy; after a refresh, they re-encode the stored histories and rebuild their correction heads. The small setting allows 64 shared geometric references and 128 replay rows per horizon; the large setting allows 128 references and 256 replay rows. A Gaussian kernel ridge regression (KRR) comparator maintains separate sliding-window supports for each horizon using recursive add/delete inverse updates. Its forecast is a calibrated mixture of KRR and the same

raw ridge baseline. Different sharing across horizons means that these support limits do not equalize memory or computation.

The first 35% of each stream fits the initial model, the next 15% calibrates correction weights, and the final 50% evaluates forecasts. For each horizon, calibration chooses a weight from $\{0, 0.25, 0.5, 1\}$ using only targets observed by the calibration boundary. These weights remain fixed during evaluation. Previously issued forecasts are scored when their targets arrive; the newly available targets then enter the update before subsequent forecasts are issued. Bandwidths and gate temperatures are fixed throughout. This experiment therefore evaluates reference adaptation with delayed labels, rather than online training of the parameters in Section S9.

Numerical correction and evaluation outcome. The initial implementation experienced four correction-head failures and issued 8,659 framework fallback forecasts. A numerical amendment replaced accumulated normal equations with a centered, column-equilibrated augmented ridge solve using pivoted QR. The objective, penalties, inputs, scalers, reference policy, and evaluation criteria were unchanged. Repeating all 144 runs with the amended solver produced no unexpected model exceptions, framework fallbacks, or observed capacity violations. The two executions use the same streams and are not independent replications. Their protocols and source states were archived locally before their respective executions.

Table S5: Budgeted adaptive development study: scaled MSE, lower is better. Origins are averaged within horizon, horizons within series, and series within collection. The natural mean gives Dominick and Sunspot equal weight. The two capacity tiers are quotas, not equal-byte allocations. All non-ridge arms, including KRR, use calibration-selected mixing with the common ridge baseline. No losses are pooled with earlier studies.

Method	Stationary	Recurring	Dominick	Sunspot	Natural mean
Small: 64 shared references / 128 replay rows per horizon					
Raw ridge	2.2402	3.4583	1.6144	2.3697	1.9920
Fixed KAHM	2.2407	3.4583	1.6134	2.3697	1.9915
Adaptive KAHM	2.2399	3.4583	1.6135	2.3697	1.9916
Adaptive affine	2.2407	3.4583	1.6137	3.4004	2.5071
Adaptive RBF-local	2.2428	3.4583	1.6160	2.3697	1.9928
Sliding-window KRR	2.2491	3.3578	0.8685	2.3897	1.6291
Large: 128 shared references / 256 replay rows per horizon					
Raw ridge	2.0275	3.3623	0.8471	2.1045	1.4758
Fixed KAHM	2.0375	3.4031	0.8464	2.1045	1.4754
Adaptive KAHM	2.0280	1.376×10^8	0.8444	2.1045	1.4744
Adaptive affine	2.0364	1.061×10^6	0.8446	2.2085	1.5266
Adaptive RBF-local	2.0290	3.7036	0.8460	2.1045	1.4752
Sliding-window KRR	2.0551	3.2079	0.5954	2.1045	1.3499

Table S5 reports MSE normalized by a seasonal-difference scale fitted on the initial prefix. Both capacity settings fail the recurring-stream and natural-data accuracy criteria specified in Appendix SM. Adaptive KAHM's natural mean loss exceeds that of KRR by 22.26% in the small setting and 9.22% in the large setting. Only two of twelve natural horizon/series combinations select a nonzero KAHM correction at either setting; every Sunspot horizon selects zero. All small-setting recurring corrections are zero, whereas nine of twelve large-setting recurring corrections are active. The criteria were intended to decide whether this prototype warranted broader experimentation; they are descriptive thresholds, not hypothesis tests.

Feature extrapolation after an accurate solve. A large prediction error remains after the solver amendment. For recurring seed 20261101, origin 1554, and horizon 12, large-setting adaptive KAHM predicts 175,555.08 when the target is -0.9815. The largest absolute standardized query feature is approximately 3.20×10^6, compared with 392 among the currently supervised replay features. A fresh centered singular-value-decomposition solve reproduces the large correction. For the query's unnormalized signed kernel weights w_i, define $\rho = |\sum_i w_i| / \sum_i |w_i|$, with $\rho = 0$ when every weight vanishes. The examined state has $\rho = 0.354$,

above its 10^{-10} guard. Adaptive affine also exhibits severe feature extrapolation under a fresh solve. These diagnostic states show that accurate linear algebra and a nonvanishing KAHM normalizer are insufficient to control the correction. They do not separately identify the effects of scaling, delayed labels, and reference refreshes. Every issued forecast contributes to the reported loss.

Measured service cost. Every adaptive-KAHM stream meets the 20 ms screen for the 99th percentile of observation-step service time. The largest per-stream 99th percentiles are 9.89 and 11.75 ms in the small and large settings. Nevertheless, 4 of 21,662 evaluated observation steps exceed 20 ms; the percentile screen is not a per-step deadline. Service time includes updates, reference refreshes, feature and head reconstruction, and prediction. It excludes scoring and serialization. The prototype maintains geometry even when its correction weight is zero. Table S63 provides the complete resource measurements. The results support computational feasibility under this particular screen, but they do not establish a forecasting advantage for the adaptive KAHM procedure. The bounded, fixed-geometry corrections analyzed in the accompanying main paper define a different method and do not establish that this online procedure has been repaired.

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SA Supporting mathematical statements

This appendix separates information retained by a representation from the functions accessible to a specified decoder. Let $X \in \mathbb{R}^d$ be a random input and let $Y \in \mathbb{R}$ satisfy $\mathbb{E}[Y^2] < \infty$. All conditional expectations are defined up to almost-sure equality. Write $m(X) = \mathbb{E}[Y \mid X]$ for the conditional mean and $R(f) = \mathbb{E}[(Y - f(X))^2]$ for the risk of an input-based predictor f. For a measurable representation $U = \Gamma(X)$, define

$$R_X^{\text{Bayes}} = \mathbb{E}[(Y - m(X))^2], \qquad R_U^{\text{Bayes}} = \mathbb{E}[(Y - \mathbb{E}[Y \mid U])^2].$$

These are the minimum squared-error risks among all square-integrable predictors measurable with respect to X and U, respectively. Let P_X and P_U denote the distributions of X and U. For a function h of X, the notation $\|h\|_{L^2(P_X)} = (\mathbb{E}|h(X)|^2)^{1/2}$ denotes its population root mean square; the analogous definition applies to functions of U. We omit the distribution in this norm when the argument determines it. If the representation is learned, the statements apply conditional on its fitted state and under the resulting evaluation distribution.

Proposition S1 (Information retained by a representation)**.** *For every measurable representation $U = \Gamma(X)$,*

$$R_U^{\text{Bayes}} - R_X^{\text{Bayes}} = \mathbb{E}[(m(X) - \mathbb{E}[m(X) \mid U])^2].$$

For every square-integrable prediction $f(U)$,

$$R(f \circ \Gamma) - R_X^{\text{Bayes}} = \|m - f \circ \Gamma\|^2_{L^2(P_X)}.$$

Proof. Decompose the prediction error as $Y - f(U) = (Y - m(X)) + (m(X) - f(U))$. Because U is a function of X, the second term is measurable with respect to X. The cross term therefore vanishes:

$$\mathbb{E}[(Y - m(X))(m(X) - f(U))] = \mathbb{E}[\mathbb{E}[Y - m(X) \mid X](m(X) - f(U))] = 0.$$

Expanding the squared error gives the second identity. Among all U-measurable predictions, the second term is minimized by the conditional expectation $\mathbb{E}[m(X) \mid U]$. The tower property gives $\mathbb{E}[m(X) \mid U] = \mathbb{E}[Y \mid U]$, which proves the first identity. □

Proposition S2 (Restoring the outer score normalizer)**.** *Let C be the number of scores in one partition. For $\mathbf{T} = (T_1, \ldots, T_C)^\top \in [0,1]^C$, $\omega > 0$, and $\tau > 0$, define*

$$\ell_c = \omega \log(1 - T_c + \tau), \qquad \zeta = \log \sum_{j=1}^{C} \exp(\ell_j), \qquad \phi_c = \exp(\ell_c - \zeta).$$

The vector $\mathbf{T}$ and the pair (ϕ, ζ) generate the same sigma-algebra in exact arithmetic. The inverse is

$$T_c = 1 + \tau - \exp\{(\log \phi_c + \zeta)/\omega\}, \qquad c = 1, \ldots, C.$$

For several separately normalized partitions, the inverse requires one log normalizer ζ per partition.

Proof. Since $1 - T_c + \tau \geq \tau > 0$, each ℓ_c is finite and each ϕ_c is strictly positive. Taking logarithms gives $\log \phi_c + \zeta = \ell_c = \omega \log(1 - T_c + \tau)$, from which the inverse follows. Both the forward and inverse maps are continuous on their stated domains and hence measurable. Each object is therefore a measurable function of the other, which proves equality of their generated sigma-algebras. Applying the same argument separately to each partition proves the final statement. □

If scores are clipped before normalization, the inverse restores the clipped scores. It does not recover discarded reconstruction vectors. Moreover, the inverse is nonlinear, so linear decoders on $\mathbf{T}$ and on (ϕ, ζ) need not express the same functions or achieve the same fitted risk. The proposition concerns the outer affinity normalizer; the signed denominator inside a KAHM reconstruction is a different quantity.

Proposition S3 (A sufficient condition for an injective local code)**.** *Let $\mathcal{X} \subseteq \mathbb{R}^d$ be an input domain. For $c = 1, \dots, C$, fix a center $\mu_c \in \mathbb{R}^d$, a basis matrix $P_c \in \mathbb{R}^{d \times p_c}$, and a measurable gate $\phi_c : \mathcal{X} \to (0, \infty)$. Form $A \in \mathbb{R}^{(\sum_c p_c) \times d}$ by stacking the matrices $P_c^\top$. If A has full column rank d, then the code*

$$\Gamma_{\text{local}}(z) = \operatorname{col}\big(\phi(z), \{\phi_c(z) P_c^\top (z - \mu_c)\}_{c=1}^{C}\big)$$

is injective on $\mathcal{X}$ and has a measurable inverse on its image. Here col *concatenates column-vector blocks and $\phi = (\phi_1, \dots, \phi_C)^\top$. For a random input in $\mathcal{X}$, this code and the input generate the same sigma-algebra and consequently have the same Bayes risk for every square-integrable target.*

Proof. The code contains each strictly positive $\phi_c(z)$ and the vector $\phi_c(z) P_c^\top (z - \mu_c)$. Dividing that vector by its gate and adding the known vector $P_c^\top \mu_c$ recovers $P_c^\top z$. Stacking the recovered vectors yields Az. Full column rank implies that $A^\top A$ is invertible, so $z = (A^\top A)^{-1} A^\top (Az)$. This explicitly defines an inverse, proving injectivity. Division by a positive coordinate and multiplication by fixed matrices are measurable operations. Hence the inverse is measurable on the code's image. The code itself is a measurable function of z, so the generated sigma-algebras coincide. Their classes of measurable predictors, and thus their Bayes risks, coincide as well. □

The gates in Equation (S6) are positive in exact arithmetic because $\tau > 0$. Full rank of the basis stack is sufficient, not necessary: the gates may encode information absent from a rank-deficient linear stack. The condition must be assessed for each fitted bank. Very small gates and an ill-conditioned A can make the displayed inverse sensitive to numerical perturbations. Existence of the inverse also does not ensure that a particular finite decoder can approximate it.

Proposition S4 (A score symmetry and its prediction consequence)**.** *Let $Q : \mathbb{R}^d \to \mathbb{R}^d$ be an orthogonal linear map on the preprocessed input space. Suppose each reconstruction satisfies $r_c(Qx) = Q r_c(x)$ and Q preserves the identities of the local models and their partitions. Then each radial–angular score, each block affinity, and each block log normalizer in Section S3 is invariant under Q.*

For the branch distribution in Equation (S9)*, let $Q(x_1, x_2) = (-x_1, x_2)$. Any measurable representation $U = \Gamma(X)$ satisfying $\Gamma(QX) = \Gamma(X)$ almost surely obeys*

$$R_U^{\text{Bayes}} = \frac{1}{300} + \lambda + (1 - \lambda)\mathbb{E}[(S - \mathbb{E}[S \mid U])^2].$$

In particular, $R_U^{\text{Bayes}} \geq 1/300 + \lambda$, with equality whenever U determines S almost surely.

Proof. Orthogonality preserves Euclidean norms and inner products. Equivariance of r_c therefore preserves both $\|x - r_c(x)\|_2$ and the cosine between x and $r_c(x)$, including the specified zero-norm convention. Each radial and angular discrepancy, and hence their combined score, is unchanged. The affinities and log normalizers are fixed functions of the scores within each partition, so they are also unchanged.

For the branch distribution, $X_1 = T$ and $S = 2\{X_2 - c\cos(\pi X_1)\}$. Thus X determines both latent signal variables, and $m(X) = \sqrt{1 - \lambda}\, S + \sqrt{3\lambda}\, T$. Replacing T by $-T$ preserves its joint law with S and leaves U unchanged. For every bounded measurable function a of U, this symmetry gives $\mathbb{E}[Ta(U)] = -\mathbb{E}[Ta(U)]$, so $\mathbb{E}[T \mid U] = 0$. Since S is unchanged by the reflection, the same argument gives $\mathbb{E}[(S - \mathbb{E}[S \mid U])T] = 0$. It follows that

$$m(X) - \mathbb{E}[m(X) \mid U] = \sqrt{1 - \lambda}\,(S - \mathbb{E}[S \mid U]) + \sqrt{3\lambda}\, T.$$

The two terms on the right are orthogonal in L^2. Their squared norms are $(1 - \lambda)\mathbb{E}[(S - \mathbb{E}[S \mid U])^2]$ and $3\lambda\mathbb{E}[T^2] = \lambda$. Finally, the independent noise has variance $1/300$, so Proposition S1 yields the claimed identity and lower bound. □

The first assertion also applies when Q permutes reconstructors whose scores are combined by a minimum within a fixed group. Permuting distinct groups can instead permute the output coordinates and need not leave the representation unchanged. Moreover, a reflection of raw inputs need not commute with a fitted scaler. Finite fitted partitions and reference sets need not have the symmetry assumed by the proposition. A

measured small reflection difference is consequently a diagnostic, rather than a proof of the exact Bayes lower bound. An unrestricted decoder could in principle use arbitrarily small nonzero feature differences.

Proposition S5 (Approximate symmetry under a decoder budget)**.** *Let Q be a measurable, distribution-preserving involution on the input space: $Q(QX) = X$ almost surely and QX has distribution P_X. Let d_U be a measurable pseudometric on the representation space. For $\kappa \geq 0$, suppose the decoder g satisfies $|g(u) - g(v)| \leq \kappa d_U(u, v)$ for all u, v. Assume $m, g \circ \Gamma \in L^2(P_X)$ and*

$$\delta_\Gamma^2 = \mathbb{E}[d_U(\Gamma(X), \Gamma(QX))^2] < \infty.$$

Define the odd part of the conditional mean by $m_o = (m - m \circ Q)/2$. Then

$$R(g \circ \Gamma) - R_X^{\text{Bayes}} \geq \left(\|m_o\|_{L^2(P_X)} - \kappa\delta_\Gamma/2\right)_+^2,$$

where $(a)_+ = \max\{a, 0\}$. For the branch distribution, $\|m_o\|_{L^2(P_X)} = \sqrt{\lambda}$.

Proof. For $h \in L^2(P_X)$, define $Vh = h \circ Q$. Distribution preservation implies $\|Vh\|_{L^2} = \|h\|_{L^2}$, and the involution assumption implies $V^2 h = h$. Set $h_e = (h + Vh)/2$ and $h_o = (h - Vh)/2$. Then $Vh_e = h_e$ and $Vh_o = -h_o$. If a_e is any even function and b_o any odd function, distribution preservation gives $\mathbb{E}[a_e(X)b_o(X)] = \mathbb{E}[a_e(QX)b_o(QX)] = -\mathbb{E}[a_e(X)b_o(X)]$, proving orthogonality. For $f = g \circ \Gamma$, this yields

$$\|m - f\|_{L^2}^2 = \|m_e - f_e\|_{L^2}^2 + \|m_o - f_o\|_{L^2}^2 \geq \|m_o - f_o\|_{L^2}^2.$$

The Lipschitz assumption bounds the odd component of the prediction:

$$|f_o(X)| = \tfrac{1}{2}|g(\Gamma(X)) - g(\Gamma(QX))| \leq \tfrac{\kappa}{2} d_U(\Gamma(X), \Gamma(QX)).$$

Therefore $\|f_o\|_{L^2} \leq \kappa\delta_\Gamma/2$. The reverse triangle inequality gives $\|m_o - f_o\|_{L^2} \geq (\|m_o\|_{L^2} - \kappa\delta_\Gamma/2)_+$. Combine the last three inequalities with Proposition S1. For the branch law, $m_o(X) = \sqrt{3\lambda}\,T$ and $\mathbb{E}[T^2] = 1/3$, which proves the final assertion. □

Using this proposition as a numerical risk lower bound requires a population upper bound for δ_Γ and a verified Lipschitz constant for the decoder in the same metric. An empirical reflection distance alone provides neither. The experimental diagnostic decoders are not constrained to a common Lipschitz budget.

Proposition S6 (Reconstruction through the predictor's representation)**.** *Let $U = \Gamma(X)$, and let h be a measurable map from the representation space into $\mathbb{R}^d$. Suppose a version of m is L-Lipschitz in Euclidean distance on a domain containing X and $h(U)$ almost surely. Assume*

$$\mathbb{E}\|X - h(U)\|_2^2 \leq \varepsilon_{\text{rec}}^2 < \infty.$$

Then

$$R_U^{\text{Bayes}} - R_X^{\text{Bayes}} \leq L^2 \varepsilon_{\text{rec}}^2.$$

If h additionally satisfies $\|h(u) - h(v)\|_2 \leq L_h d_U(u, v)$ for a representation metric d_U, then for every $\kappa \geq LL_h$,

$$\inf_{\text{Lip}_{d_U}(g) \leq \kappa} \{R(g \circ \Gamma) - R_X^{\text{Bayes}}\} \leq L^2 \varepsilon_{\text{rec}}^2.$$

The infimum is over square-integrable decoders with Lipschitz constant at most κ relative to d_U. Finally, if a particular decoder f satisfies $\|m \circ h - f\|_{L^2(P_U)} \leq \delta_{\text{head}}$, then

$$R(f \circ \Gamma) - R_X^{\text{Bayes}} \leq (L\varepsilon_{\text{rec}} + \delta_{\text{head}})^2.$$

Proof. The Lipschitz assumption gives $\|m(X) - m(h(U))\|_{L^2} \leq L\varepsilon_{\text{rec}}$. Since $m(X)$ is square-integrable, this also proves that $m(h(U))$ is a square-integrable U-measurable prediction. Conditional expectation minimizes squared error among such predictions, so

$$\mathbb{E}[(m(X) - \mathbb{E}[m(X) \mid U])^2] \leq \mathbb{E}[(m(X) - m(h(U)))^2] \leq L^2 \varepsilon_{\text{rec}}^2.$$

The first assertion follows from Proposition S1. If h is L_h-Lipschitz, then $|m(h(u)) - m(h(v))| \le L L_h d_U(u, v)$. Thus $m \circ h$ is an admissible decoder in the constrained infimum whenever $\kappa \ge L L_h$, proving the second assertion. For the specified decoder f, the triangle inequality gives

$$\|m - f \circ \Gamma\|_{L^2(P_X)} \le \|m(X) - m(h(U))\|_{L^2} + \|m(h(U)) - f(U)\|_{L^2} \le L\varepsilon_{\mathrm{rec}} + \delta_{\mathrm{head}}.$$

Squaring and applying Proposition S1 proves the last bound. □

The reconstruction must be a function of the retained code U. A KAHM map evaluated directly on X need not have this property, even if its reconstruction error is small. The domain condition is also substantive: a conditional mean defined only on the support of X must admit the stated Lipschitz version at the reconstructed points $h(U)$.

Proposition S7 (Common scaling of an inner kernel vector)**.** *Let $k \in \mathbb{R}^n$ be a finite kernel vector and let $B \in \mathbb{R}^{n \times n}$ be a fixed fitted matrix. If $\mathbf{1}_n^\top Bk \neq 0$, then for every real $a > 0$,*

$$\frac{B(ak)}{\mathbf{1}_n^\top B(ak)} = \frac{Bk}{\mathbf{1}_n^\top Bk}.$$

Consequently, evaluating a Gaussian query vector with entries $\exp(\ell_j - \max_i \ell_i)$ preserves its normalized signed weights in exact arithmetic whenever all log-kernel values ℓ_j are finite and the original denominator is nonzero.

Proof. Linearity gives $B(ak) = aBk$ and $\mathbf{1}_n^\top B(ak) = a\mathbf{1}_n^\top Bk$. The nonzero factor a cancels. For finite log-kernel values, each centered exponential is the original exponential multiplied by the common positive real factor $a = \exp(-\max_i \ell_i)$. □

This identity concerns evaluation of the query vector with the training kernel and fitted matrix B held fixed. It does not resolve a genuinely zero denominator or prevent cancellation between signed weights. Its numerical purpose is to avoid rounding an entire Gaussian query vector to zero before normalization.

SB Generators, shared components, and decoders

SB.1 Forecasting laws and target construction

The nonlinear autoregressive (AR) generator starts with independent $u_0, u_1 \sim \mathrm{Unif}[-0.8, 0.8]$ and evolves according to

$$u_{t+1} = \tanh\{1.25u_t - 0.65u_{t-1} + 0.25\sin(3u_t) + \eta_{t+1}\}, \qquad \eta_t \overset{\mathrm{iid}}{\sim} \mathrm{Unif}[-0.1, 0.1].$$

Here t is a discrete time index, and the innovations η_t are independent of the initial conditions. The first 80 stored values are discarded and the next 72 form one episode. Episodes use independent initial states and innovation sequences.

The Van der Pol and Duffing generators use position x and velocity v, with derivatives taken with respect to continuous time:

$$\dot{x} = v, \qquad \dot{v} = (1 - x^2)v - x,$$
$$\dot{x} = v, \qquad \dot{v} = -0.2v + x - x^3 + 0.3\cos(1.2t + \theta),$$

respectively. Initial position and velocity are independent $\mathrm{Unif}[-2, 2]$ draws for each episode. The forcing phase $\theta \sim \mathrm{Unif}[0, 2\pi]$ is also sampled independently for each episode. The common random-number procedure draws this phase for both systems, although it is used only by Duffing. Integration uses classical fourth-order Runge–Kutta in 64-bit floating-point precision, with step 0.025, burn-in time 10, and observation interval 0.1. At the 72 retained observation times, the sensor is

$$u_j = 0.95\tanh(x_j) + \eta_j, \qquad \eta_j \overset{\mathrm{iid}}{\sim} \mathrm{Unif}[-0.02, 0.02],$$

where x_j is the simulated position at observation j. The sensor noise is independent of the initial conditions and is drawn before integration in the specified random-number sequence. These procedures define numerical observation laws. The samples are not assumed to be exact continuous-time solutions or stationary after burn-in.

For all three laws, zero-based forecast origins are $t \in \{11, 15, \ldots, 59\}$. At origin t, the input column is $(u_{t-11}, \ldots, u_t)^\top$ and the target column is $(u_{t+1}, u_{t+4}, u_{t+12})^\top$. Each episode provides 13 origins. Training, validation, and test sets contain 128, 128, and 512 independent episodes, respectively, yielding 1,664, 1,664, and 6,656 rows. Rows from the same episode are dependent.

The branch distribution is defined in Equation (S9). The primary training set contains 1,024 independent points; its first 256 points form the secondary, nested training-size condition. Validation and test contain 512 and 2,048 independent points. Within a repetition, the same draws of S, T, ϵ generate both curvatures. For each curvature and training size, one bank is fitted and reused for all three target mixtures. The latent branch labels are not used to fit the partitions. Since $T = X_1$ and $S = 2\{X_2 - c\cos(\pi X_1)\}$, the raw input determines the signal. Its variance is $(1-\lambda) + 3\lambda/3 = 1$, while the independent noise variance is $1/300$. For every $\lambda \in [0,1]$, $|Y_\lambda| \le \sqrt{1-\lambda} + \sqrt{3\lambda} + 0.1 \le 2.1$, which justifies a common clipping interval across target mixtures.

The master random seed is 20261030. Phase identifiers and child streams determine separate pilot, first-stage confirmation, natural-data fitting, and capacity-study inputs and fits. The capacity study uses phase identifier 4. The executable stream rules and realized identifiers are recorded in the archive. A repeated replication label across phases does not imply repeated observations. Shared randomness within a repetition permits paired comparisons, but does not increase the number of independent repetitions.

SB.2 The common bank and the locality control

Input means and standard deviations are fitted to all training rows, using the population variance divisor $1/n$ for n training observations. A standard deviation at most 10^{-12} is replaced by one. Clustering and reference selection use at most 1,600 training rows, sampled without replacement. Each of the three k-means fits uses five initializations and at most 150 iterations; the cluster counts are 2, 4, and 8. Within each cluster, at most 24 reference rows are sampled without replacement and converted to 32-bit floating-point precision before fitting the local KAHM map. The requested local rank is $\min(2, d-1, n_c-1)$, where d is the input dimension and n_c the number of retained references. The reference mean and the principal basis obtained by this fit are also used by the affine control.

The local fit is defined as follows. Let $H \in \mathbb{R}^{d\times n_c}$ contain the reference columns, so that $H = V_c^\top$ in the notation of Section S3. A principal basis is obtained from the centered reference covariance with divisor $n_c - 1$, retaining eigenvalues above machine epsilon. The requested rank is reduced whenever a retained projected coordinate has range below 10^{-3}, until the condition is removed or rank one is reached. Let p_c be the resulting rank, let $P_c \in \mathbb{R}^{d\times p_c}$ be the retained basis, and write $q_i = P_c^\top H_{:i} \in \mathbb{R}^{p_c}$ for projected reference i. Let Σ_c be the projected sample covariance with divisor $n_c - 1$. The training Gram matrix is

$$K_{ij} = \exp\left\{-\frac{(q_i - q_j)^\top \Sigma_c^{-1}(q_i - q_j)}{2p_c}\right\}, \qquad i, j = 1, \ldots, n_c.$$

The regularizer is determined from the input references alone. Define $\nu = \|H\|_F^2/(dn_c)$, where $\|H\|_F^2$ is the sum of the squares of all entries of H. Starting at $\rho_0 = 2.5\nu$, the algorithm iterates

$$B_j = (K + \rho_j I_{n_c})^{-1}, \qquad \rho_{j+1} = 2\nu + \frac{\|H - H(KB_j)^\top\|_F^2}{dn_c}, \qquad j = 0, 1, \ldots.$$

It stops when $|\rho_{j+1} - \rho_j| \le 10^{-3}$ or after 100 iterations and returns the last computed inverse B_j as B_c. Future targets do not enter this iteration. Query kernels use the same projected covariance and Gaussian scale $1/(2p_c)$. If a cluster has fewer than two usable references or no positive empirical-variance direction, its reconstruction is the reference center and its local coordinate block is empty. Other numerical failures are recorded as failures of the prescribed procedure.

Although the two reconstructions share their input basis, their output ranges differ. KAHM combines full-dimensional reference vectors with nonlinear query-dependent weights and need not remain in the truncated affine principal subspace. The affine reconstruction always remains in that subspace. The comparison therefore isolates a reconstruction operator within a common fitted bank, rather than equating the functional capacity of the two maps.

The locality-aware affine score gives a separate control for position along an otherwise well-reconstructed affine set. For model c, let $z_{c1}, \ldots, z_{cn_c} \in \mathbb{R}^d$ be its standardized reference vectors. Define

$$s_c = \max\left\{\frac{1}{n_c}\sum_{i=1}^{n_c}\|z_{ci}-\mu_c\|_2^2, 10^{-6}\right\},$$
$$v_c = \max\left\{\frac{1}{n_c}\sum_{i=1}^{n_c}\|(I_d - P_cP_c^\top)(z_{ci}-\mu_c)\|_2^2, 0.05s_c, 10^{-6}\right\}.$$

The score and its within-partition affinities are

$$\beta_c(z) = \frac{\|z - r_c^{\mathrm{A}}(z)\|_2^2}{v_c} + 0.1\frac{\|z-\mu_c\|_2^2}{s_c}, \qquad \psi_c(z) = \frac{\exp\{-\beta_c(z)\}}{\sum_{j:b(j)=b(c)}\exp\{-\beta_j(z)\}},$$

where $b(c)$ identifies the partition. The center-distance term can distinguish positions along a line even when affine reconstruction is exact. This secondary control changes the score rather than the reconstruction alone. A radial-basis locality control instead normalizes $\exp\{-\|z-\mu_c\|_2^2/(2\,\mathrm{median}_{j:b(j)=b(c)}\, s_j)\}$ within each partition, where median is the sample median of the local scales in that partition.

The representation has 14 stored affinity coordinates and three within-partition sum constraints, so its affine dimension is at most 11. Appending the three log normalizers gives 17 stored coordinates. Local augmentation has 28 coordinates for the two-dimensional branch inputs and 42 for the twelve-dimensional forecasting histories. The additional coordinates are the entries of $\phi_c(z)P_c^\top(z-\mu_c)$, computed directly from the query in the shared basis. Full reconstruction vectors are not inputs to these controlled-study decoders.

Reconstruction diagnostics divide each query's squared reconstruction error by the input dimension. The best-model statistic takes the minimum error over local models within each partition. The weighted statistic averages those individual errors with the reconstruction's own affinity weights. Both statistics then average over queries, partitions, and repetitions. In particular, the weighted statistic is not the error of an affinity-weighted reconstruction vector. These geometric summaries are descriptive and are separate from the primary forecast contrasts.

SB.3 Ridge and finite nonlinear decoders

Each input feature column is standardized using training data alone, replacing standard deviations below 10^{-8} by one. The ridge decoder minimizes the average sum of squared target-coordinate errors plus $\alpha\|W\|_F^2$, where W is the coefficient matrix from standardized features to target coordinates. Its intercept is unpenalized. Validation selects the penalty from

$$\alpha \in \{10^{-8}, 10^{-6}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1\}.$$

Forecast horizons share the selected penalty, whereas the three branch target mixtures are selected separately.

The nonlinear decoder selects M training observations without replacement as landmarks in the standardized feature space. For a fixed M, a common seed selects the same row indices across representations, although the landmark feature values differ. Let b^2 be the median pairwise squared landmark distance among distances greater than 10^{-14}; set $b^2 = 1$ if no such pair exists. For each bandwidth multiplier $a \in \{0.25, 1, 4\}$, define $\sigma_a^2 = \max(b^2a^2, 10^{-12})$ and the kernel $k_a(u,v) = \exp\{-\|u-v\|_2^2/(2\sigma_a^2)\}$. Let K_{MM} be the corresponding $M \times M$ landmark Gram matrix. After numerical symmetrization, retain eigenvalues exceeding $\max(10^{-10}\lambda_{\max}, 10^{-12})$, where $\lambda_{\max}$ is the largest eigenvalue. Let V contain the retained eigenvectors and Λ be the diagonal matrix of retained positive eigenvalues. If $k_M(u)$ is the column of similarities from a standardized query u to the landmarks, its Nyström coordinates are $\Lambda^{-1/2}V^\top k_M(u)$.

These coordinates are concatenated with the standardized linear input features. The resulting feature columns are standardized again using training data, and the same ridge objective is fitted. Validation selects the bandwidth–penalty pair. Candidates are evaluated in ascending bandwidth multiplier and then ascending penalty order; the first minimizing pair resolves a tie. The first-stage study uses $M = 64$, and the fresh capacity study uses both $M = 64$ and $M = 256$.

Standardizing the basis columns and imposing an ordinary Euclidean coefficient penalty produces a finite nonlinear decoder with a linear component. It does not reproduce the usual reproducing-kernel Hilbert space norm of exact kernel ridge regression. The realized dimension may be smaller than the input dimension plus M because of the eigenvalue cutoff. Changing M can also change the landmark sample, median-distance bandwidth, retained rank, and validation-selected parameters. The capacity comparison therefore concerns complete procedures at different landmark budgets.

Predictions are clipped for validation and test evaluation to $[-1, 1]$ for the three forecasting laws and to $[-2.1, 2.1]$ for the branch targets. These intervals contain the corresponding target supports. Clipping is not part of the training least-squares objective. Coefficients remain fitted to training data after validation; there is no combined training–validation refit. A secondary selector chooses the lowest validation MSE among eight complete representation–budget procedures. Its tie order is increasing budget followed by the configured feature order. Test outcomes enter neither parameter selection nor selection of the complete procedure.

SB.4 Airfoil grouping and the interpretation of repetitions

The Airfoil snapshot has 1,503 observations, five covariates, and sound pressure as the target. Exact joint equality of the four non-frequency covariates defines 106 operating-setting groups. A permutation with seed 2026103007 allocates 64, 21, and 21 groups to training, validation, and test, yielding 905, 295, and 303 observations. All frequencies from one setting remain in the same split. Training weights observations equally. Validation and test first average losses within a setting and then weight settings equally. Reported errors are in squared decibels.

The 24 fitting seeds use this same cohort and split. They quantify fitting variability conditional on these observations and do not constitute 24 independent natural tasks, test cohorts, or populations of physical operating conditions. The synthetic comparison intervals are not applied to Airfoil. A claim about a population of operating conditions would require an appropriate sampling model or independently sampled cohorts.

SC Estimands, multiplicity, and descriptive diagnostics

SC.1 The unit of paired inference

For method a and repetition r, the forecasting loss is

$$\widehat{L}_{r,a} = \frac{1}{512} \sum_{j=1}^{512} \frac{1}{13 \cdot 3} \sum_{o=1}^{13} \sum_{h \in \{1,4,12\}} (\widehat{y}_{r,a,j,o,h} - y_{r,j,o,h})^2.$$

Here j indexes test episodes, o indexes forecast origins within an episode, h is the forecast horizon, $\widehat{y}$ is the fitted prediction, and y is the observed target. For the branch experiments, $\widehat{L}_{r,a}$ is instead the average noisy-target squared error over 2,048 independent test points. Loss against the branch conditional mean is reported as a separate diagnostic. Methods within a repetition share the same test draws, so their losses can be differenced in pairs. The 24 complete training–validation–test repetitions are the independent units used for primary inference. Overlapping windows, horizons, local models, candidate fits, and target mixtures are not additional replication units.

For a primary contrast, let D_r be its paired loss difference in repetition r. Define $\overline{D} = 24^{-1} \sum_{r=1}^{24} D_r$ and $s_D^2 = 23^{-1} \sum_{r=1}^{24} (D_r - \overline{D})^2$. For a family of F contrasts, the reported interval is

$$\overline{D} \ \pm \ t_{23,\, 1-0.05/(2F)} \, \frac{s_D}{\sqrt{24}},$$

where $t_{\nu,p}$ denotes the p quantile of the Student t distribution with ν degrees of freedom. The critical values are approximately 3.545808 for $F = 29$ and 3.180663 for $F = 12$. For independent normally distributed paired differences, the marginal Student intervals have exact nominal coverage, and the Bonferroni inequality gives simultaneous coverage of at least 95% within a family. For other distributions, the Student coverage is approximate; 24 repetitions do not by themselves ensure a reliable normal approximation. Bonferroni permits dependence between contrasts and does not require independence of the windows within a repetition.

The synthetic estimand averages over training, validation, and finite test-sample randomness under each specified numerical generator. It does not average over an independently sampled population of forecasting tasks. The replication count was set by the computational budget rather than a prospective power calculation.

SC.2 Primary controlled contrasts

The first-stage experiment has nine settings: six curvature–target combinations for the branch law and the three forecasting laws. Let $\widehat{L}_r(\mathcal{F}, H)$ denote the test loss from features $\mathcal{F}$ and decoder type H in repetition r of a specified setting. The paired contrasts are

$$\begin{aligned} G_r &= \widehat{L}_r(\phi^{\mathrm{A}}, \mathrm{ridge}) - \widehat{L}_r(\phi^{\mathrm{K}}, \mathrm{ridge}), \\ N_r &= \widehat{L}_r(\phi^{\mathrm{K}}, \mathrm{ridge}) - \widehat{L}_r([\phi^{\mathrm{K}}, \log Z^{\mathrm{K}}], \mathrm{ridge}), \\ L_r &= \widehat{L}_r(\phi^{\mathrm{K}}, \mathrm{ridge}) - \widehat{L}_r(\Gamma^{\mathrm{K}}_{\mathrm{local}}, \mathrm{ridge}). \end{aligned}$$

The superscripts identify the reconstruction used to form the features; $\log Z^{\mathrm{K}}$ denotes the vector of three partition log normalizers. Thus G_r tests the reconstruction substitution, N_r tests retention of the score normalizers, and L_r tests retention of local coordinates. A positive value favors the second procedure in each difference. Let $G(c, \lambda) = \mathbb{E}[G_r(c, \lambda)]$ for a branch setting and define the target interaction as $I_c = G(c, 0) - G(c, 1)$. Its paired estimator uses $G_r(c, 0) - G_r(c, 1)$ within each repetition. The 27 setting-specific contrasts and the two curvature-specific interactions form the 29-comparison family.

On straight branches, the estimated geometry advantage for the position target is 0.978026, with simultaneous interval $[0.96255, 0.99350]$. The estimated target interaction is -0.978039, with interval $[-0.99351, -0.96257]$. The negative interaction means the advantage is larger for position than for branch identity; it does not imply a KAHM advantage on the branch-identity target.

The normalizer inverse is numerically accurate to about 2.8×10^{-16}. Nevertheless, the first-stage Van der Pol MSE with a nonlinear decoder is 0.00158 for full scores and 0.01273 for the information-equivalent $[\phi, \log Z]$ features. These secondary comparisons illustrate differences in finite decoding and regularization. They do not estimate the corresponding Bayes risks.

SC.3 Comparison families and practical margins

The 29-comparison family uses a setting-specific practical margin of 5% of the mean raw-ridge MSE from two pilot repetitions. An interaction uses the mean of the margins for its two endpoint settings. Pilot outcomes are excluded from the confirmatory estimates. These margins are design thresholds with a scale estimated from two pilot observations; they are not externally validated application tolerances. A positive interaction I_c would mean a larger reconstruction advantage for branch identity, so a negative interaction has the opposite interpretation.

The capacity study has a separate family of 12 differences: raw features minus KAHM local features and affine local features minus KAHM local features, each evaluated on three systems at two landmark budgets. For each law, the practical margin is 5% of the first-stage mean raw-RBF MSE across all 24 first-stage repetitions. The margins, candidates, and 24 fresh replication streams were fixed before capacity-study observations were generated. Thus they are fixed conditional on the first-stage results that motivated the follow-up. The capacity study has its own 95% simultaneous interval family; this does not imply joint 95% coverage across both stages.

We report a benefit at the declared practical threshold when the simultaneous lower endpoint exceeds its fixed margin δ. An interval above zero whose lower endpoint does not exceed δ supports a positive difference

under the interval assumptions but leaves that practical threshold unresolved. An entirely negative interval favors the comparator, and an interval containing zero leaves the sign unresolved. An interval contained in $[-\delta, \delta]$ supports that practical band; merely failing to exclude zero does not establish equivalence. Percentage changes are descriptive ratios of mean losses, without ratio confidence intervals.

There is no primary interaction test for a change in advantage with landmark budget. A positive interval at one budget and an unresolved interval at another would therefore not establish such an interaction. All specified settings are reported, including adverse and unresolved comparisons, and every prescribed repetition completed successfully.

SC.4 Reflection, reconstruction, and numerical checks

The reflection diagnostic applies Q to the raw branch input before input scaling and feature construction. For a p-coordinate feature vector, it reports

$$\left\{\frac{1}{p}\mathbb{E}_n\|\Gamma(X) - \Gamma(QX)\|_2^2\right\}^{1/2},$$

where $\mathbb{E}_n$ denotes the average over the diagnostic sample. The standardized version first divides feature coordinates by their training-set standard deviations, replacing values at most 10^{-12} by one. This differs from the 10^{-8} threshold used to standardize decoder inputs. The statistic is a sample per-coordinate root mean square under this chosen scaling. It is not the population quantity δ_Γ in Proposition S5 with an independently verified decoder Lipschitz constant. Values in Table S13 average the root mean squares computed separately for each repetition.

The numerical sensitivity analysis reuses all 24 capacity-study Van der Pol banks and their training, validation, and test observations. It changes only evaluation of the Gaussian query vector according to Proposition S7, before multiplication by the same fitted inverse B. The training kernel, regularizer, principal basis, reference values, covariance metric, partition, and input scaler are held fixed. Decoders are fitted to the stabilized training features and selected using the same validation observations. The 96 outcomes comprise two encodings, two landmark budgets, and 24 repetitions. Because the analysis reuses observations examined in the capacity study, its comparisons are descriptive and are outside the two primary interval families.

The test calculations contain 99,762 zero reconstruction normalizers among 2,236,416 query–local-model evaluations. Of these, 99,687 have all Gaussian entries rounded to zero before normalization, and the remaining 75 arise in subsequent arithmetic. After stabilized query evaluation, the zero count is zero. Local-feature mean losses are effectively unchanged, whereas full-score losses and some individual predictions change. These results assess this specific underflow mechanism as an explanation for the local-feature benefit; they do not establish numerical robustness for every KAHM encoding or for signed normalization in general.

SC.5 Retrospective influence and coordinate-recovery diagnostics

These analyses were conducted after the primary evaluation and use the same paired repetitions. They assess sensitivity of the reported estimates and coordinate recovery without defining an additional confirmatory comparison family. For each of the 41 primary contrasts, we compute bias-corrected sample skewness and delete each repetition once. Each deletion recomputes the Student interval with 23 remaining repetitions, 22 degrees of freedom, and the original family size (29 or 12). The range of deletion endpoints is an influence diagnostic, not a confidence interval or an additional simultaneous-coverage claim. Table S6 shows the four Van der Pol comparisons; the complete descriptive results accompany the executable artifact. All four Van der Pol comparisons retain lower endpoints above the fixed practical margin after any one-repetition deletion. This does not establish a distribution-free mean guarantee or remove the normal-approximation qualification of the primary analysis. We also inspect the 120 first-stage primary banks and 72 capacity banks from a completed computational reproduction using the published seeds. No bank or head is refitted for these diagnostics. Replayed test inputs, targets, and groups are checked against their recorded hashes before the original feature map is evaluated. Let A vertically stack the $P_c^\top$ matrices. Proposition S3 shows that positive gates and full column rank of A suffice to recover the scaled input from the gated local coordinates and

Table S6: Retrospective Van der Pol influence diagnostics. Lower endpoints are multiplied by 10^3; the fixed margin on this scale is 0.148907. Skewness is the bias-corrected Fisher–Pearson sample coefficient. The minimum deletion lower endpoint is the smallest over all 24 one-repetition deletions. The final column counts individual paired differences exceeding the margin; it is not a mean-effect confidence statement.

Control	M	Skewness	Original lower	Min. deletion lower	$D_r > \delta$
Raw	64	-0.07	1.7369	1.7278	24/24
Affine local	64	2.69	0.2755	0.2685	24/24
Raw	256	0.01	0.7550	0.7524	24/24
Affine local	256	0.33	0.1810	0.1786	23/24

gates: division by ϕ_c recovers $P_c^\top(z - \mu_c)$, after which one solves for z. A deficient stack does not establish information loss in the full code, because the gates can carry additional information. Numerical ranks use the threshold $\sigma_1 \max\{n_A, d\}\epsilon_{64}$, where σ_1 is the largest singular value of A, n_A is its number of rows, and $\epsilon_{64} = 2^{-52}$ is the machine precision for 64-bit floating-point arithmetic. These ranks are finite-precision diagnostics rather than exact rank proofs. Small singular values and small gates can make coordinate recovery sensitive to perturbations. Test-sample gate minima are not population lower bounds, and these computations do not guarantee a well-conditioned inverse or easy decoding by any particular fitted head.

Table S7: Retrospective diagnostics over 24 stored banks per row. Rank ranges concern the stacked local bases in input dimension d. The smallest singular value and gate entries are minima over the inspected banks (and, for gates, their original test observations). The condition number κ_2 is the ratio of the largest to the smallest singular value for a full-column-rank stack; the table reports its maximum over the inspected banks. A dash means that not every stack in that row has full numerical column rank. KAHM and affine representations share the basis stack. Computational reproduction supplies no new independent statistical replications.

Phase / setting	Rank / d	Min. σ_d	Max. κ_2	Min. ϕ^{K}	Min. ϕ^{A}
First: Straight branch	1/2	0	—	3.36×10^{-25}	2.56×10^{-20}
First: Curved branch	2/2	2.19×10^{0}	1.39×10^{0}	3.55×10^{-28}	8.97×10^{-29}
First: Nonlinear AR	12/12	6.20×10^{-2}	4.90×10^{1}	2.21×10^{-35}	2.36×10^{-34}
First: Van der Pol	12/12	1.60×10^{-2}	2.20×10^{2}	2.84×10^{-45}	9.43×10^{-20}
First: Duffing	12/12	1.33×10^{-2}	2.80×10^{2}	3.92×10^{-43}	3.11×10^{-7}
Capacity: Nonlinear AR	12/12	7.31×10^{-2}	4.19×10^{1}	9.97×10^{-37}	7.75×10^{-37}
Capacity: Van der Pol	12/12	1.60×10^{-2}	2.19×10^{2}	6.21×10^{-44}	1.56×10^{-16}
Capacity: Duffing	12/12	1.32×10^{-2}	2.84×10^{2}	5.38×10^{-43}	1.86×10^{-8}

SD Complete controlled results and diagnostics

Table S8: The six primary branch settings at 1,024 training points: mean ridge test MSE over 24 repetitions, with the primary KAHM advantage G and its interval from the 29-comparison family. The Bayes MSE is 1/300. The affine-locality scoring control reuses the shared references and is secondary.

c	λ	Affine ϕ	KAHM ϕ	Affine locality	G [interval]
0	0	0.003334	0.003347	0.003344	−0.000013 [−0.000031, 0.000005]
0	0.5	0.503448	0.013995	0.003991	0.489453 [0.481962, 0.496944]
0	1	1.002694	0.024668	0.004641	0.978026 [0.962552, 0.993500]
1	0	0.282887	0.281126	0.314394	0.001761 [−0.071260, 0.074781]
1	0.5	0.160813	0.165207	0.167971	−0.004395 [−0.043672, 0.034882]
1	1	0.062366	0.051042	0.045555	0.011324 [0.000882, 0.021767]

Table S9: All nine first-stage primary settings: mean test MSE over 24 repetitions for ridge decoders. The local-coordinate improvement L and its interval belong to the 29-comparison family. Retaining local position substantially improves KAHM prediction in eight settings; KAHM local coordinates do not dominate affine local coordinates.

Setting	KAHM ϕ	KAHM local	Affine local	Raw	L [interval]
$c = 0, \lambda = 0$	0.003347	0.003378	0.003337	0.003338	−0.000031 [−0.000043, −0.000020]
$c = 0, \lambda = 0.5$	0.013995	0.003379	0.003333	0.003334	0.010616 [0.010033, 0.011200]
$c = 0, \lambda = 1$	0.024668	0.003380	0.003333	0.003330	0.021287 [0.020199, 0.022376]
$c = 1, \lambda = 0$	0.281126	0.017982	0.048916	0.671776	0.263144 [0.217187, 0.309101]
$c = 1, \lambda = 0.5$	0.165207	0.010497	0.026756	0.337620	0.154710 [0.122867, 0.186552]
$c = 1, \lambda = 1$	0.051042	0.004188	0.004656	0.003329	0.046854 [0.040400, 0.053307]
Nonlinear AR	0.005300	0.004867	0.004867	0.005053	0.000433 [0.000368, 0.000498]
Van der Pol	0.111108	0.014475	0.011582	0.041340	0.096633 [0.092436, 0.100830]
Duffing	0.053729	0.007560	0.004448	0.006764	0.046168 [0.041729, 0.050608]

Table S10: Post-hoc Van der Pol numerical sensitivity on the same 24 second-stage banks and samples. Stable query evaluation removes observed zero denominators while leaving local-coordinate mean MSE effectively unchanged. Δ_r is the original-minus-stable MSE difference within repetition r; wins count stable predictions beating the raw and affine local controls at the same width. These comparisons are outside the two primary interval families.

Encoding	M	Original MSE	Stable MSE	Max. $\lvert\Delta_r\rvert$	Wins raw; affine
Local	64	0.001026627	0.001026627	1.01×10^{-12}	24/24; 24/24
Local	256	0.000871969	0.000871969	1.69×10^{-11}	24/24; 24/24
Scores	64	0.001571085	0.001565219	8.74×10^{-5}	24/24; 4/24
Scores	256	0.001049909	0.001045801	3.53×10^{-5}	24/24; 17/24

Tables S12 and S13 give descriptive input-geometry summaries. The two complete primary comparison families follow. Their values are generated from the archived outcomes, with each difference, standard error, interval, and prespecified margin recomputed from the archived per-repetition losses by the table-generation program.

Table S11: Secondary Airfoil static-regression results on one fixed group split, averaged over 24 fitting seeds. Units are dB^2. These seeds do not constitute independent natural-data replications, and the comparison is descriptive.

Representation	Ridge MSE	RBF_{64} MSE
Raw	19.4726	15.6898
Affine ϕ	34.1630	36.0476
KAHM ϕ	31.8586	34.3338
Affine local	22.7200	22.4050
KAHM local	20.9040	20.9313
KAHM scores	24.2168	21.7070

Table S12: Held-out reconstruction errors per standardized input coordinate, averaged over the three blocks and 24 full repetitions. "Best" takes the best local reconstruction separately in each block; "weighted" averages the individual squared errors with that method's affinities. The affine value is lower in all 24 repetitions of each of these five input settings, for both summaries. These are descriptive diagnostics, not the supervised primary endpoint.

Setting	KAHM best	Affine best	KAHM weighted	Affine weighted
Branch $c = 0$	3.276×10^{-3}	2.156×10^{-16}	4.006×10^{-3}	1.548×10^{-8}
Branch $c = 1$	3.583×10^{-2}	2.932×10^{-2}	3.793×10^{-2}	3.238×10^{-2}
Nonlinear AR	1.166×10^{-2}	1.084×10^{-2}	1.305×10^{-2}	1.214×10^{-2}
Van der Pol	4.353×10^{-3}	1.683×10^{-3}	4.495×10^{-3}	1.916×10^{-3}
Duffing	4.877×10^{-3}	5.613×10^{-4}	5.239×10^{-3}	1.226×10^{-3}

Table S13: Branch reflection differences at training size 1,024, averaged over 24 banks. The statistic is a per-coordinate RMS, with and without training-only coordinate scaling. The raw reflection is applied before the fitted input scaling and feature construction. These values neither establish exact information loss for the learned code nor certify a decoder-specific population risk bound.

c	Encoding	Stored-coordinate RMS	Standardized RMS
0	KAHM ϕ	4.894×10^{-1}	1.402×10^{0}
0	Affine ϕ	1.451×10^{-9}	7.533×10^{-9}
0	KAHM local	5.289×10^{-1}	1.418×10^{0}
0	Affine local	4.993×10^{-1}	1.236×10^{0}
1	KAHM ϕ	5.380×10^{-1}	1.427×10^{0}
1	Affine ϕ	5.321×10^{-1}	1.471×10^{0}
1	KAHM local	4.854×10^{-1}	1.442×10^{0}
1	Affine local	5.522×10^{-1}	1.473×10^{0}

Table S14: Complete representation means in the fresh capacity study, with MSE multiplied by 10^3. The full-score KAHM representation is a prespecified secondary control. These descriptive point estimates are reported alongside the separate 12-comparison primary family.

System	M	Raw	Affine local	KAHM local	KAHM scores
Nonlinear AR	64	4.4238	4.9120	4.8795	4.7984
Nonlinear AR	256	4.6958	5.2347	5.1749	4.8256
Van der Pol	64	2.8586	1.3767	1.0266	1.5711
Van der Pol	256	1.6541	1.0780	0.8720	1.0499
Duffing	64	3.4597	3.0549	4.2154	4.7005
Duffing	256	2.8364	2.7981	4.0736	3.3808

SD.1 Selected pipelines and realized dimensions

The selected procedure identifies which representation and landmark budget minimize validation error. The retained nonlinear basis records the realized feature dimension after the eigenvalue cutoff. Both are reported because a nominal landmark budget does not uniquely determine the fitted feature dimension.

Table S15: Secondary validation-only selection among the eight complete capacity-study pipelines. Mean test MSE is multiplied by 10^3. Counts concern independent complete repetitions under each generator, not a new population of forecasting tasks.

System	Selector MSE	Validation selections
Nonlinear AR	4.4238	Raw, 64: 24/24
Van der Pol	0.8720	KAHM local, 256: 24/24
Duffing	2.7923	Affine local, 256: 23/24; Raw, 256: 1/24

Table S16: Retained RBF-plus-linear dimensions after the Nyström eigenvalue cutoff in the fresh capacity study. Means and ranges cover the 24 repetitions in each cell. The same landmark count M does not imply the same realized dimension or function class.

System	M	Representation	Mean dimension	Range
Nonlinear AR	64	Raw	74.88	61–76
Nonlinear AR	64	Affine local	106.00	106–106
Nonlinear AR	64	KAHM local	106.00	106–106
Nonlinear AR	64	KAHM scores	76.71	62–78
Nonlinear AR	256	Raw	189.04	78–268
Nonlinear AR	256	Affine local	297.62	296–298
Nonlinear AR	256	KAHM local	297.50	293–298
Nonlinear AR	256	KAHM scores	148.88	77–219
Van der Pol	64	Raw	76.00	76–76
Van der Pol	64	Affine local	106.00	106–106
Van der Pol	64	KAHM local	106.00	106–106
Van der Pol	64	KAHM scores	78.00	78–78
Van der Pol	256	Raw	268.00	268–268
Van der Pol	256	Affine local	298.00	298–298
Van der Pol	256	KAHM local	272.38	261–285
Van der Pol	256	KAHM scores	231.88	215–251
Duffing	64	Raw	76.00	76–76
Duffing	64	Affine local	106.00	106–106
Duffing	64	KAHM local	105.46	98–106
Duffing	64	KAHM scores	77.96	77–78
Duffing	256	Raw	268.00	268–268
Duffing	256	Affine local	296.92	272–298
Duffing	256	KAHM local	240.38	176–298
Duffing	256	KAHM scores	214.33	164–270

Table S17: Complete first-stage family (29 contrasts), in original MSE units. Intervals use the two-sided 95% Bonferroni paired-replicate Student procedure with 23 degrees of freedom. Under paired normality, each t interval is exact and Bonferroni family coverage is at least 95%; otherwise coverage is approximate. The two stages have separate families.

Setting / budget	Contrast	$\overline{D}$	Simultaneous interval	δ	Assessment
$c = 0, \lambda = 0$	G	-0.000013	$[-0.000031, 0.000005]$	0.000168	unresolved †
$c = 0, \lambda = 0$	N	-0.000008	$[-0.000016, -1.85 \times 10^{-7}]$	0.000168	negative †
$c = 0, \lambda = 0$	L	-0.000031	$[-0.000043, -0.000020]$	0.000168	negative †
$c = 0, \lambda = 0.5$	G	0.489453	[0.481962, 0.496944]	0.000168	practical +
$c = 0, \lambda = 0.5$	N	0.000944	[0.000088, 0.001799]	0.000168	positive
$c = 0, \lambda = 0.5$	L	0.010616	[0.010033, 0.011200]	0.000168	practical +
$c = 0, \lambda = 1$	G	0.978026	[0.962552, 0.993500]	0.000168	practical +
$c = 0, \lambda = 1$	N	0.001938	[0.000255, 0.003621]	0.000168	practical +
$c = 0, \lambda = 1$	L	0.021287	[0.020199, 0.022376]	0.000168	practical +
$c = 1, \lambda = 0$	G	0.001761	$[-0.071260, 0.074781]$	0.033542	unresolved
$c = 1, \lambda = 0$	N	0.069128	[0.028663, 0.109593]	0.033542	positive
$c = 1, \lambda = 0$	L	0.263144	[0.217187, 0.309101]	0.033542	practical +
$c = 1, \lambda = 0.5$	G	-0.004395	$[-0.043672, 0.034882]$	0.016863	unresolved
$c = 1, \lambda = 0.5$	N	0.040663	[0.019789, 0.061538]	0.016863	practical +
$c = 1, \lambda = 0.5$	L	0.154710	[0.122867, 0.186552]	0.016863	practical +
$c = 1, \lambda = 1$	G	0.011324	[0.000882, 0.021767]	0.000168	practical +
$c = 1, \lambda = 1$	N	0.012111	[0.005647, 0.018574]	0.000168	practical +
$c = 1, \lambda = 1$	L	0.046854	[0.040400, 0.053307]	0.000168	practical +
Nonlinear AR	G	-1.89×10^{-7}	$[-0.000050, 0.000049]$	0.000253	unresolved †
Nonlinear AR	N	-0.000016	$[-0.000062, 0.000029]$	0.000253	unresolved †
Nonlinear AR	L	0.000433	[0.000368, 0.000498]	0.000253	practical +
Van der Pol	G	-0.049419	$[-0.060532, -0.038307]$	0.002036	negative
Van der Pol	N	0.013443	[0.005084, 0.021803]	0.002036	practical +
Van der Pol	L	0.096633	[0.092436, 0.100830]	0.002036	practical +
Duffing	G	-0.030948	$[-0.035414, -0.026482]$	0.000338	negative
Duffing	N	0.005558	[0.001721, 0.009395]	0.000338	practical +
Duffing	L	0.046168	[0.041729, 0.050608]	0.000338	practical +
$c = 0$ interaction	I_0	-0.978039	$[-0.993512, -0.962566]$	0.000168	opposite sign
$c = 1$ interaction	I_1	-0.009564	$[-0.082839, 0.063712]$	0.016855	unresolved

†: the entire interval lies within the fixed practical band $[-\delta, \delta]$. An unresolved sign alone does not establish equivalence.

Table S18: Complete fresh capacity family (12 contrasts). All numerical loss quantities are multiplied by 10^3. Intervals use the two-sided 95% Bonferroni paired-replicate Student procedure with 23 degrees of freedom. Under paired normality, each t interval is exact and Bonferroni family coverage is at least 95%; otherwise coverage is approximate. The two stages have separate families.

Setting / budget	Contrast	$\overline{D}$	Simultaneous interval	δ	Assessment
Nonlinear AR / 64	Raw–K	-0.4556	$[-0.5235, -0.3878]$	0.2201	negative
Nonlinear AR / 64	Affine–K	0.0325	$[-0.0031, 0.0680]$	0.2201	unresolved †
Nonlinear AR / 256	Raw–K	-0.4791	$[-0.6013, -0.3569]$	0.2201	negative
Nonlinear AR / 256	Affine–K	0.0598	$[-0.0131, 0.1327]$	0.2201	unresolved †
Van der Pol / 64	Raw–K	1.8320	$[1.7369, 1.9271]$	0.1489	practical +
Van der Pol / 64	Affine–K	0.3501	$[0.2755, 0.4247]$	0.1489	practical +
Van der Pol / 256	Raw–K	0.7822	$[0.7550, 0.8094]$	0.1489	practical +
Van der Pol / 256	Affine–K	0.2060	$[0.1810, 0.2311]$	0.1489	practical +
Duffing / 64	Raw–K	-0.7557	$[-1.0831, -0.4282]$	0.1727	negative
Duffing / 64	Affine–K	-1.1605	$[-1.4707, -0.8503]$	0.1727	negative
Duffing / 256	Raw–K	-1.2372	$[-1.5242, -0.9503]$	0.1727	negative
Duffing / 256	Affine–K	-1.2756	$[-1.5702, -0.9810]$	0.1727	negative

†: the entire interval lies within the fixed practical band $[-\delta, \delta]$. An unresolved sign alone does not establish equivalence.

SE Representations and fitted forecasting rules

A history $x \in \mathbb{R}^d$ contains the observations available at a forecast origin; its response $y \in \mathbb{R}^H$ contains the next H observations. For the univariate tasks below, the input dimension d equals the history length L. We fit direct multioutput regressors to historical pairs $\{(x_i, y_i)\}_{i=1}^n$. The construction below specifies which part of the regressor uses KAHM geometry. It does not assume that overlapping forecasting windows are independent. All input transformations, reference sets, and decoder bases are fitted on the training portion of each fit.

SE.1 Shared reference geometry

Write $z(x) = D^{-1}(x - \bar{x})$, where $\bar{x}$ and the diagonal scales D are training means and population standard deviations. A standard deviation at most 10^{-12} is replaced by one; all entries of D are therefore positive. The symbol $\mathbf{1}$ denotes a vector of ones of the required dimension. Unless specified otherwise, $\|\cdot\|$ is the Euclidean norm for vectors, $\|\cdot\|_F$ is the Frobenius norm for matrices, and I is an identity matrix of the dimension required by its summand. Thus the standardization retains every raw coordinate, including coordinates constant in the observed training set. Three independently seeded k-means partitions, with $k \in \{2, 4, 8\}$, use the same seeded subset of at most 512 training histories. Each partition has five initializations and at most 150 iterations. Each cluster supplies at most 24 references sampled without replacement; all its rows are used if there are fewer. For a common seed and training set, the alternative gates use identical partition and reference-selection rules. They are nevertheless fitted as separate computational pipelines.

Index the 14 local models by c, with the partition blocks denoted $\mathcal{C}_b$. Let $Q_c = [q_{c1}^\top; \ldots; q_{cm_c}^\top] \in \mathbb{R}^{m_c \times d}$ contain the training-standardized references, with each coordinate rounded to float32 before subsequent double-precision calculations, and let $\mu_c = m_c^{-1} Q_c^\top \mathbf{1}$. Each $q_{cj} \in \mathbb{R}^d$ is a column vector. The orthonormal columns of $P_c \in \mathbb{R}^{d \times r_c}$ are leading eigenvectors of the centered reference covariance. The requested rank is $\min(2, d-1, m_c-1)$; only eigenvalues above double-precision machine epsilon are retained. The implementation further reduces a rank greater than one if the smallest projected coordinate range is below 10^{-3}. A reference set with no usable direction has $r_c = 0$ and uses its center as its reconstruction. Other numerical failures remain recorded failures.

The affine reconstruction and centered local coordinates are

$$a_c(z) = \mu_c + P_c P_c^\top (z - \mu_c), \qquad u_c(z) = P_c^\top (z - \mu_c). \tag{S17}$$

The affine and locality controls compute these quantities without fitting or querying any KAHM kernel.

SE.2 Reconstruction scores and local gates

KAHM reconstruction. Kernel Affine Hull Machines construct nonlinear reconstructions from local kernel weights (Kumar et al., 2024). The local reconstruction parameters are fitted from the reference histories without forecast targets. For $r_c > 0$, let Σ_c be the sample covariance of the rows of $Q_c P_c$. Its Gaussian kernel is

$$k_c(z, q) = \exp\left\{ -\frac{(P_c^\top (z - q))^\top \Sigma_c^{-1} P_c^\top (z - q)}{2 r_c} \right\}, \qquad (K_c)_{ij} = k_c(q_{ci}, q_{cj}). \tag{S18}$$

The local regularizer is obtained by the following fixed iteration. Put $\nu_c = \|Q_c\|_F^2/(d m_c)$ and initialize $\lambda_0 = 5\nu_c/2$. Its iteration is

$$B_t = (K_c + \lambda_t I)^{-1}, \qquad \lambda_{t+1} = 2\nu_c + \frac{\|Q_c - (K_c B_t) Q_c\|_F^2}{d m_c}. \tag{S19}$$

It stops after an update of size at most 10^{-3} or 100 updates, and retains the last computed B_t as B_c. In particular, it does not perform an extra inverse at the newly updated λ_{t+1}. For the column vector $k_c(z) = (k_c(z, q_{cj}))_{j=1}^{m_c} \in \mathbb{R}^{m_c}$, define

$$v_c(z) = B_c^\top k_c(z), \qquad \widehat{z}_c(z) = \frac{Q_c^\top v_c(z)}{\mathbf{1}^\top v_c(z)}. \tag{S20}$$

The matrix B_c is symmetric in exact arithmetic, so $B_c^\top k_c(z) = B_c k_c(z)$. The weights may be signed. The reconstruction is therefore not generally a convex average. If the computed denominator is exactly zero, the implementation replaces it by one; that convention is an implementation choice, not a definition of a continuous extension at a singularity.

Stable Gaussian evaluation. For queries we compute the analytic log kernel $\ell_{cj}(z) = -\|S_c P_c^\top (z - q_{cj})\|^2/(2r_c)$, where S_c is the stored symmetric square root of the inverse covariance, and use $\widetilde{k}_{cj}(z) = \exp\{\ell_{cj}(z) - \max_j \ell_{cj}(z)\}$ in Equation (S20). In exact arithmetic, $S_c^\top S_c = \Sigma_c^{-1}$, and multiplying every kernel entry for one query by the same positive constant cancels from the normalized reconstruction whenever its denominator is nonzero. This calculation avoids underflow caused solely by a large common negative log-kernel offset. It does not prevent signed cancellation, guarantee a well-conditioned denominator, or guarantee equivalence at an exact zero. The implementation retains the zero-denominator convention, records numerical diagnostics, and rejects nonfinite outputs. The unshifted evaluation is used only for the numerical comparison in Section S4; the natural-data study uses shifted kernels.

Matched folding gates. Given either the KAHM reconstruction $\widehat{z}_c$ or the affine reconstruction a_c, denote it by $r_c^{\mathrm{rec}}(z)$. Define

$$\rho_c(z) = 1 - \exp\{-\|z - r_c^{\mathrm{rec}}(z)\|\}, \qquad \theta_c(z) = \frac{1}{\pi} \arccos\left(\mathrm{clip}_{[-1,1]} \frac{z^\top r_c^{\mathrm{rec}}(z)}{\|z\| \, \|r_c^{\mathrm{rec}}(z)\|} \right),$$
$$s_c(z) = \sqrt{\{\rho_c(z)^2 + \theta_c(z)^2\}/2}, \qquad \phi_c(z) = \frac{[1 - \mathrm{clip}_{[0,1]} s_c(z) + \tau]^\omega}{\sum_{j \in \mathcal{C}_b} [1 - \mathrm{clip}_{[0,1]} s_j(z) + \tau]^\omega}, \quad c \in \mathcal{C}_b. \tag{S21}$$

When either norm in the cosine denominator is zero, its cosine is set to zero, so that $\theta_c = 1/2$. The operator $\mathrm{clip}_{[a,b]}(u) = \min\{b, \max\{a, u\}\}$ truncates a scalar to $[a, b]$. The fixed values are $\omega = 16$ and $\tau = 10^{-6}$; normalization is evaluated with log-sum-exp. The comparison between these two gates changes the reconstruction while retaining the score, reference geometry, and local-coordinate construction.

Locality control. The third gate combines distance from the affine reconstruction with distance from the local reference center. With $\epsilon = 10^{-6}$, set

$$\begin{aligned} t_c &= \max\left\{ m_c^{-1} \sum_j \|q_{cj} - \mu_c\|^2, \epsilon \right\}, \\ e_c &= \max\left\{ m_c^{-1} \sum_j \|q_{cj} - a_c(q_{cj})\|^2, 0.05 t_c, \epsilon \right\}, \\ b_c(z) &= \frac{\|z - a_c(z)\|^2}{e_c} + 0.1 \frac{\|z - \mu_c\|^2}{t_c}, \\ \phi_c(z) &= \frac{\exp[-b_c(z)]}{\sum_{j \in \mathcal{C}_b} \exp[-b_j(z)]}. \end{aligned} \tag{S22}$$

This distinguishes proximity to the local center from projection residual. It provides a non-KAHM locality control using the same reference observations and principal directions.

SE.3 Local codes and finite forecasting heads

For each gate family g, concatenate all gates and gated local coordinates:

$$T_g(x) = \big((\phi_c(z(x)))_c, (\phi_c(z(x))u_c(z(x)))_c\big), \qquad T_{\mathrm{raw}+g}(x) = (z(x), T_g(x)). \tag{S23}$$

The stored local width is $14 + \sum_c r_c \le 42$; the actual width is recorded. The raw augmentation prepends all d history coordinates, with no truncation when $d > 28$. The combination of local association and local position is related to coordinate and tangent coding (Yu et al., 2009; Yu & Zhang, 2010); our gates and decoder are the explicit constructions above. The matrix formed by vertically stacking the fourteen matrices $P_c^\top$ has rank at most 28. A deficient stack alone does not prove information loss in T_g, because the gates also depend on the query. Raw augmentation gives the simpler unconditional reconstruction property stated below, without a rank assumption.

Nyström plus linear basis. Let $v(x)$ denote $z(x)$, $T_g(x)$, or $T_{\mathrm{raw}+g}(x)$, and let $q(x)$ be its coordinatewise training standardization. Select M training rows without replacement as landmarks $a_1, \ldots, a_M$ in q-space. The requested M is 32 or 64, capped by the training size. Set σ^2 to the median positive squared inter-landmark distance and define

$$k_\sigma(q, a) = \exp\{-\|q - a\|^2/(2\sigma^2)\}, \qquad b(x) = \Big(q(x),\ \mathrm{diag}(\eta_J^{-1/2}) U_J^\top (k_\sigma(q(x), a_j))_{j=1}^M\Big). \tag{S24}$$

Here the landmark Gram matrix has entries $k_\sigma(a_i, a_j)$ and eigendecomposition $U\,\mathrm{diag}(\eta)U^\top$. The columns of U are orthonormal eigenvectors and η contains their nonnegative eigenvalues. The index set J retains eigenvalues exceeding $\max(10^{-10}\eta_{\max}, 10^{-12})$, where $\eta_{\max} = \max_j \eta_j$ and U_J contains the retained columns (Williams & Seeger, 2001). The operator diag forms a diagonal matrix from its argument. Distances at most 10^{-14} are excluded from the bandwidth median; if none remain the squared bandwidth is one, and in all cases it is floored at 10^{-12}. The bandwidth multiplier is fixed to one. Finally, standardize b on training rows to obtain $f(x)$ and the design $F_{i,:} = f(x_i)^\top$. Both head standardizations replace scales below 10^{-8} by one. Raw ridge omits the Nyström block and uses the same centering and scaling convention.

For each target coordinate $h \in \{1, \ldots, H\}$, let y_{ih} denote that coordinate of training response y_i. With feature width $p = \dim f(x)$, intercept $a_h \in \mathbb{R}$, coefficient $\beta_h \in \mathbb{R}^p$, and penalty $\alpha > 0$, the fitted decoder solves

$$\min_{\beta_h, a_h} \ \frac{1}{n} \sum_{i=1}^n [y_{ih} - a_h - f(x_i)^\top \beta_h]^2 + \alpha \|\beta_h\|^2. \tag{S25}$$

Thus the normal equations use $F^\top F/n + \alpha I$, and the intercept is unpenalized. Internally, targets are standardized using training statistics and transformed back after prediction. For fixed α, linearity makes this

exactly equivalent in real arithmetic to Equation (S25) in the original target units. This is a Euclidean ridge penalty on a standardized finite basis, rather than the canonical Gaussian RKHS norm. Adding local features also changes the Gaussian distances and the resulting Nyström basis; the raw and augmented nonlinear decoder classes need not be nested.

One common α from the ordered grid $(0.01, 1, 100)$ is selected by validation MSE averaged over origins and horizons in original target units, with the first grid entry winning ties. Within a method, geometry, training and validation features, basis construction, and the Gram eigensystem are computed once; all three alphas reuse those quantities. Different method variants share no fitted bank or free preprocessing. The selected configuration is then fitted anew on the admissible pretest training data. Search designs and eigensystems are not retained in the serialized final model. This gives an explicit efficient search whose full computation can be measured.

SE.4 Information and finite-decoder risk

The following information and risk identities explain why the representation can affect a finite fitted decoder while preserving the information in the raw history. Both identities apply beyond KAHM.

Proposition S8 (Information preserved by raw augmentation)**.** *Let $X \in \mathbb{R}^d$ and $Y \in \mathbb{R}^H$ be random history and response vectors, and let Θ denote a fitted state. Suppose Θ specifies $\bar{x} \in \mathbb{R}^d$, a diagonal matrix D with strictly positive entries, and a measurable feature map $T_g : \mathbb{R}^d \to \mathbb{R}^p$ whose values are finite on the support of X. Define $T_{\mathrm{raw}+g}(x) = (D^{-1}(x - \bar{x}), T_g(x))$ as a concatenated column vector. Then*

$$\sigma(T_{\mathrm{raw}+g}(X), \Theta) = \sigma(X, \Theta),$$

where $\sigma(\cdot)$ is the sigma-field generated by its arguments. If $\mathbb{E}\|Y\|_2^2 < \infty$, the conditional mean and optimal squared-loss risk based on either input are equal, conditional on Θ.

Proof. The fitted map is measurable, so $\sigma(T_{\mathrm{raw}+g}(X), \Theta) \subseteq \sigma(X, \Theta)$. Conversely, the first d coordinates of the augmented vector are $z(X) = D^{-1}(X - \bar{x})$. Because D is invertible and belongs to the fitted state, they recover $X = \bar{x} + Dz(X)$ measurably. This proves the reverse inclusion and therefore equality of the two sigma-fields. The squared-loss Bayes predictor is conditional expectation of Y onto the available sigma-field. Equal sigma-fields give the same conditional mean, the same residual $Y - \mathbb{E}[Y \mid X, \Theta]$, and hence the same conditional mean squared error. □

This is an exact-arithmetic statement about the representation, not a conditioning guarantee or a claim that a particular fitted decoder uses every retained coordinate. Empirical gains from raw augmentation must therefore be explained through finite fitting, representation-dependent regularization, or numerical behavior, rather than increased Bayes information relative to the same raw input.

Proposition S9 (Fixed-feature, fixed-penalty risk identity)**.** *Consider one scalar response, n training rows, and penalty $\alpha > 0$. Let $F \in \mathbb{R}^{n \times p}$ be the training feature matrix after centering, so that $F^\top \mathbf{1} = 0$. Let $f_* \in \mathbb{R}^p$ be the query features centered with the same training means. Let $\mathcal{G}$ be the sigma-field generated by the training and query inputs and the randomness used to construct the unsupervised feature map; thus F and f_* are $\mathcal{G}$-measurable. Define*

$$w_* = \frac{\mathbf{1}}{n} + F(F^\top F + n\alpha I_p)^{-1} f_* \in \mathbb{R}^n. \tag{S26}$$

Let $\mathbf{Y} \in \mathbb{R}^n$ contain the training responses and let Y_* be the scalar query response. Assume their conditional second moments are finite, and put

$$\mu = \mathbb{E}[\mathbf{Y} \mid \mathcal{G}], \quad \mu_* = \mathbb{E}[Y_* \mid \mathcal{G}], \quad \Sigma = \mathrm{Cov}(\mathbf{Y} \mid \mathcal{G}), \quad c = \mathrm{Cov}(\mathbf{Y}, Y_* \mid \mathcal{G}), \quad v_* = \mathrm{Var}(Y_* \mid \mathcal{G}).$$

Here Cov *and* Var *denote conditional covariance and variance; $\Sigma \in \mathbb{R}^{n \times n}$ and $c \in \mathbb{R}^n$. The ridge predictor with a free intercept is $\widehat{Y}_* = w_*^\top \mathbf{Y}$, and its conditional risk satisfies*

$$\mathbb{E}[(\widehat{Y}_* - Y_*)^2 \mid \mathcal{G}] = (w_*^\top \mu - \mu_*)^2 + w_*^\top \Sigma w_* + v_* - 2w_*^\top c. \tag{S27}$$

Table S19: Fixed source allocation and forecast regimes. "Selected/all" counts series, not independent source families. Each selected series uses both regimes for its frequency. L is history length, H forecast horizon, and s the declared seasonal period.

Phase	Source snapshot	Frequency	Selected/all
Development	Electricity (Godahewa et al., 2020a)	weekly	4/321
	NN5, with missing values (Godahewa et al., 2020c)	daily	4/111
	Tourism (Godahewa et al., 2020e)	monthly	4/366
	FRED-MD (Godahewa et al., 2020k)	monthly	4/107
Confirmation	Traffic (Godahewa et al., 2020f)	weekly	4/862
	Hospital (Godahewa et al., 2020b)	monthly	4/767
	Saugeen River flow (Godahewa et al., 2020d)	daily	1/1
	US births (Godahewa et al., 2020g)	daily	1/1

Frequency	Short (L, H)	Long (L, H)	s
Monthly	$(12, 3)$	$(24, 6)$	12
Weekly	$(13, 4)$	$(26, 8)$	52
Daily	$(28, 7)$	$(56, 14)$	7

Proof. Centering gives the intercept $\bar{Y} = n^{-1}\mathbf{1}^\top\mathbf{Y}$. The ridge normal equations for the mean squared loss are $(F^\top F + n\alpha I_p)\widehat{\beta} = F^\top\mathbf{Y}$. The matrix is positive definite because $\alpha > 0$, so the solution exists and is unique. Substitution into $\widehat{Y}_* = \bar{Y} + f_*^\top\widehat{\beta}$ gives Equation (S26). Conditional on $\mathcal{G}$, w_* is fixed. Thus the prediction error has conditional mean $w_*^\top\mu - \mu_*$ and conditional variance

$$\mathrm{Var}(w_*^\top\mathbf{Y} - Y_* \mid \mathcal{G}) = w_*^\top\Sigma w_* + v_* - 2w_*^\top c.$$

The second moment is the sum of this variance and the squared conditional mean, which proves the stated identity. □

The identity holds without independent training responses or an independent future response. Overlapping windows and serial dependence can affect both Σ and c; neither may be discarded without additional assumptions. Averaging the identity over horizons gives the corresponding average risk. Changing geometry changes w_* and can change all its bias and covariance terms, without a universal ordering between representations. The identity does not automatically describe the full validation-selected procedure: after penalty selection, the smoother depends on observed validation responses. No generalization or selection guarantee is inferred from it.

SF Natural-data protocol

Source allocation, task roster, model grids, and analysis are fixed before comparative outcomes. The evaluation compares matched representations, accuracy and computational cost, and prospective model selection on reserved source families. The reproduction materials include specifications, input checksums, and source checkpoints. The study was not externally preregistered.

SF.1 Sources and chronological tasks

Table S19 gives eight versioned Monash snapshots (Godahewa et al., 2021), chosen by provenance, application diversity, length, and size. Related series, channels, and aggregations share a source family and cannot cross the boundary between development and confirmation. Models are local to each series, including differently dated Tourism series. The supplement retains the records' CC BY 4.0 licenses, versions, attribution, original-source references, and archive hashes.

Training-prefix eligibility requires at least 16 finite windows in each regime and a positive training error scale. A fixed salted hash of source and series ID orders eligible series; we retain the first four, or all if fewer exist,

Table S20: Capacity caps. These are limits on examples and model size, not equal realized time budgets. A shorter source uses its available examples.

Setting	Training windows	RBF landmarks	Tree leaves/iterations	MLP width/iterations
Small	512	32	15/100	32/200
Medium	1,024	64	31/150	64/300

across regimes and capacities. No later loss or availability criterion selects replacements. The 26 series give 52 base tasks; two capacities and seeds 17 and 43 give 208 runs: 128 development and 80 confirmation, with four source families per phase.

For a length-n series indexed from zero, let $a = \lfloor 3n/5 \rfloor$ and $b = \lfloor 4n/5 \rfloor$. Training, validation, and test intervals are $[0, a)$, $[a, b)$, and $[b, n)$. An origin t is the last observed index, with

$$x_t = (y_{t-L+1}, \ldots, y_t)^\top, \qquad y_t^+ = (y_{t+1}, \ldots, y_{t+H})^\top. \tag{S28}$$

The complete target interval must lie within its split; contexts may include earlier observed values. Stateful controls receive $t + 1$, their first-future index. We retain raw snapshot values, including NN5 missing markers, and discard windows with an unavailable context or target without interpolation. Evenly spaced indices among available origins give at most 64 validation and 64 test windows, and the training cap in Table S20. Every method receives the same evaluation origins.

Before outcomes, length checks reduced the proposed monthly $(48, 6)$ and weekly $(52, 8)$ regimes, which yield zero Hospital and three Traffic training windows. The adopted long regimes yield 21 and 29, respectively. Context and horizon are paired, so their separate effects are not identified. NN5 T88 at $(56, 14)$ retains only seven test origins out of 146 candidates. All task availability counts are recorded.

All candidate preprocessing is fitted on training data, and validation MSE selects hyperparameters. The selected pipeline is then fitted anew on all eligible windows with targets preceding b, subject to its training cap. This refit includes windows crossing the former training/validation boundary because all their targets precede the test segment. Parameters are fixed thereafter; earlier realized observations may enter later contexts or state updates. Test outcomes never retune parameters.

SF.2 Comparators, capacity, and loss

The twelve methods comprise raw ridge and raw RBF heads; affine-, locality-, and KAHM-gated local RBF pipelines; each local pipeline augmented by the full raw history; and seasonal naive, Holt, histogram gradient boosting, and a one-hidden-layer MLP. Section SE defines the feature maps and finite RBF head. The primary matched pair is raw RBF (R) versus raw-history plus KAHM RBF (K). The raw-history controls retain the inputs independently of local-basis rank (Proposition S8). The evaluation reports every method in this twelve-method portfolio.

The geometric bank uses cluster counts $(2, 4, 8)$, requested rank two, at most 24 references per local model, a 512-row fitting cap, $\omega = 16$, and $\tau = 10^{-6}$. Reduced effective ranks and numerical fallbacks are recorded. Each alternative constructs its own representation; affine/locality controls do not fit a KAHM kernel. Within a method, the fixed representation and head statistics are reused across ridge penalties $\{0.01, 1, 100\}$; the bandwidth multiplier is one. Boosting/MLP regularization is chosen from $\{0.01, 1\}$, Holt selects damped or undamped trend, and seasonal naive has no tuned parameter. These modest declared grids do not imply optimal tuning of every comparator.

Seasonal naive can use its declared lag from the observed past even when it exceeds L, using the last finite value for an unavailable lag and period one before a full seasonal lag exists. Holt fits a capped historical segment, forward-fills only within that segment, and assimilates observations strictly before each forecast; an absent online observation causes a state prediction without measurement update. Boosting and MLP fit direct multi-output forecasts without a random internal validation split.

For a nonempty set $\mathcal{O}$ of evaluated forecast origins, the scaled task loss is

$$\ell = \frac{1}{|\mathcal{O}|Hq} \sum_{t \in \mathcal{O}} \sum_{h=1}^{H} (\widehat{y}_{t+h} - y_{t+h})^2, \tag{S29}$$

Here q is fitted on the uncapped initial training prefix. For seasonal period s, let $\mathcal{P}_s$ contain indices for which both y_i and y_{i-s} are finite observations in that prefix. The seasonal scale is $q_s = |\mathcal{P}_s|^{-1} \sum_{i \in \mathcal{P}_s} (y_i - y_{i-s})^2$ when at least two such pairs exist. We use $q = q_s$ if this value is positive and finite; otherwise we apply the same rule with $s = 1$. A series with neither valid scale is ineligible. This denominator remains fixed during refitting. Raw MSE, q, predictions, and origin losses are saved and independently checked.

For each source family, we average seeds and capacities within each history–horizon regime, then regimes within each series, and finally series within the family. The overall mean weights source families equally. These nested observations are not independent source replications. We report family outcomes, dispersion, and leave-one-family-out sensitivity, without broad-population confidence intervals or window-level significance claims.

SF.3 Measured pipeline and selector costs

Methods execute sequentially in fresh child processes with one BLAS thread and deterministically shuffled order, limited to 120 seconds per method and 1,800 per task. Wall and process CPU time are separate. Each warm-library scientific pipeline includes data/checksum preparation, array loading, configuration fitting and validation, selection bookkeeping, selected refitting, forecasting, and model serialization. Source preparation is equally allocated across its tasks and charged consistently to alternatives. Shared imports are initialized before measurement; initialization, process launch/IPC, and launch-to-result duration are recorded separately rather than allocated arbitrarily to policies.

Let S_m be the set of computational steps for searching method m and F_m the steps for its final refit, forecasts, and serialization. Let $\mathcal{D}_{\text{data}}$ contain shared data preparation steps, c_v be the measured cost of step v, and c_{decision} the selection bookkeeping cost. A validation selector over candidate set $\mathcal{A}$ chooses method m^* and incurs

$$\mathcal{D}_{\text{val}} = \mathcal{D}_{\text{data}} \cup \bigcup_{m \in \mathcal{A}} S_m \cup F_{m^*}, \qquad C = \sum_{v \in \mathcal{D}_{\text{val}}} c_v + c_{\text{decision}}. \tag{S30}$$

Set unions count shared dependencies once. A learned selector instead pays for its required metadata/geometry/probe nodes plus $S_{m^*} \cup F_{m^*}$, even when it rejects KAHM. Geometry-only acquisition is conservatively billed through its measured fitted-probe component. Thus policy costs are counterfactual sums of measured required computations, not the cost of running every alternative in the benchmark.

Development-label acquisition also costs the R, K searches, refits, and forecasts, plus each variant's feature/probe nodes. We report this investment separately from regression fitting, online policy costs, and the full research benchmark's elapsed time, without assuming a deployment count for amortization. Summed standalone task costs are conservative where sharing across seeds or capacities could reduce work.

We also record serialized model bytes and peak process resident set size (RSS). RSS includes the interpreter, arrays present at worker initialization, and earlier process history; it is neither incremental memory nor an additive exact policy-memory reconstruction. No GPU measurement is inferred from CPU timing. Failed attempts remain visible and retain the available timing records. Method failures are included in the coverage accounting before losses are aggregated over the specified task and method roster.

SF.4 Prospective benefit diagnostic

The diagnostic predicts benefit before a task's test outcomes, while using its training history and permitted validation information. Its development label is $\log[(\ell_R + \epsilon)/(\ell_K + \epsilon)]$, with $\epsilon = 10^{-12}$. Development test errors may train this cross-task rule; confirmation errors cannot train, standardize, tune, or select it.

We retain four weighted ridge rules: metadata M, M plus geometry G, M plus a validation probe P, and $M + P + G$. With N_{train} denoting the number of retained training windows, the metadata features are

$\log(1 + N_{\text{train}})$, $\log(1 + H/L)$, and training-prefix lag-one autocorrelation, computed from finite adjacent observation pairs and set to zero if either variance is zero or fewer than three pairs are available. Geometry adds normalized KAHM gate entropy and $\log(1 + r)$. For n_p probe-training inputs $z_i \in \mathbb{R}^d$ and the three partition blocks $\mathcal{C}_b$, these quantities are

$$E_{\text{gate}} = -\frac{1}{3n_p} \sum_{b=1}^{3} \sum_{i=1}^{n_p} \frac{\sum_{c \in \mathcal{C}_b} \phi_c(z_i) \log \phi_c(z_i)}{\log |\mathcal{C}_b|}, \qquad r = \frac{1}{3n_p d} \sum_{b=1}^{3} \sum_{i=1}^{n_p} \sum_{c \in \mathcal{C}_b} \phi_c(z_i) \|z_i - a_c(z_i)\|_2^2.$$

The gates and affine reconstructions are defined in Appendix SE. A singleton block would contribute zero normalized entropy. The probe fits the matched R, K pair with at most 128 training windows, 16 landmarks, a 128-row bank cap, and penalty one; its feature is the log ratio of scaled validation losses with the same loss floor.

The diagnostic penalty is fixed at one, with an unpenalized intercept. Source-balanced feature standardization and fitting occur inside each development fold, holding out an entire source family. All four variants are then fitted on all development families and frozen before confirmation. The primary incremental comparison is $M + P + G$ against $M + P$, testing whether geometry adds value beyond metadata and a reduced-capacity validation probe. Controls include always R, always K, direct probe selection, validation selection within the matched pair, the six-method raw/practical portfolio, the two-method KAHM portfolio, and the full twelve-method portfolio.

A positive predicted log benefit selects K; zero or negative estimates select R. A fixed 5% relative MSE change defines secondary practical benefit/harm, not a tuned selection threshold. All four variants are reported without choosing one from confirmation performance. A test-set oracle is only a hindsight regret reference. Utility requires improved held-out loss–cost behavior against metadata and probe controls, not merely correlation with retrospective gains.

Before confirmation, a verification program checks development predictions against their targets and reported errors. The archived checkpoint records the code, dependencies, task roster, development records, and diagnostic coefficients. Confirmation evaluates the complete specified roster. This is a selected, small collection, not a random sample of forecasting domains or a state-of-the-art survey. Seasonal periods are conventions rather than fitted optimal cycles. In particular, FRED-MD is a fixed-vintage snapshot: its metadata describe transformed series without a complete per-series preprocessing audit (Godahewa et al., 2020k). Our train-only guarantees apply to processing in this extension, not to unverified upstream decisions, historical revisions, or release-time availability. The conclusions must remain conditional on these sources, contexts, grids, and resource settings.

SG Additional natural-source results and diagnostics

The following tables report all methods, selection policies, source means, capacity settings, diagnostic variants, and fitting warnings. These are descriptive summaries under the same fixed source hierarchy. They do not pool the natural-source results with the controlled-study confidence families. Full precision, forecasts, targets, origins, selected parameters, and cost components are retained in the anonymous supplement.

Table S21: Standalone confirmation methods, equally weighted across 4 original source families using the frozen within-source hierarchy. Loss is training-scale-normalized MSE. Wall and CPU seconds cover complete warm-library scientific pipelines. RSS is the averaged maximum recorded process high-water mark, not incremental memory; model size is serialized bytes in KiB. No uncertainty intervals are inferred from dependent tasks.

Method	Scaled MSE	Wall (s)	CPU (s)	RSS (MiB)	Model (KiB)
Raw ridge	0.7798	0.188	0.1879	171.5	5.655
Raw RBF	0.8075	0.1688	0.1687	171.5	37.32
Affine local	0.9449	0.5002	0.4897	171.5	93.2
Locality local	0.8964	0.2898	0.2898	171.5	93.4
KAHM local	0.9354	0.3218	0.3214	171.5	164.6
Raw + affine	0.8458	0.3379	0.3265	171.5	107.8
Raw + locality	0.8462	0.2791	0.279	171.5	107.8
Raw + KAHM	0.8419	0.3874	0.3814	171.5	179.2
Seasonal naive	1.05	0.1614	0.1613	171.5	0.1382
Holt	2.373	0.3449	0.3446	171.5	0.4219
Histogram boosting	0.9555	8.91	8.901	171.5	2591
MLP	0.8785	1.463	1.462	171.5	71.62

Table S22: Confirmation cost transparency for the primary matched pair, with equal source-family weighting. Total is the complete warm-library scientific pipeline. Cleanup is its measured selection-overhead component, including explicit garbage collection; the percentage is the ratio of aggregated component and total times. Launch separately reports the measured isolated-child launch-to-result interval. These are implementation-specific costs, not intrinsic kernel arithmetic or universal complexity ratios.

Method	Total (s)	Cleanup (s)	Cleanup (%)	Launch (s)
Raw RBF	0.16879	0.13808	81.8	0.17748
Raw + KAHM	0.38741	0.18253	47.11	0.40097

Table S23: Frozen-policy confirmation outcomes over 4 source families with the same hierarchical weighting. Gain is raw-RBF scaled MSE minus chosen loss. Harm is the weighted fraction with loss more than 5% above raw RBF. The final column counts selection of the specific raw-plus-KAHM pair member, including in portfolios. Wall cost includes all required probes/searches and decision inference, then the chosen final refit, forecast and model serialization. Offline evidence acquisition and diagnostic fitting are separate.

Policy	Scaled MSE	Wall (s)	Gain	Harm (%)	Raw+K (%)
Always raw	0.8075	0.1688	0	0	0
Always raw + KAHM	0.8419	0.3874	-0.0344	25	100
Validation: matched pair	0.8097	0.4812	-0.002159	5.47	32
Validation: raw portfolio	0.8639	8.338	-0.05638	10.9	0
Validation: KAHM family	0.8406	0.6242	-0.03306	26.6	73.4
Validation: full portfolio	0.8764	9.948	-0.0689	14.8	14.1
Cheap probe	0.8143	0.3465	-0.006771	9.38	50
Metadata	0.816	0.2976	-0.008445	15.6	50
Metadata + geometry	0.816	0.3695	-0.008445	15.6	50
Metadata + probe	0.8071	0.3309	0.0004363	3.12	31.2
Metadata + probe + geometry	0.8071	0.3309	0.0004363	3.12	31.2

Table S24: Development scaled MSE by original source family. The final column gives equal-source aggregation; source columns retain the declared within-source weights. Development diagnostics use leave-one-source-family-out fits; confirmation diagnostics use the frozen development fit. Always-policy rows duplicate their corresponding standalone methods deliberately.

Method or policy	FRED-MD	NN5	Tourism	Electricity	Equal-source
Raw ridge	0.5601	0.8584	2.865	0.4243	1.177
Raw RBF	10.46	0.7552	3.476	0.4388	3.783
Affine local	2.548	0.8795	3.769	0.4535	1.912
Locality local	23.83	0.8148	3.778	0.4112	7.209
KAHM local	12.23	0.9163	3.636	0.4525	4.309
Raw + affine	1.409	0.7614	3.172	0.4481	1.448
Raw + locality	10.18	0.7474	3.046	0.4418	3.603
Raw + KAHM	4.254	0.7704	3.229	0.4539	2.177
Seasonal naive	2.269	1.089	2.982	0.5507	1.723
Holt	0.2409	1.358	8.228	0.6984	2.631
Histogram boosting	34.02	0.8032	4.786	0.4404	10.01
MLP	1.022	0.8475	2.77	0.4715	1.278
Always raw	10.46	0.7552	3.476	0.4388	3.783
Always raw + KAHM	4.254	0.7704	3.229	0.4539	2.177
Validation: matched pair	5.267	0.7335	3.229	0.4415	2.418
Validation: raw portfolio	0.2409	0.8021	3.014	0.4528	1.127
Validation: KAHM family	4.104	0.7792	3.223	0.4572	2.141
Validation: full portfolio	0.2409	0.8052	3.092	0.4675	1.152
Cheap probe	5.106	0.742	3.25	0.4217	2.38
Metadata	4.254	0.7704	3.229	0.4539	2.177
Metadata + geometry	4.254	0.7704	3.394	0.4539	2.218
Metadata + probe	4.157	0.759	3.338	0.4342	2.172
Metadata + probe + geometry	4.215	0.7704	3.338	0.4342	2.189

Table S25: Development warm-library wall seconds by original source family. The final column gives equal-source aggregation; source columns retain the declared within-source weights. Development diagnostics use leave-one-source-family-out fits; confirmation diagnostics use the frozen development fit. Always-policy rows duplicate their corresponding standalone methods deliberately.

Method or policy	FRED-MD	NN5	Tourism	Electricity	Equal-source
Raw ridge	0.3731	0.1631	0.1888	0.1873	0.2281
Raw RBF	0.1929	0.1912	0.1834	0.1692	0.1842
Affine local	0.3285	0.2805	0.2983	0.2863	0.2984
Locality local	0.5431	0.29	0.2784	0.2649	0.3441
KAHM local	0.3705	0.3696	0.3735	0.3369	0.3626
Raw + affine	0.2923	0.2998	0.2979	0.288	0.2945
Raw + locality	0.3185	0.2967	0.2988	0.2533	0.2918
Raw + KAHM	0.4275	0.3807	0.3252	0.3024	0.3589
Seasonal naive	0.2125	0.177	0.1834	0.1878	0.1902
Holt	0.3224	0.42	0.3682	0.316	0.3567
Histogram boosting	3.086	4.954	1.729	1.819	2.897
MLP	1.278	0.9551	0.8521	0.5508	0.9089
Always raw	0.1929	0.1912	0.1834	0.1692	0.1842
Always raw + KAHM	0.4275	0.3807	0.3252	0.3024	0.3589
Validation: matched pair	0.5562	0.5313	0.4929	0.4301	0.5026
Validation: raw portfolio	4.066	5.306	2.641	2.719	3.683

Method or policy	FRED-MD	NN5	Tourism	Electricity	Equal-source
Validation: KAHM family	0.692	0.6568	0.6115	0.5511	0.6278
Validation: full portfolio	5.861	6.513	4.138	4.075	5.147
Cheap probe	0.4739	0.4615	0.4004	0.3328	0.4172
Metadata	0.4284	0.3815	0.3258	0.3031	0.3597
Metadata + geometry	0.5195	0.465	0.3474	0.3744	0.4266
Metadata + probe	0.5029	0.474	0.2891	0.3411	0.4018
Metadata + probe + geometry	0.5284	0.4828	0.2891	0.341	0.4103

Table S26: Confirmation scaled MSE by original source family. The final column gives equal-source aggregation; source columns retain the declared within-source weights. Development diagnostics use leave-one-source-family-out fits; confirmation diagnostics use the frozen development fit. Always-policy rows duplicate their corresponding standalone methods deliberately.

Method or policy	Traffic	Hospital	Saugeen	US births	Equal-source
Raw ridge	0.5737	0.8461	0.522	1.177	0.7798
Raw RBF	0.5739	0.8228	0.6106	1.223	0.8075
Affine local	0.5796	0.928	0.6462	1.626	0.9449
Locality local	0.5719	0.7702	0.6504	1.593	0.8964
KAHM local	0.5775	0.9029	0.6652	1.596	0.9354
Raw + affine	0.5764	0.9198	0.6068	1.28	0.8458
Raw + locality	0.572	0.9356	0.6002	1.277	0.8462
Raw + KAHM	0.5768	0.9238	0.6012	1.266	0.8419
Seasonal naive	0.6484	0.7848	1.023	1.743	1.05
Holt	0.6842	1.003	0.6079	7.197	2.373
Histogram boosting	0.6524	1.025	0.5386	1.606	0.9555
MLP	0.6811	1.15	0.5365	1.147	0.8785
Always raw	0.5739	0.8228	0.6106	1.223	0.8075
Always raw + KAHM	0.5768	0.9238	0.6012	1.266	0.8419
Validation: matched pair	0.5738	0.8236	0.6042	1.237	0.8097
Validation: raw portfolio	0.6087	1.116	0.5639	1.167	0.8639
Validation: KAHM family	0.577	0.9182	0.6012	1.266	0.8406
Validation: full portfolio	0.6122	1.099	0.5695	1.225	0.8764
Cheap probe	0.5761	0.8211	0.6253	1.235	0.8143
Metadata	0.5739	0.8228	0.6012	1.266	0.816
Metadata + geometry	0.5739	0.8228	0.6012	1.266	0.816
Metadata + probe	0.5761	0.8189	0.6106	1.223	0.8071
Metadata + probe + geometry	0.5761	0.8189	0.6106	1.223	0.8071

Table S27: Confirmation warm-library wall seconds by original source family. The final column gives equal-source aggregation; source columns retain the declared within-source weights. Development diagnostics use leave-one-source-family-out fits; confirmation diagnostics use the frozen development fit. Always-policy rows duplicate their corresponding standalone methods deliberately.

Method or policy	Traffic	Hospital	Saugeen	US births	Equal-source
Raw ridge	0.1454	0.1959	0.1905	0.2204	0.188
Raw RBF	0.1644	0.1636	0.1693	0.1779	0.1688
Affine local	0.2424	0.2732	0.4038	1.082	0.5002
Locality local	0.2311	0.254	0.4021	0.2721	0.2898
KAHM local	0.2579	0.2764	0.3496	0.4035	0.3218

Method or policy	Traffic	Hospital	Saugeen	US births	Equal-source
Raw + affine	0.3969	0.2809	0.354	0.3199	0.3379
Raw + locality	0.2223	0.2752	0.3166	0.3022	0.2791
Raw + KAHM	0.2674	0.4235	0.4676	0.3911	0.3874
Seasonal naive	0.1551	0.164	0.1874	0.1393	0.1614
Holt	0.2759	0.3535	0.4118	0.3386	0.3449
Histogram boosting	1.774	1.39	17.59	14.89	8.91
MLP	0.4046	0.392	2.264	2.79	1.463
Always raw	0.1644	0.1636	0.1693	0.1779	0.1688
Always raw + KAHM	0.2674	0.4235	0.4676	0.3911	0.3874
Validation: matched pair	0.3909	0.5408	0.5273	0.4656	0.4812
Validation: raw portfolio	2.276	2.19	15.94	12.95	8.338
Validation: KAHM family	0.4622	0.6403	0.7116	0.6829	0.6242
Validation: full portfolio	3.595	3.661	17.63	14.9	9.948
Cheap probe	0.3029	0.2768	0.4534	0.3531	0.3465
Metadata	0.1651	0.1644	0.4686	0.3923	0.2976
Metadata + geometry	0.2216	0.2212	0.5797	0.4555	0.3695
Metadata + probe	0.2958	0.457	0.3033	0.2673	0.3309
Metadata + probe + geometry	0.2958	0.457	0.3033	0.2673	0.3309

Table S28: Development capacity comparison. Each capacity changes specified training-window and model-size caps; it is not an equal-time budget. Values are equal-source weighted and retain all prescribed context–horizon settings and seeds.

Method or policy	medium MSE	medium wall (s)	small MSE	small wall (s)
Raw ridge	1.177	0.28	1.177	0.1761
Raw RBF	4.582	0.1846	2.984	0.1838
Affine local	1.989	0.3292	1.836	0.2676
Locality local	8.238	0.4065	6.18	0.2817
KAHM local	3.855	0.3779	4.763	0.3473
Raw + affine	1.394	0.3071	1.501	0.282
Raw + locality	3.879	0.3038	3.327	0.2798
Raw + KAHM	2.332	0.3819	2.022	0.3359
Seasonal naive	1.723	0.2078	1.723	0.1725
Holt	2.631	0.345	2.632	0.3683
Histogram boosting	9.795	3.468	10.23	2.326
MLP	1.253	1.108	1.302	0.7098
Always raw	4.582	0.1846	2.984	0.1838
Always raw + KAHM	2.332	0.3819	2.022	0.3359
Validation: matched pair	2.342	0.5295	2.493	0.4758
Validation: raw portfolio	1.128	4.223	1.127	3.142
Validation: KAHM family	2.261	0.6594	2.021	0.5963
Validation: full portfolio	1.137	5.805	1.166	4.488
Cheap probe	2.297	0.4321	2.463	0.4022
Metadata	2.332	0.3828	2.022	0.3367
Metadata + geometry	2.364	0.4492	2.072	0.404
Metadata + probe	2.337	0.419	2.007	0.3846
Metadata + probe + geometry	2.348	0.4327	2.03	0.388

Table S29: Confirmation capacity comparison. Each capacity changes specified training-window and model-size caps; it is not an equal-time budget. Values are equal-source weighted and retain all prescribed context–horizon settings and seeds.

Method or policy	medium MSE	medium wall (s)	small MSE	small wall (s)
Raw ridge	0.7635	0.1561	0.7961	0.2199
Raw RBF	0.7767	0.1719	0.8383	0.1657
Affine local	0.916	0.3088	0.9738	0.6917
Locality local	0.8908	0.3176	0.902	0.2621
KAHM local	0.9243	0.334	0.9466	0.3097
Raw + affine	0.8233	0.3859	0.8684	0.2899
Raw + locality	0.8363	0.2831	0.8561	0.275
Raw + KAHM	0.8288	0.3732	0.8551	0.4016
Seasonal naive	1.05	0.1621	1.05	0.1607
Holt	2.373	0.333	2.373	0.3569
Histogram boosting	0.9215	12.73	0.9896	5.086
MLP	0.8712	1.956	0.8859	0.9695
Always raw	0.7767	0.1719	0.8383	0.1657
Always raw + KAHM	0.8288	0.3732	0.8551	0.4016
Validation: matched pair	0.7847	0.4538	0.8347	0.5085
Validation: raw portfolio	0.8404	11.6	0.8874	5.074
Validation: KAHM family	0.8281	0.6116	0.8531	0.6369
Validation: full portfolio	0.8541	12.98	0.8987	6.918
Cheap probe	0.7943	0.3749	0.8343	0.3182
Metadata	0.8009	0.3187	0.8311	0.2764
Metadata + geometry	0.8009	0.3779	0.8311	0.3611
Metadata + probe	0.7765	0.2805	0.8377	0.3812
Metadata + probe + geometry	0.7765	0.2805	0.8377	0.3812

Table S30: Estimated-benefit diagnostics with frozen hierarchical weights. These are descriptive log-benefit prediction errors and classification accuracies; they are not calibrated probabilities or interval coverage. No recalibration is fitted on confirmation data.

Diagnostic	Log RMSE	Log bias	Sign (%)	5% class (%)
Development: Metadata	1.035	0.01412	57	49.2
Development: Metadata + geometry	1.037	0.03491	47.7	43
Development: Metadata + probe	1.012	0.03064	52.3	53.9
Development: Metadata + probe + geometry	1.02	0.05091	50.8	43
Confirmation: Metadata	0.1643	0.05835	55.5	59.4
Confirmation: Metadata + geometry	0.1612	0.05269	55.5	75
Confirmation: Metadata + probe	0.1845	0.04069	58.6	73.4
Confirmation: Metadata + probe + geometry	0.1847	0.04196	58.6	71.9

Table S31: One-off diagnostic development costs. Acquisition sums the union of both matched predictors' complete search/final nodes and each variant's metadata/probe/geometry dependencies within every development task. Summing standalone tasks can conservatively repeat preparation across seeds/tasks. Acquire and total denote wall time; CPU covers acquisition. The rows share development labels and some features, so their costs are not additive. Ridge fitting is measured separately; no deployment-count amortization is assumed.

Variant	Acquire (s)	CPU (s)	Ridge (ms)	Total (s)
Metadata	68.49	68.45	6.657	68.5
Metadata + geometry	78.6	78.56	10.04	78.61
Metadata + probe	80.83	80.79	9.035	80.84
Metadata + probe + geometry	80.83	80.79	6.146	80.84

Table S32: Frozen development-only standardization and ridge coefficients, rounded for display. G denotes geometry and P the cheap probe. An absent feature is shown by a dash. Predictions use the unrounded values in the frozen diagnostic JSON and the fixed zero decision threshold.

Feature	Mean	Scale	Meta	Meta+G	Meta+P	Meta+P+G
Intercept	–	–	0.023966	0.023966	0.023966	0.023966
log(1 + training windows)	5.1417	0.66192	0.017096	0.013047	-0.0099182	-0.0089504
log(1 + horizon/history)	0.23442	0.019538	-0.012299	-0.013178	-0.0043432	-0.0045441
Lag-one autocorrelation	0.7189	0.24263	0.0069415	0.001093	-0.014509	-0.012806
KAHM gate entropy	0.074168	0.063288	–	-0.010312	–	-0.0026401
log(1 + affine residual)	0.18213	0.12986	–	-0.013232	–	0.0063025
Probe log-loss ratio	0.25347	0.67921	–	–	0.14611	0.1467

Table S33: Recorded fitting behavior. Search and selected-refit warning events are separated. Counts describe task fits, which include dependent seeds and settings. Warned counts fits with recorded warnings; at cap counts selected refits reaching the recorded iteration limit. Full warning text is retained in the JSON report. No warning-triggered post hoc retuning is applied.

Method	Fits	Warned	Search	Refit	At cap
Development: Raw ridge	128	0	0	0	0
Development: Raw RBF	128	0	0	0	0
Development: Affine local	128	0	0	0	0
Development: Locality local	128	0	0	0	0
Development: KAHM local	128	0	0	0	0
Development: Raw + affine	128	0	0	0	0
Development: Raw + locality	128	0	0	0	0
Development: Raw + KAHM	128	0	0	0	0
Development: Seasonal naive	128	0	0	0	0
Development: Holt	128	0	0	0	0
Development: Histogram boosting	128	0	0	0	0
Development: MLP	128	128	232	105	105
Confirmation: Raw ridge	80	0	0	0	0
Confirmation: Raw RBF	80	0	0	0	0
Confirmation: Affine local	80	0	0	0	0
Confirmation: Locality local	80	0	0	0	0
Confirmation: KAHM local	80	0	0	0	0

Method	Fits	Warned	Search	Refit	At cap
Confirmation: Raw + affine	80	0	0	0	0
Confirmation: Raw + locality	80	0	0	0	0
Confirmation: Raw + KAHM	80	0	0	0	0
Confirmation: Seasonal naive	80	0	0	0	0
Confirmation: Holt	80	0	0	0	0
Confirmation: Histogram boosting	80	0	0	0	0
Confirmation: MLP	80	79	137	68	68

Table S34: Selection at numerical search-grid endpoints. Counts are descriptive; a two-point grid necessarily chooses an endpoint. These observations do not establish the benefit of a larger grid, and confirmation outcomes are not used to retune any method.

Method	Parameter	Choices	Lower	Upper	Declared grid
Development: Raw ridge	alpha	128	84	4	0.01,1,100
Development: Raw RBF	alpha	128	35	4	0.01,1,100
Development: Affine local	alpha	128	45	8	0.01,1,100
Development: Locality local	alpha	128	35	9	0.01,1,100
Development: KAHM local	alpha	128	51	7	0.01,1,100
Development: Raw + affine	alpha	128	32	4	0.01,1,100
Development: Raw + locality	alpha	128	25	5	0.01,1,100
Development: Raw + KAHM	alpha	128	30	4	0.01,1,100
Development: Histogram boosting	regularization	128	78	50	0.01,1
Development: MLP	regularization	128	26	102	0.01,1
Confirmation: Raw ridge	alpha	80	20	56	0.01,1,100
Confirmation: Raw RBF	alpha	80	13	56	0.01,1,100
Confirmation: Affine local	alpha	80	9	58	0.01,1,100
Confirmation: Locality local	alpha	80	9	60	0.01,1,100
Confirmation: KAHM local	alpha	80	8	59	0.01,1,100
Confirmation: Raw + affine	alpha	80	12	59	0.01,1,100
Confirmation: Raw + locality	alpha	80	12	61	0.01,1,100
Confirmation: Raw + KAHM	alpha	80	12	60	0.01,1,100
Confirmation: Histogram boosting	regularization	80	34	46	0.01,1
Confirmation: MLP	regularization	80	5	75	0.01,1

SH Reconstructing the numerical evidence

The reproduction materials provide separate controlled and natural forecasting entry points. The accompanying instructions describe how to verify file checksums, reconstruct archived losses, replay diagnostic models, regenerate tables and figures, and compile the manuscript. Source checkpoints identify the exact implementations used for each experiment. Recalculating results from those records verifies the same observations; it does not add independent replications.

For the controlled experiments, the runner can regenerate datasets and fits from the specified configurations and random seeds. A shorter verification path checks the archived compact outcome records. The natural study supplies all 208 completed runs, prepared windows, prediction arrays, source hashes, and environment and cost measurements. Its verification program checks the twelve-method roster, targets, forecast origins, all 2,496 losses, and diagnostic decisions without refitting the forecasting models. Replaying a diagnostic ridge regression may produce different elapsed times; the reported resource comparison uses the original measurements.

Dataset attribution, licenses, implementation notices, and the TMLR template license accompany the corresponding materials. The review version omits author-identifying project descriptions and links.

SI Sensitivity to geometry scaling

SI.1 Intervention and scope

The original experiments in Sections S4 and S5 use training z-score geometry scaling. The sensitivity analysis was specified after inspecting those experiments. Its protocol and execution sources were published and hashed before the sensitivity fits, and both fixed scaler arms were retained. All sensitivity configurations were specified before their own evaluation, but the source data and simulation draws had already been examined. The resulting comparisons are therefore exploratory. No adoption diagnostic or best-scaler selection rule was refitted.

For a history $x \in \mathbb{R}^d$ and coordinate $j \in \{1, \dots, d\}$, the intervention replaces the geometry input scaler by

$$z_j(x) = 2\frac{x_j - m_j}{M_j - m_j} - 1, \qquad m_j = \min_{i \in \mathcal{T}} x_{ij}, \quad M_j = \max_{i \in \mathcal{T}} x_{ij}.$$

The fitting set $\mathcal{T}$ is the exact training input set for that fit, before its geometry subsample. Coordinates with exactly zero training range are mapped to zero for every geometry query, as specified by the implementation. Validation and test queries are not clipped; values can exceed $[-1, 1]$. The raw-input comparator and raw-history component retain their training z-score standardization. Feature standardization, target standardization, loss scales, forecast horizons, data and model seeds, hyperparameter grids, query arithmetic, and the controlled study's known-support output clipping are unchanged.

Changing the scaler inside the geometry pipeline matters. It can alter partitions, truncated principal directions, absolute reconstruction distances, and angles relative to the origin. The KAHM kernel fit also uses a regularization level depending on the squared reference-coordinate magnitude. Downstream feature standardization does not reverse these nonlinear changes. By contrast, a nonconstant coordinate's positive affine min–max transform followed by refitted z-score standardization cancels in exact arithmetic when threshold-based scale fallbacks are inactive: $[(ax + b) - (a\mu + b)]/(a\sigma) = (x - \mu)/\sigma$ for $a > 0$. The experiment therefore replaces the geometry scaler itself.

SI.2 Natural-source execution

All 208 original configurations are reused: 128 on the original development sources and 80 on the four formerly reserved sources. Each configuration runs raw plus affine, raw plus locality, and raw plus KAHM under both scalers, together with unchanged raw ridge and raw RBF references: 1,664 standalone method runs. The local-only natural architectures and the previously fitted benefit diagnostic are not reevaluated in this sensitivity analysis. Each fixed arm uses the original validation-only alpha selection; the selected model is then refitted on the original admissible pretest windows. Its scaler is fitted anew on that refit set, without test observations.

The primary descriptive contrast is min–max minus standard raw-plus-KAHM scaled MSE on the four formerly reserved source families. Each mean uses the original source/series/regime/capacity/seed hierarchy. The original development sources are reported separately. Source-wise and leave-one-source-out results describe heterogeneity; neither seeds nor overlapping histories supply additional independent source families. There are no new confirmatory p-values or simultaneous superiority claims.

Methods execute serially in one benchmark worker with one BLAS thread; method order is shuffled deterministically by task identity, independently of outcomes. New warm-library costs include alpha search, selection cleanup, selected refit, prediction, and serialization. Current data loading, launch-to-result wall time, and the additional sensitivity diagnostics are recorded separately; the original source preparation is reused, rather than freshly timed. The launch-to-result field includes child execution and communication, rather than isolating startup overhead. The recorded import timer starts after some adapter dependencies have loaded, so it does not measure full cold startup. All cost ratios use baseline references remeasured in the same execution, so their numerator and denominator share the timing environment.

Method	Standard MSE	Min–max MSE	Change (%)
Raw ridge	1.1769	1.1769	fixed
Raw RBF	3.7827	3.7827	fixed
Raw + affine	1.4475	1.3306	-8.08
Raw + locality	3.6030	3.3309	-7.55
Raw + KAHM	2.1768	2.3444	+7.70

Table S35: The same fixed sensitivity comparison on the four original development sources. Their outcomes remain separate from the formerly reserved sources. Scaled MSE uses the original equal-source hierarchy; change is min–max versus standard in percent. Raw references are fixed, and no scaler is selected from these test outcomes.

SI.3 Controlled execution

The sensitivity recreates the 24 complete capacity-study data and fit repetitions for each of autoregression, Van der Pol, and Duffing, at 64 and 256 landmarks. All four original representations are retained under both geometry scalers: raw, affine local coordinates, KAHM local coordinates, and KAHM scores. A locality-coordinate control uses the locality gates in Equation (S22) with the same local principal coordinates. Thus 72 paired jobs produce 1,440 head fits. The raw representation is identical across scaler arms. The original target bounds and validation selection rules remain in force.

For each law/capacity cell, the unit of paired dispersion is a complete data/fit draw. Episode losses and horizon errors are retained but are not counted as additional independent fitted-method repetitions. The analysis reports means and paired standard deviations/standard errors descriptively; the original contrast families' confidence statements are not extended to these post-hoc comparisons. Controlled banks are shared across their representations, so controlled execution times are not standalone deployment cost estimates.

SI.4 Numerical verification and retained evidence

The standard natural arms and fixed references are checked against all 1,040 corresponding task–method prediction arrays and selected hyperparameters. All 576 original standard controlled method/capacity fits are checked against archived predictions, losses, and selected hyperparameters; 144 controlled raw cross-scaler pairs are checked for equality. A separate verification program reconstructs the sensitivity losses for natural data from per-origin forecasts and all controlled losses from the full saved prediction arrays before compact export.

The review supplement retains every natural sensitivity forecast and all controlled per-episode losses, execution records, selected settings, scaler statistics, out-of-range diagnostics, and file digests. Full controlled predictions and fitted banks are regenerated by the supplied runner; the archived verification report compares their full prediction arrays before the compact results are exported. Replaying the compact controlled losses is distinguished from independently recreating and checking those full prediction arrays. The sensitivity outputs are archived separately from the original evaluations.

SI.5 Natural heterogeneity and costs

Table S35 keeps the original development outcomes separate. KAHM's development loss increases by 7.70%; the source changes are -1.99% on Electricity, $+2.06\%$ on NN5, -5.20% on Tourism, and $+19.54\%$ on FRED-MD. Removing FRED-MD changes the aggregate scaler effect to a 3.62% reduction. These alternative aggregates describe sensitivity to source composition; they do not validate a scaler-selection rule. Table S36 reports both original role groups, and Tables S37 and S38 retain every fixed method.

Source	K_s	K_m	$\Delta\%$	R	A_m	L_m
Original development sources						
Electricity	0.4539	0.4448	-1.99	0.4388	0.4584	0.4387
NN5	0.7704	0.7863	+2.06	0.7552	0.7824	0.7516
Tourism	3.2286	3.0606	-5.20	3.4759	3.0768	3.0986
FRED-MD	4.2545	5.0857	+19.54	10.4610	1.0050	9.0346
Formerly reserved sources						
Traffic	0.5768	0.5759	-0.15	0.5739	0.5750	0.5722
Hospital	0.9238	0.9158	-0.86	0.8228	0.9157	0.8371
Saugeen	0.6012	0.6067	+0.90	0.6106	0.6646	0.6773
US births	1.2659	1.2704	+0.35	1.2228	1.2830	1.3085

Table S36: Source-level scaling sensitivity. K_s and K_m denote raw + KAHM with standard and min–max geometry; R, A_m, and L_m denote raw RBF, raw + min–max affine, and raw + min–max locality. Entries are original-hierarchy scaled MSE; $\Delta\% = 100(K_m/K_s - 1)$. Original role labels identify the historical design, while every newly computed comparison is post-hoc.

Method / geometry scaler	Electricity	NN5	Tourism	FRED-MD
Raw ridge	0.4243	0.8584	2.8648	0.5601
Raw RBF	0.4388	0.7552	3.4759	10.4610
Raw + affine / std.	0.4481	0.7614	3.1717	1.4090
Raw + affine / min–max	0.4584	0.7824	3.0768	1.0050
Raw + locality / std.	0.4418	0.7474	3.0458	10.1769
Raw + locality / min–max	0.4387	0.7516	3.0986	9.0346
Raw + KAHM / std.	0.4539	0.7704	3.2286	4.2545
Raw + KAHM / min–max	0.4448	0.7863	3.0606	5.0857

Table S37: Every fixed sensitivity method on the original development sources. Entries are source-level scaled MSE using the original nested task weights. The two raw references retain their original standardization.

Method / geometry scaler	Traffic	Hospital	Saugeen	US births
Raw ridge	0.5737	0.8461	0.5220	1.1775
Raw RBF	0.5739	0.8228	0.6106	1.2228
Raw + affine / std.	0.5764	0.9198	0.6068	1.2802
Raw + affine / min–max	0.5750	0.9157	0.6646	1.2830
Raw + locality / std.	0.5720	0.9356	0.6002	1.2770
Raw + locality / min–max	0.5722	0.8371	0.6773	1.3085
Raw + KAHM / std.	0.5768	0.9238	0.6012	1.2659
Raw + KAHM / min–max	0.5759	0.9158	0.6067	1.2704

Table S38: Every fixed sensitivity method on the formerly reserved sources. Entries are source-level scaled MSE using the original nested task weights. The two raw references retain their original standardization.

The source-balanced fraction of test input coordinates outside the final refit range is 3.77% on the formerly reserved sources and 21.45% on the original development sources. The corresponding fractions of histories with at least one such coordinate are 33.08% and 49.26%. The largest absolute transformed test coordinate is 3.1990 and 1.8638, respectively. No final natural refit has a constant input coordinate. These diagnostics confirm that unclipped min–max scaling is distinct from enforcing bounded query inputs; a clipped alternative was not tested.

Table S39 reports warm-library computation times for each stage. The retained KAHM min–max Saugeen configuration with history 28, horizon 7, medium capacity, and seed 17 takes 6.2914 seconds, including 4.7618 seconds of search and 1.3632 seconds of cleanup. It is included at its original hierarchical weight, without trimming or a timing-based rerun. The single execution cannot distinguish algorithmic cost changes from runtime variation with the precision of a separate repeated timing benchmark.

Method / geometry scaler	Search	Cleanup	Refit	Forecast	Serialize	Total
Raw ridge	0.0065	0.1770	0.0022	0.0001	0.0005	0.1863
Raw RBF	0.0136	0.1557	0.0062	0.0003	0.0005	0.1762
Raw + affine / std.	0.2316	0.1554	0.0530	0.0036	0.0021	0.4457
Raw + affine / min–max	0.0909	0.1639	0.0593	0.0025	0.0012	0.3178
Raw + locality / std.	0.0887	0.1671	0.0612	0.0022	0.0018	0.3210
Raw + locality / min–max	0.0825	0.1557	0.0672	0.0025	0.0029	0.3108
Raw + KAHM / std.	0.1162	0.1594	0.0939	0.0094	0.0015	0.3804
Raw + KAHM / min–max	0.2682	0.1934	0.0845	0.0084	0.0016	0.5561

Table S39: Contemporaneous, source-balanced warm method times (seconds) on the four formerly reserved sources. Each method independently performs its alpha search, selected refit, prediction, serialization, and cleanup. These totals exclude reused data preparation, current input loading, process startup, and additional sensitivity audits. Launch-to-result and audit elapsed times are recorded separately; the recorded import interval covers only part of initialization. Every timing, including outliers, is retained. These single-execution observations are implementation-specific. Cost ratios use reference methods measured in this same execution.

SI.6 Complete controlled sensitivity

Table S40 includes all ten representation/scaler arms in every law/capacity cell. KAHM local-coordinate mean error changes by $-0.24/-0.32\%$ on autoregression, $-0.18/+0.23\%$ on Van der Pol, and $-2.85/-7.35\%$ on Duffing at 64/256 landmarks. The main mean rankings are unchanged. The score-only KAHM representation is more sensitive on Van der Pol, where min–max raises its error by 6.03% and 4.78%. A secondary ordering consequently reverses at 256 landmarks: standardized KAHM scores have mean MSE 1.0499×10^{-3}, below affine-local 1.0780×10^{-3}, whereas their min–max errors are 1.1001×10^{-3} and 1.0812×10^{-3}, respectively. The persistence claim concerns the main KAHM-local versus raw/control comparisons, not every pair of methods. The repetition-level CSV files supply all paired differences and standard errors; the displayed standard deviations describe variability without a simultaneous confidence interpretation.

Table S40: Every controlled sensitivity arm, including raw and KAHM scores. Each entry is mean MSE with between-repetition SD in parentheses, multiplied by 10^3, over the same 24 complete draws. Δ is paired min–max-minus-standard mean difference; its parentheses contain paired SD. These are descriptive dispersions on reused draws, not confidence intervals. The raw arms are identical.

Law	M	Features	Standard (SD)	Min–max (SD)	Δ (paired SD)
Nonlinear AR	64	Raw	4.4238 (0.2161)	4.4238 (0.2161)	+0.0000 (0.0000)
Nonlinear AR	64	Affine local	4.9120 (0.3059)	4.8940 (0.2749)	-0.0180 (0.1507)
Nonlinear AR	64	Locality local	4.9306 (0.3008)	4.9414 (0.2705)	+0.0109 (0.1362)
Nonlinear AR	64	KAHM local	4.8795 (0.2769)	4.8678 (0.2601)	-0.0116 (0.0929)
Nonlinear AR	64	KAHM scores	4.7984 (0.2734)	4.8042 (0.2862)	+0.0058 (0.1198)
Nonlinear AR	256	Raw	4.6958 (0.3312)	4.6958 (0.3312)	+0.0000 (0.0000)
Nonlinear AR	256	Affine local	5.2347 (0.3501)	5.1807 (0.4489)	-0.0540 (0.2677)
Nonlinear AR	256	Locality local	5.2540 (0.3511)	5.2577 (0.3890)	+0.0036 (0.1347)
Nonlinear AR	256	KAHM local	5.1749 (0.3303)	5.1584 (0.3832)	-0.0165 (0.1586)
Nonlinear AR	256	KAHM scores	4.8256 (0.2710)	4.8538 (0.3025)	+0.0282 (0.1257)
Van der Pol	64	Raw	2.8586 (0.1574)	2.8586 (0.1574)	+0.0000 (0.0000)
Van der Pol	64	Affine local	1.3767 (0.1578)	1.3482 (0.1225)	-0.0286 (0.0855)
Van der Pol	64	Locality local	1.1324 (0.0737)	1.1240 (0.0805)	-0.0084 (0.0363)
Van der Pol	64	KAHM local	1.0266 (0.0656)	1.0248 (0.0678)	-0.0018 (0.0198)
Van der Pol	64	KAHM scores	1.5711 (0.2109)	1.6658 (0.2221)	+0.0947 (0.1891)
Van der Pol	256	Raw	1.6541 (0.0402)	1.6541 (0.0402)	+0.0000 (0.0000)
Van der Pol	256	Affine local	1.0780 (0.0501)	1.0812 (0.0525)	+0.0032 (0.0297)
Van der Pol	256	Locality local	0.9413 (0.0363)	0.9448 (0.0391)	+0.0036 (0.0191)
Van der Pol	256	KAHM local	0.8720 (0.0293)	0.8740 (0.0277)	+0.0020 (0.0060)
Van der Pol	256	KAHM scores	1.0499 (0.0600)	1.1001 (0.0794)	+0.0502 (0.0735)
Duffing	64	Raw	3.4597 (0.1516)	3.4597 (0.1516)	+0.0000 (0.0000)
Duffing	64	Affine local	3.0549 (0.1764)	2.9981 (0.1357)	-0.0568 (0.1262)
Duffing	64	Locality local	2.8016 (0.1346)	2.8227 (0.1447)	+0.0211 (0.1128)
Duffing	64	KAHM local	4.2154 (0.4775)	4.0951 (0.4723)	-0.1203 (0.4225)
Duffing	64	KAHM scores	4.7005 (0.5431)	4.5547 (0.3330)	-0.1458 (0.4739)
Duffing	256	Raw	2.8364 (0.1029)	2.8364 (0.1029)	+0.0000 (0.0000)
Duffing	256	Affine local	2.7981 (0.1245)	2.7742 (0.1318)	-0.0238 (0.1180)
Duffing	256	Locality local	2.7006 (0.1398)	2.6921 (0.1321)	-0.0085 (0.1001)
Duffing	256	KAHM local	4.0736 (0.4520)	3.7741 (0.3794)	-0.2995 (0.3555)
Duffing	256	KAHM scores	3.3808 (0.2940)	3.3354 (0.2419)	-0.0454 (0.2415)

SJ Additive forecasting: theory and experimental protocol

SJ.1 Oracle residual structure and finite correction

Condition on the fitted baseline $f_0 : \mathbb{R}^d \to \mathbb{R}^{d_y}$ and representation $\Gamma : \mathbb{R}^d \to \mathbb{R}^p$. Let (X, Y) be a fresh history–response pair under the conditional evaluation distribution, let $e = Y - f_0(X)$, and assume $\mathbb{E}\|e\|_2^2 < \infty$. Write $Z = \Gamma(X)$ and $q^*(Z) = \mathbb{E}[e \mid Z]$. For any measurable $q : \mathbb{R}^p \to \mathbb{R}^{d_y}$ with finite second moment, the difference $q^*(Z) - q(Z)$ is measurable with respect to Z. Consequently,

$$\mathbb{E}\langle e - q^*(Z), q^*(Z) - q(Z)\rangle = \mathbb{E}\langle \mathbb{E}[e - q^*(Z) \mid Z], q^*(Z) - q(Z)\rangle = 0.$$

Here $\langle a, b\rangle = a^\top b$ is the Euclidean inner product. Expanding $e - q = (e - q^*) + (q^* - q)$ gives

$$\mathbb{E}\|e - q(Z)\|_2^2 = \mathbb{E}\|e - q^*(Z)\|_2^2 + \mathbb{E}\|q^*(Z) - q(Z)\|_2^2.$$

Conditional Jensen's inequality gives $\mathbb{E}\|q^*(Z)\|_2^2 \le \mathbb{E}\|e\|_2^2$, so q^* is an admissible square-integrable correction. The second term is nonnegative and vanishes for $q = q^*$. Taking $q = 0$ and then subtracting the minimum risk proves Equation (S1). This identity describes residual structure available to an unrestricted predictor of Z. It does not show that a finite fitted ridge correction can estimate q^* reliably. The comparison is with the chosen baseline f_0; a deterministic map of X cannot supply information absent from the raw history.

SJ.2 Residual alignment for a fixed correction

Condition on all data and random choices used to fit a baseline f_0 and a correction $q : \mathbb{R}^d \to \mathbb{R}^{d_y}$. In this subsection, q denotes the complete map from history to correction, including any fitted feature transformation. Assume $\mathbb{E}\|Y - f_0(X)\|_2^2 < \infty$ and $\mathbb{E}\|q(X)\|_2^2 < \infty$, and define

$$R(f) = \mathbb{E}\|Y - f(X)\|_2^2, \qquad A = \mathbb{E}\langle Y - f_0(X), q(X)\rangle, \qquad B = \mathbb{E}\|q(X)\|_2^2.$$

The expectation is over a fresh evaluation pair conditional on the fitted maps. Expanding the squared error yields, for every scalar mixing weight γ,

$$R(f_0 + \gamma q) = R(f_0) - 2\gamma A + \gamma^2 B.$$

If $B = 0$, then $q(X) = 0$ almost surely; the Cauchy–Schwarz inequality also gives $A = 0$, so every weight has the same risk. If $B > 0$, completing the square gives

$$-2\gamma A + \gamma^2 B = B(\gamma - A/B)^2 - A^2/B.$$

Thus the risk-minimizing weight over $[0, 1]$ is $\gamma^* = \min\{1, \max\{0, A/B\}\}$. A weight $\gamma \in (0, 1]$ strictly improves the baseline exactly when $A > 0$ and $\gamma < 2A/B$. For mean squared loss averaged over target coordinates, the same identities hold after dividing R, A, and B by d_y.

The fitted selector does not observe the population values A and B. Its residual estimates, regularization, and validation choices introduce sampling error. Including $\gamma = 0$ therefore makes the baseline an available candidate, but does not guarantee that the selected correction improves future risk. Positive alignment with training residuals alone is insufficient to establish $A > 0$ on evaluation data.

SJ.3 Distinction from representation-class certificates

The empirical difference in Equation (S2) compares two fitted forecasts held fixed during evaluation. A class-level risk bound instead concerns the infimum of risk over a specified representation–decoder class. A positive lower bound can exclude an accuracy requirement for the entire class when the bound and its assumptions apply. The additive study does not transfer such a class-level guarantee to its forecasts. A certificate for a residual component would not automatically certify the complete additive forecast; the baseline and correction together would have to satisfy the declared representation, metric, and sensitivity constraint. Neither mean test differences nor a zero-correction option establishes necessity of the additional features among all possible decoders.

SJ.4 Protocol and source roster

The executable configuration and protocol for this extension are stored in `extensions/forecasting_utility`. These files specify the additive experiment independently of the controlled, natural-source, and scaling evaluations. A pre-evaluation checkpoint records hashes of the configuration, protocol, source and dependency files, archive metadata, and roster selected from training-prefix information. The checkpoint records the procedure used for evaluation and permits verification of later changes. The study was not externally preregistered.

Weather contains daily Australian weather-station variables, COVID-19 Deaths contains the archived Johns Hopkins death-count series, and Car Parts contains intermittent monthly sales extracted from the `expsmooth` collection. The exact Monash records are 10.5281/zenodo.4654822, 10.5281/zenodo.4656009, and 10.5281/zenodo.4656022, respectively. Solar Weekly is record 10.5281/zenodo.4656151 and contains

Table S41: Additional archive collections and the fixed hash-selected roster. Each selected series is evaluated at $M = 16$ and 64. L is history length, H is the number of consecutive direct target steps, and p is the declared seasonal period. The Solar collection has simulation provenance. Source record DOIs, byte hashes, and licenses are retained in the artifact.

Collection	Frequency	L	H	p	Selected series
Weather	Daily	28	7	7	T1476, T1967, T416
Solar Weekly	Weekly	6	2	52	T88, T94, T8
COVID-19 Deaths	Daily	28	7	7	T42, T146, T96
Car Parts, missing	Monthly	6	2	12	T1324, T2368, T857

aggregated simulated NREL production series. All four downloaded records declare CC BY 4.0. The use of a historical public snapshot does not establish real-time revision-aware evaluation; upstream release vintages and the original producers' measurement pipelines are not reconstructed here.

Training-prefix eligibility requires at least 16 complete available training windows and a positive finite error scale. Hash ordering uses the fixed salt `forecasting-utility-natural-v1` and the source and series identifiers. Complete contexts and targets are required; no future-aware missing-value filling is applied. This defines an available-window forecasting estimand. The scaling denominator uses the uncapped initial training prefix and is unchanged by the selected pretest refit. Periods are stated in Table S41; when the seasonal-difference mean square is unavailable or zero, the first-difference mean square is used. This scale is not a fitted baseline's test error.

The naive forecast uses the seasonal observation when available and the last available observation otherwise; synthetic episodes use persistence. The seasonal lag may extend beyond the finite context supplied to the learned models, particularly for weekly Solar and monthly Car Parts. Thus this practical control shares their targets and forecast origins, but not necessarily their identical finite-history information set; every observation it accesses is available at or before the forecast origin. The practical nonlinear control fits an independent histogram-gradient regressor per target coordinate, with 100 iterations, at most 15 leaves, L_2 penalty 0.01, and early stopping disabled. It receives the same raw histories and split origins, without a separate hyperparameter search. This is one fixed practical control, not a comprehensive comparison with modern neural, pretrained, probabilistic, or multivariate forecasters.

For every model, preprocessing is fitted on the permitted fitting inputs only. An additive correction first receives residuals from the selected raw predictor on those same fitting rows. A natural-data pretest refit rebuilds both the raw predictor and the correction using the chosen settings; thus the final raw component differs from the validation-stage fit but is again held fixed while its correction is fitted. Synthetic fits retain training-only inputs. A final zero-weight correction bypasses geometry at prediction time, but its already incurred selection costs remain charged. Additional candidate searches are therefore observable costs even when the deployed forecast equals the raw baseline. The synthetic zero-weight fit retains its searched correction bank in memory, because the selected training-only object is retained. The natural zero-weight refit omits that bank. Reported serialized bytes describe this actual stored state, not the smallest equivalent deployable model; neither case evaluates the correction during a zero-weight prediction.

Full validation-candidate records, chosen penalties and mixing weights, warnings, split origins, saved forecasts, and wall/CPU timing components are retained. Saved test forecasts permit independent reconstruction of both episode-level synthetic losses and original-unit and scaled archive-source losses. A task or method failure is recorded with its exception, without replacing its source, seed, or model. The completeness audit distinguishes requested configurations, successful outputs, and failures before producing the manuscript assets.

SJ.5 Complete controlled comparisons and source-level results

Table S42 retains every incomplete configuration. The failed series is also retained in the original roster and window table. Full-roster geometry and policy losses remain undefined. A post-hoc matched-success analysis

uses the 22 configurations and 11 series for which every method completed. The archive displays below use that diagnostic support unless explicitly marked as full-roster coverage.

Table S42: Every incomplete configuration in the frozen execution. The shared bank requests eight nonempty groups but these training inputs have only seven distinct histories. Concatenated KAHM and additive KAHM, affine, and locality each fail; both dependent policies have no decision. The specified method reports a fitting failure when the requested groups cannot be formed. The four baseline forecasts are retained.

Source	Series	M	Train rows	Distinct	Failed fits	Failed policies
Car Parts	T1324	16	23	7	4	2
Car Parts	T1324	64	23	7	4	2

The controlled means in Table S1 are supplemented by every practical control and fixed policy in Table S43. Table S44 gives all 24 primary interval endpoints. The raw and concatenated contrasts form a single family for this experiment. Its simultaneous coverage does not extend to interval families from the other studies.

Table S43: Remaining controls and declared policies in the same synthetic factorial. Mean MSE is multiplied by 10^3. The two policies use the fixed 5% validation margin; their test losses are descriptive.

Law	N	M	Raw ridge	Persistence	Boosting	KAHM rule	Geometry rule
Nonlinear AR	32	16	5.4248	8.8523	5.6510	4.8163	4.8163
Nonlinear AR	32	64	5.4248	8.8523	5.6510	5.0133	5.0133
Nonlinear AR	128	16	5.3733	8.8523	4.8297	4.6648	4.6570
Nonlinear AR	128	64	5.3733	8.8523	4.8297	4.4806	4.4806
Van der Pol	32	16	61.0069	290.7804	3.9407	15.6100	12.8539
Van der Pol	32	64	61.0069	290.7804	3.9407	8.2892	7.9561
Van der Pol	128	16	60.8383	290.7804	1.4755	14.4955	11.5577
Van der Pol	128	64	60.8383	290.7804	1.4755	7.4507	7.0588
Duffing	32	16	15.3586	38.7289	7.4754	6.6748	5.2717
Duffing	32	64	15.3586	38.7289	7.4754	5.3159	4.7580
Duffing	128	16	14.8690	38.7289	4.7537	5.8873	4.5908
Duffing	128	64	14.8690	38.7289	4.7537	4.3081	3.6958

Table S44: All primary synthetic contrasts. Differences and interval endpoints are multiplied by 10^3; positive differences favor additive KAHM. The 24 intervals form their own 95% Bonferroni paired Student family over 12 complete repetitions. The normality approximation is not a distribution-free guarantee or joint coverage claim with earlier studies.

Law	N	M	Control	Δ	Simultaneous interval
Nonlinear AR	32	16	Raw RBF	0.1190	[-0.0325, 0.2706]
Nonlinear AR	32	16	Concat.	0.2326	[0.0750, 0.3902]
Nonlinear AR	32	64	Raw RBF	-0.0024	[-0.0155, 0.0107]
Nonlinear AR	32	64	Concat.	0.3230	[0.0983, 0.5478]
Nonlinear AR	128	16	Raw RBF	0.0947	[-0.0443, 0.2337]
Nonlinear AR	128	16	Concat.	0.0891	[0.0195, 0.1587]
Nonlinear AR	128	64	Raw RBF	0.0000	[-0.0051, 0.0051]
Nonlinear AR	128	64	Concat.	0.2201	[0.1437, 0.2965]
Van der Pol	32	16	Raw RBF	6.4813	[3.8242, 9.1385]
Van der Pol	32	16	Concat.	-8.5175	[-13.2045, -3.8305]
Van der Pol	32	64	Raw RBF	0.9173	[0.3991, 1.4356]
Van der Pol	32	64	Concat.	-5.6375	[-6.7666, -4.5083]
Van der Pol	128	16	Raw RBF	6.0508	[4.0580, 8.0435]
Van der Pol	128	16	Concat.	-6.9265	[-11.4856, -2.3674]
Van der Pol	128	64	Raw RBF	0.9524	[0.6890, 1.2159]
Van der Pol	128	64	Concat.	-5.0420	[-5.7154, -4.3685]
Duffing	32	16	Raw RBF	0.6926	[-0.0149, 1.4001]
Duffing	32	16	Concat.	1.9464	[1.1163, 2.7766]
Duffing	32	64	Raw RBF	0.0200	[-0.0226, 0.0625]
Duffing	32	64	Concat.	-0.2612	[-1.0430, 0.5206]
Duffing	128	16	Raw RBF	0.8065	[0.3672, 1.2458]
Duffing	128	16	Concat.	1.7380	[0.9065, 2.5694]
Duffing	128	64	Raw RBF	0.0548	[-0.0243, 0.1339]
Duffing	128	64	Concat.	-0.2138	[-0.5053, 0.0777]

The matched-success archive comparison is reported by source in Table S45. Window counts and effective decoder sizes appear in Tables S46 and S47. Counts are shared across methods, and requested Nyström sizes can be capped by short fitting segments. The selected weights and policy decisions in Tables S48 and S49 describe related task configurations, not independent draws of a universal adoption probability. Table S50 reports all four leave-one-source-out aggregates.

Table S45: Post-hoc matched-success scaled MSE by archive source, averaging capacities within series and then series. The 22 configurations contain three Weather, three Solar, three COVID, and two Car Parts series; the failed Car Parts series is excluded only from this diagnostic. Solar Weekly has simulation provenance. All procedures and both specified policies are shown.

Procedure	Weather	Solar	COVID	Car Parts
Raw ridge	0.3363	2.1992	0.6404	0.2106
Raw RBF	0.3181	3.0358	33.1371	0.2010
Concatenated KAHM	0.3178	2.9745	1.3490	0.2031
Additive KAHM	0.3213	3.0358	33.0164	0.2038
Additive affine	0.3189	3.0358	33.7662	0.2038
Additive locality	0.3189	2.9891	34.3905	0.5510
Persistence / seasonal naive	0.5793	1.4511	2.4268	0.3676
Histogram gradient boosting	0.4482	1.9788	38.5859	0.2567
KAHM 5% rule	0.3213	3.0358	33.4246	0.2024
All-geometry 5% rule	0.3192	2.9891	33.1917	0.5496

Table S46: Exact available-window counts for all twelve originally selected series, including Car Parts T1324 whose geometric fits fail, after the fixed caps. Inputs and targets are shared by the two capacities and every procedure. Initial training observations set the scale; refit windows stop before the reserved test boundary. These windows are related observations, not independent source replications.

Source	Series	Length	Train	Valid.	Refit	Test	Scale
Weather	T1476	14262	512	64	512	64	Seasonal
Weather	T1967	6810	512	64	512	64	Seasonal
Weather	T416	6811	512	64	512	64	Seasonal
Solar Weekly	T88	52	24	9	34	10	First diff.
Solar Weekly	T94	52	24	9	34	10	First diff.
Solar Weekly	T8	52	24	9	34	10	First diff.
COVID-19 Deaths	T42	212	93	36	135	37	Seasonal
COVID-19 Deaths	T146	212	93	36	135	37	Seasonal
COVID-19 Deaths	T96	212	93	36	135	37	Seasonal
Car Parts	T1324	51	23	9	33	10	Seasonal
Car Parts	T2368	51	23	9	33	10	Seasonal
Car Parts	T857	51	23	9	33	10	Seasonal

Table S47: Actual archive decoder sizes on post-hoc common support after the selected pretest refit. Entries are ranges across three retained series per source, or two for Car Parts. Requested landmarks are capped by available fitting histories; the retained Nyström rank can be smaller still. Raw basis width includes the appended history coordinates. Additive methods preserve this same raw head and add a separate ridge correction.

Source	Requested M	Raw landmarks	Raw rank	Raw width	Concat. rank
Weather	16	16	4–16	32–44	4–16
Weather	64	64	15–60	43–88	15–60
Solar Weekly	16	16	16	22	16
Solar Weekly	64	34	34	40	34
COVID-19 Deaths	16	16	5–16	33–44	5–16
COVID-19 Deaths	64	64	16–62	44–90	16–62
Car Parts	16	16	10–12	16–18	10–12
Car Parts	64	33	17–18	23–24	17–18

Table S48: Selected correction weights before the additional 5% policy gate. Synthetic counts concern 144 configurations and archive counts the 22 successful geometric configurations. Each archive geometry additionally has two failed fits with no selected weight. They characterize the frozen selector; frequencies across related configurations are not independent Bernoulli observations.

Study	Correction	0	0.25	0.5	1
Synthetic	Additive KAHM	10	8	18	108
Synthetic	Additive affine	14	6	11	113
Synthetic	Additive locality	14	6	8	116
Archive	Additive KAHM	13	2	0	7
Archive	Additive affine	11	4	2	5
Archive	Additive locality	6	5	0	11

Table S49: Choices made by both declared 5% validation policies. Archive decisions cover the 22 completed evaluations; each policy also has two failures and no decision on those configurations. Selection cost includes the correction searches even when the policy chooses the raw baseline.

Study	Rule	Raw	+KAHM	+Affine	+Locality
Synthetic	KAHM 5% rule	72	72	0	0
Synthetic	All-geometry 5% rule	48	5	39	52
Archive	KAHM 5% rule	15	7	0	0
Archive	All-geometry 5% rule	10	1	3	8

Table S50: Descriptive leave-one-source-out sensitivity of the post-hoc common-support additive KAHM comparison. Relative change is additive/raw minus one. These re-aggregations reuse the same four sources and do not create additional evidence or select a preferred aggregate.

Omitted source	Raw RBF	Additive KAHM	Relative change
Car Parts	12.1637	12.1245	-0.32%
COVID-19 Deaths	1.1850	1.1870	+0.17%
Solar Weekly	11.2188	11.1805	-0.34%
Weather	12.1246	12.0853	-0.32%

SJ.6 Measured accuracy and resource profiles

On this same matched-success support, Table S51 separates model wall time, model CPU time, and data preparation. Table S51 normalizes each procedure's mean loss and model wall time by the raw RBF comparator measured in this same execution. It does not combine timings from different study stages. The displayed costs refer to complete required model dependencies, including validation selection, and are distinct from a single already-fitted prediction latency.

Table S51: Post-hoc common-support archive resource components, in source-balanced mean seconds per successful task. Model wall time is the main cost measure; preparation-plus-model adds common data preparation. Downloads, imports, artifact I/O, and research analysis remain outside these model measurements. The node-based policy totals charge shared work once. Model KiB describe actual stored state, not minimal storage. These diagnostic averages do not remove failed-attempt records from the artifact.

Procedure	Model wall	Model CPU	Prep.	Prep. + model	Model KiB
Raw ridge	0.0015	0.0015	0.0018	0.0033	3.5
Raw RBF	0.0037	0.0038	0.0018	0.0055	17.9
Concatenated KAHM	0.0691	0.0691	0.0018	0.0709	102.3
Additive KAHM	0.0498	0.0498	0.0018	0.0516	55.6
Additive affine	0.0369	0.0369	0.0018	0.0386	40.7
Additive locality	0.0379	0.0379	0.0018	0.0396	45.1
Persistence / seasonal naive	0.0012	0.0012	0.0018	0.0030	0.1
Histogram gradient boosting	0.6380	0.6378	0.0018	0.6398	616.0
KAHM 5% rule	0.0467	0.0467	0.0018	0.0484	48.0
All-geometry 5% rule	0.0869	0.0869	0.0018	0.0886	39.0

SJ.7 Realized residual alignment

For each saved additive forecast, let $d = \widehat{f}_{\mathrm{add}}(X) - \widehat{f}_{\mathrm{raw}}(X)$, which incorporates the chosen mixing weight. The prespecified descriptive analysis reconstructs $2\,\mathrm{mean}\langle Y - \widehat{f}_{\mathrm{raw}}(X), d\rangle$ and $\mathrm{mean}\|d\|^2$ directly from test predictions, averaging target coordinates and dividing by the fixed training scale for each archive task in the common-support diagnostic. Their difference equals the realized raw-minus-additive MSE before rounding. The analysis code and independent manuscript builder both verify this equality. Table S52 reports the same source hierarchy used for the main loss comparison.

This decomposition describes how the realized correction aligned with the test residual, including its amplitude cost. It does not identify the oracle conditional mean in Equation (S1), explain which representation component caused the alignment, or supply a new out-of-sample guarantee. Test-derived terms are never used to retune the correction, its mixing coefficient, or either adoption policy.

Table S52: Descriptive test residual decomposition on the post-hoc common-support archive subset. For $d = \widehat{f}_{\text{add}} - \widehat{f}_{\text{raw}}$, alignment is $2\,\text{mean}\langle Y - \widehat{f}_{\text{raw}}, d\rangle$ and energy is $\text{mean}\|d\|^2$, averaged over target coordinates and divided by the fixed training scale before hierarchical aggregation. Raw-minus-additive MSE equals alignment minus energy before rounding. These are identities of the saved predictions, not fresh inference or a population certificate.

Source	Correction	Alignment	Energy	Gain
Weather	Additive KAHM	-0.0024712	0.00068791	-0.0031591
Weather	Additive affine	0.00021308	0.00097397	-0.00076088
Weather	Additive locality	-0.0004239	0.00036141	-0.00078531
Solar Weekly	Additive KAHM	0	0	0
Solar Weekly	Additive affine	0	0	0
Solar Weekly	Additive locality	0.10669	0.059975	0.04671
COVID-19 Deaths	Additive KAHM	0.1667	0.045967	0.12074
COVID-19 Deaths	Additive affine	-0.59915	0.029908	-0.62906
COVID-19 Deaths	Additive locality	-1.1003	0.15312	-1.2534
Car Parts	Additive KAHM	-0.00065859	0.0020919	-0.0027505
Car Parts	Additive affine	-0.00064528	0.0021013	-0.0027466
Car Parts	Additive locality	-0.058009	0.29197	-0.34998
Equal-source mean	Additive KAHM	0.040893	0.012187	0.028707
Equal-source mean	Additive affine	-0.1499	0.0082457	-0.15814
Equal-source mean	Additive locality	-0.26301	0.12636	-0.38936

SK Target-adapted geometry: theory, protocol, and complete results

SK.1 Residual-target transport between fitted baselines

The following projection identity distinguishes residuals of a donor baseline from residuals of the final baseline. Condition on the fitted state, including a measurable representation $\Gamma : \mathbb{R}^d \to \mathbb{R}^p$ and two predictors $f_0, f_d : \mathbb{R}^d \to \mathbb{R}^H$. Let (X, Y) be a fresh evaluation pair, write $Z = \Gamma(X)$, and assume that Y, $f_0(X)$, and $f_d(X)$ have finite second moments. All expectations below refer to this conditional evaluation distribution. For a fixed symmetric positive-definite matrix $W \in \mathbb{R}^{H\times H}$, define

$$\langle a, b\rangle_W = a^\top W b, \qquad \|a\|_W^2 = a^\top W a, \qquad R_W(f) = \mathbb{E}\|Y - f(X)\|_W^2.$$

Here $a, b \in \mathbb{R}^H$. The equal-coordinate mean squared loss uses $W = I_H/H$, where I_H is the H-dimensional identity matrix.

Proposition S10 (Correction gain and donor mismatch)**.** *Define the conditional residual and conditional donor discrepancy by*

$$m_0(Z) = \mathbb{E}[Y - f_0(X) \mid Z], \qquad b_d(Z) = \mathbb{E}[f_0(X) - f_d(X) \mid Z].$$

For every measurable correction $q : \mathbb{R}^p \to \mathbb{R}^H$ satisfying $\mathbb{E}\|q(Z)\|_2^2 < \infty$,

$$R_W(f_0) - R_W(f_0 + q \circ \Gamma) = \mathbb{E}\|m_0(Z)\|_W^2 - \mathbb{E}\|q(Z) - m_0(Z)\|_W^2. \tag{S31}$$

The donor's optimal population correction is $q_d(Z) = \mathbb{E}[Y - f_d(X) \mid Z] = m_0(Z) + b_d(Z)$. Hence

$$R_W(f_0) - R_W(f_0 + q_d \circ \Gamma) = \mathbb{E}\|m_0(Z)\|_W^2 - \mathbb{E}\|b_d(Z)\|_W^2, \tag{S32}$$

$$R_W(f_0) - R_W(f_0 + \widehat{q} \circ \Gamma) = \mathbb{E}\|m_0(Z)\|_W^2 - \mathbb{E}\|b_d(Z) + a(Z)\|_W^2, \tag{S33}$$

where $\widehat{q}$ is any square-integrable fitted correction and $a(Z) = \widehat{q}(Z) - q_d(Z)$. Moreover,

$$\mathbb{E}\|b_d(Z)\|_W^2 \le \mathbb{E}\|f_0(X) - f_d(X)\|_W^2.$$

Proof. Set $\epsilon = Y - f_0(X) - m_0(Z)$. By the definition of conditional expectation, $\mathbb{E}[\epsilon \mid Z] = 0$. Because $m_0(Z) - q(Z)$ is measurable with respect to Z and W is fixed,

$$\mathbb{E}\langle \epsilon, m_0(Z) - q(Z)\rangle_W = \mathbb{E}\langle \mathbb{E}[\epsilon \mid Z], m_0(Z) - q(Z)\rangle_W = 0.$$

Expanding the squared norm of $Y - f_0(X) - q(Z) = \epsilon + m_0(Z) - q(Z)$ therefore gives

$$R_W(f_0 + q \circ \Gamma) = \mathbb{E}\|\epsilon\|_W^2 + \mathbb{E}\|m_0(Z) - q(Z)\|_W^2.$$

Set $q = 0$ in this equality and subtract the general expression to obtain Equation (S31). Linearity of conditional expectation gives $q_d(Z) = m_0(Z) + b_d(Z)$. Substituting this correction proves Equation (S32); substituting $\widehat{q}(Z) = q_d(Z) + a(Z)$ proves Equation (S33). Finally, the function $v \mapsto v^\top W v$ is convex. Conditional Jensen's inequality yields

$$\|\mathbb{E}[f_0(X) - f_d(X) \mid Z]\|_W^2 \le \mathbb{E}[\|f_0(X) - f_d(X)\|_W^2 \mid Z].$$

Taking expectations proves the bound on donor discrepancy. □

The final term in Equation (S33) contains $2\mathbb{E}\langle b_d, a\rangle_W$. It cannot be replaced by separate nonnegative mismatch and estimation penalties without another orthogonality argument: approximation error may reinforce or partially cancel the donor discrepancy. Even an exact donor correction can harm the final base when its conditional mismatch exceeds the final base's predictable residual signal.

For fixed square-integrable donors $f_1, \ldots, f_K$ and deterministic weights $\pi_k \ge 0$ summing to one, let $\bar{f} = \sum_k \pi_k f_k$. Expanding around $\bar{f}$ gives

$$\sum_k \pi_k \mathbb{E}\|Y - f_k(X) - q(Z)\|_W^2 = \mathbb{E}\|Y - \bar{f}(X) - q(Z)\|_W^2 + \sum_k \pi_k \mathbb{E}\|f_k(X) - \bar{f}(X)\|_W^2.$$

The cross term vanishes because the weighted donor deviations sum to zero. Hence the common-law population optimizer is $\mathbb{E}[Y - \bar{f}(X) \mid Z]$, and the proposition applies with $f_d = \bar{f}$. This does not identify a finite OOF fit with that optimizer. Chronological folds can have different laws and sample sizes; a donor index depending on the observation must retain that joint law rather than being replaced by constant mixture weights.

For a fixed realized correction $\delta(X)$ satisfying $\mathbb{E}\|\delta(X)\|_2^2 < \infty$, direct expansion also gives

$$R_W(f_0) - R_W(f_0 + \delta) = 2\mathbb{E}\langle Y - f_0(X), \delta(X)\rangle_W - \mathbb{E}\|\delta(X)\|_W^2. \tag{S34}$$

A separately implemented verification program checks its empirical counterpart from saved test predictions. It does not estimate the oracle m_0 or b_d. Donor/final disagreement is recorded separately on correction-training rows. The final baseline has seen those rows, so this diagnostic is not an unbiased estimate of future disagreement or predictive risk.

SK.2 Evaluation protocol and source roster

The executable protocol is `extensions/target_geometry/PROTOCOL.md`; its configuration, numerical dependencies, implementation, and prepared input roster were hashed before the full evaluation. Development checks use separate seeds and do not enter confirmation. The synthetic seeds are 11200–11211. For law index l and split index $s = 0, 1, 2$, the generator in Appendix SB receives the first state value from `SeedSequence([seed,1,s,9027])`. Each task has 128/64/128 training/validation/test episodes of length 72, history 12, offsets $1, 4, 12$ and stride four, yielding 13 origins per episode. Observation generation follows the same law-specific conventions as the controlled study.

For synthetic OOF targets, episode index modulo four assigns folds. Each donor excludes every row of the queried episodes. Within its remaining episodes, the first 80% of ordered episode IDs fit its inner candidate models and the rest select its penalty. The selected donor is then fitted on all donor episodes. No queried episode or outer-validation target selects that donor's penalty.

For natural series, the first $\lceil 0.4n \rceil$ capped training origins initialize the donor history; the remainder form three contiguous query blocks. If t_b is the first query origin, only donor rows with $t + H \leq t_b$ are allowed. An 80/20 inner chronological split applies the same target-availability condition before inner validation. Fewer than eight inner-fitting rows invokes the predefined donor penalty one. Fewer than eight outer donor rows makes that query block ineligible for *both* residual-target procedures. The fitting sample is otherwise fixed by indices, not by residual values or model success. Fewer than eight eligible correction rows retains the available baselines and sends the correction arms to their declared ridge fallback. An eligible donor fit failure sends its dependent methods to that fallback. An unavailable primary base or required evaluation split is retained as a task failure.

Table S53: Archive collections and fixed target-geometry roster. L is history length, H the number of consecutive target steps, and p the seasonal period. Selection uses only training-prefix eligibility and identifier hashes. These are 12 series from four selected collections, not 12 independent application domains.

Collection	Frequency	L	H	p	Selected series
Australian electricity	Half-hourly	48	12	48	T3, T5, T1
Bitcoin indicators	Daily	28	7	7	sent_addresses, confirmation_time, block_size
Melbourne pedestrian	Hourly	48	6	24	T10, T14, T20
KDD2018 air quality	Hourly	48	6	24	T239, T99, T233

The source records are respectively 10.5281/zenodo.4659727, 10.5281/zenodo.5121965, 10.5281/zenodo.4656626, and 10.5281/zenodo.4656719. All selected archive versions declare CC BY 4.0; archive/member hashes, attribution, source exclusions and metadata are retained in `extensions/target_geometry/sources/manifest.json`. Bitcoin selections are activity indicators, not price forecasts. These collections broaden source coverage but are not sampled randomly from application domains. The study uses fixed archive versions and does not reconstruct historical data-release vintages.

Training-prefix eligibility requires 96 finite windows and a positive seasonal-difference mean-square scale, with first-difference fallback. The selection salt is `kahm-target-geometry-new-collections-v1`. Each series uses chronological 60/20/20 splits. Training is capped at 512 windows and validation/test at 64 origins per segment by even index spacing. All targets lie inside their segment; later test histories can contain already observed test values. Missing windows are excluded without interpolation, defining an available-window estimand. The loss scale uses the initial training prefix only. The naive control uses the declared seasonal lag with an available-value fallback; synthetic naive prediction is persistence.

The 18 primary contrasts use 12 paired repetition-level means each. For positive differences favoring the second method, they compare ridge with learned-OOF KAHM; identity-OOF KAHM with learned-OOF KAHM; identity-in-sample KAHM with identity-OOF KAHM; and learned-OOF affine, locality, and RBF-local with learned-OOF KAHM. For paired differences $d_1, \ldots, d_{12}$, let $\bar{d}$ be their mean and s_d their sample standard deviation. The interval is $\bar{d} \pm t_{11,1-0.05/(2\cdot 18)} s_d/\sqrt{12}$, where $t_{\nu,p}$ is the p quantile of Student's t distribution with ν degrees of freedom. Simultaneous coverage uses the stated normal approximation. No joint coverage is asserted with prior study families. Archive results and leave-one-source-out re-aggregations are descriptive; no window-based confidence interval is attached to four selected source collections.

SK.3 Implementation, safeguards, and accounting

The metric uniformly samples up to 256 correction rows without replacement using the task seed. All training inputs determine its input standardizer, matching the downstream bank; only eligible OOF residual targets fit its supervised objective. Its donor mask excludes whole synthetic episodes or natural origins separated by at most $L + H$. Queries with no admissible donor are omitted from the proxy; fewer than two admissible queries or donor groups invokes identity. For n_c eligible residual rows $r_i \in \mathbb{R}^H$, define $\bar{r}_h = n_c^{-1} \sum_{i=1}^{n_c} r_{ih}$. The pooled residual scale is $[(n_c H)^{-1} \sum_{i=1}^{n_c} \sum_{h=1}^{H} (r_{ih} - \bar{r}_h)^2]^{1/2}$, computed before row subsampling. Constant residuals or absence of positive admissible input distances also invoke identity. Log-sum-exp evaluates kernel probabilities. L-BFGS-B uses `ftol`=10^{-10}, `gtol`=10^{-6}, at most 40 iterations and 20 line-search steps. The

source retains the analytic derivative, numerical gradient check, fitted weights, objective values, donor counts, optimizer status and fallback reason. Training-objective improvement is not a forecast accuracy claim.

Each bank caps every requested group count by the number of distinct sampled partition histories. If numerical clustering still has empty groups, all counts are halved down to one, with each reduction recorded. Rank-zero local models return their reference center. Unexpected geometry or correction errors return the fitted ridge forecast with the exception and incurred work retained. A failed supervised metric uses identity; a failed secondary raw RBF also uses ridge. A numerical failure of the secondary jointly fitted head is isolated to that arm and invokes the same ridge fallback. These fallback rules are specified for this experiment; the additive study in Section S7 used a different failure policy.

KAHM, affine and locality codes import the archived reconstruction and coordinate formulas. The RBF-local gate for local model c uses $\exp\{-\|z-\mu_c\|_2^2/(2s_c)\}$, normalized within its block, where s_c is its reference mean squared radius, floored at 10^{-6}. All four multiply local coordinates by their corresponding affinities. Applying metric weights before re-standardizing each input column would cancel the intended metric; the implementation instead applies $\mathrm{diag}(e^{\theta/2})$ after the shared standardizer. Identity-weight tests recover archived features on nondegenerate data.

One task and one numerical-library thread execute at a time. Disjoint wall and process-CPU nodes record base search/prediction, donor fitting, metric learning, bank construction and transformations, head search, final prediction and selected-model serialization. Each procedure pays for its required nodes once, including a rejected correction's search. Test feature transformations are conservatively charged even when zero correction bypasses their subsequent use; they consume test inputs only. Imports, data download/preparation, artifact writes and reporting are outside model timing. Serialized bytes describe the retained model state, not an information-theoretic storage minimum.

SK.4 Complete comparisons and diagnostic records

The tables and figures report all primary contrasts and every source. The separately implemented analyzer recomputes losses from saved targets and predictions, checks the complete task/method roster, validates donor exclusions and selection records, sums charged dependencies, checks zero-correction equality, and verifies Equation (S34). Metric objectives, effective groups, selected mixing weights and donor/final disagreement remain diagnostics of the declared procedures. Neither these quantities nor a smaller proxy objective estimates unrestricted Bayes risk, identifies minimum necessary complexity, or certifies a universal KAHM adoption rule.

Primary contrasts and source sensitivity. Table S54 displays the complete 18-contrast family, with numerical endpoints in Table S54. The three ridge contrasts favor learned-OOF KAHM, while five matched-geometry contrasts favor the simpler control. None establishes a learned-metric or OOF-target increment for KAHM. Table S55 retains the source means, and Table S56 records every series's windows and training loss scale. Individual-series losses are retained in `reports/task_metrics.csv` in the target-geometry reproduction directory. Learned-OOF KAHM's source-mean reductions relative to ridge are 3.91% for Australian electricity, 1.38% for Bitcoin indicators, 4.37% for pedestrian counts, and 0.85% for air quality. The favorable source means coexist with higher loss on four individual series. The complete leave-one-source-out display in Table S59 concerns composition sensitivity of this roster, not uncertainty over new collections.

Optimization and residual diagnostics. All 36 synthetic and 12 archive metric fits report optimizer convergence; none invokes identity fallback or reaches the iteration limit. There are no recorded bank group reductions or fitting warnings. Synthetic optimization takes two to three iterations and decreases the regularized objective by a median 1.8741×10^{-6}; the learned maximum-to-minimum weight ratios span 1.0013–1.1054. The synthetic metric therefore remains close to identity under the frozen bandwidth, penalty and diagonal parameterization. Archive optimization takes two to 14 iterations, with median objective decrease 2.9692×10^{-4} and weight ratios 1.0577–16.0000. One pedestrian series reaches the declared ratio cap. These are observed solver statuses and training-objective changes, not guarantees of global optimality, forecasting improvement, or performance of an end-to-end KAHM metric.

The median correction-row donor/final disagreement divided by OOF residual MSE is 0.0024 for synthetic tasks and 0.1047 for archive tasks (Table S57). This descriptive difference documents that the two residual sources need not be interchangeable; it cannot identify the oracle transport penalty in Proposition S10 or attribute observed test differences to that penalty. Validation selects full-strength learned-OOF KAHM correction on 35 of 36 synthetic tasks, with half strength on the remaining task. Archive selections are more restrained: the counts for $\gamma = 0, 1/4, 1/2, 1$ are $1, 4, 6, 1$ (Table S58). The four archive series harmed despite nonzero validation selection confirm that the available zero option does not ensure improvement on future observations.

Table S54: The complete family of 18 prespecified synthetic comparisons. Δ is control minus candidate test MSE; positive values favor the candidate. Intervals use the paired repetition differences, Student's t_{11} quantile, and a two-sided Bonferroni allocation of 0.05/18. Family coverage is approximate because normality of repetition differences is not guaranteed. Coverage applies to this family separately from the other experiments.

Law	Control	Candidate	$\Delta \times 10^3$	Family interval
Nonlinear AR	Raw ridge	Learned, OOF KAHM	0.566	[0.459, 0.672]
Nonlinear AR	Fixed, OOF KAHM	Learned, OOF KAHM	-0.011	[-0.105, 0.083]
Nonlinear AR	Fixed, IS KAHM	Fixed, OOF KAHM	-0.003	[-0.006, 4.07×10^{-4}]
Nonlinear AR	Learned, OOF Affine	Learned, OOF KAHM	-0.032	[-0.111, 0.046]
Nonlinear AR	Learned, OOF Locality	Learned, OOF KAHM	0.038	[-0.122, 0.199]
Nonlinear AR	Learned, OOF RBF local	Learned, OOF KAHM	-0.083	[-0.182, 0.015]
Van der Pol	Raw ridge	Learned, OOF KAHM	23.510	[21.236, 25.785]
Van der Pol	Fixed, OOF KAHM	Learned, OOF KAHM	0.754	[-0.939, 2.447]
Van der Pol	Fixed, IS KAHM	Fixed, OOF KAHM	-0.003	[-0.008, 0.002]
Van der Pol	Learned, OOF Affine	Learned, OOF KAHM	-1.751	[-4.088, 0.586]
Van der Pol	Learned, OOF Locality	Learned, OOF KAHM	-6.661	[-8.765, -4.558]
Van der Pol	Learned, OOF RBF local	Learned, OOF KAHM	-5.169	[-6.229, -4.110]
Duffing	Raw ridge	Learned, OOF KAHM	1.344	[1.054, 1.634]
Duffing	Fixed, OOF KAHM	Learned, OOF KAHM	-0.103	[-0.336, 0.130]
Duffing	Fixed, IS KAHM	Fixed, OOF KAHM	-0.001	[-0.006, 0.004]
Duffing	Learned, OOF Affine	Learned, OOF KAHM	-0.836	[-1.254, -0.418]
Duffing	Learned, OOF Locality	Learned, OOF KAHM	-1.036	[-1.489, -0.583]
Duffing	Learned, OOF RBF local	Learned, OOF KAHM	-0.243	[-0.435, -0.051]

Table S55: Scaled MSE by collection, averaging its three declared series. Collections receive equal weights in the aggregate; neither series nor repeated origins increase the collection count. No confidence intervals are attached to these four descriptive collection means.

Procedure	Electricity	Bitcoin	Pedestrian	Air quality
Raw ridge	0.2879	3.0946	1.2489	0.5704
Raw RBF	0.2969	5.8096	1.0650	0.6216
Persistence / seasonal naive	0.7859	3.3622	1.5349	1.7073
Joint raw + KAHM	0.2760	3.8872	1.0713	0.5773
Fixed, IS KAHM	0.2725	3.0163	1.1943	0.5752
Fixed, IS Affine	0.2753	3.0013	1.1862	0.5708
Fixed, IS Locality	0.2678	3.0174	1.1882	0.5808
Fixed, IS RBF local	0.2885	3.0102	1.1697	0.5631
Fixed, OOF KAHM	0.2778	3.0781	1.1766	0.5741
Fixed, OOF Affine	0.2798	3.0762	1.1636	0.5728
Fixed, OOF Locality	0.2713	3.0559	1.1781	0.5887
Fixed, OOF RBF local	0.2828	3.0424	1.2035	0.5695
Learned, OOF KAHM	0.2766	3.0520	1.1943	0.5656
Learned, OOF Affine	0.2975	3.0369	1.1691	0.5711
Learned, OOF Locality	0.2751	3.0615	1.1597	0.5714
Learned, OOF RBF local	0.2838	3.0506	1.2278	0.5772

Table S56: Exact natural-series roster, selected before forecasts by a salted hash among training-prefix-eligible candidates. L is history length and H the complete forecast path length. Chronological boundaries are 60/20/20; windows with missing inputs or targets are omitted without imputation. At most 512/64/64 windows are retained by the frozen deterministic sampling rule. The loss scale is computed before validation. Bitcoin series are network indicators, not prices.

Collection	Series	L	H	Train/val/test	Loss scale
Australian electricity	T3	48	12	512/64/64	1.43×10^5
Australian electricity	T5	48	12	512/64/64	4182.897
Australian electricity	T1	48	12	512/64/64	4.50×10^5
Bitcoin indicators	sent_addresses	28	7	512/64/64	2.07×10^8
Bitcoin indicators	confirmation_time	28	7	512/64/64	163.705
Bitcoin indicators	block_size	28	7	512/64/64	3.19×10^9
Pedestrian counts	T10	48	6	512/64/64	26088.371
Pedestrian counts	T14	48	6	512/64/64	31655.986
Pedestrian counts	T20	48	6	512/64/64	27784.521
KDD air quality	T239	48	6	512/64/64	102.986
KDD air quality	T99	48	6	512/64/64	226.458
KDD air quality	T233	48	6	512/64/64	441.826

Table S57: Training diagnostics, not independent evidence of test benefit. $\mathcal{J}$ is the regularized Gaussian residual-regression proxy shared by all geometric arms. $\mathcal{J}_0$ and $\mathcal{J}_{\text{fit}}$ are the identity-metric and selected fitted-metric objectives. A finite improved iterate may be retained at an iteration limit, separately from optimizer convergence. D is mean squared disagreement between donor and final ridge predictions on the same correction rows; R_{OOF} is their out-of-fold (OOF) residual MSE. Medians summarize observed tasks and are descriptive. Exact objectives, warnings, weight ratios, missing diagnostics and residual identities are retained in the report CSV files.

Phase	Tasks	Converged	Metric fallback	Median $\mathcal{J}_0 - \mathcal{J}_{\text{fit}}$	Median D/R_{OOF}
Synthetic	36	36	0	1.874×10^{-6}	0.0024
Archive	12	12	0	0.0003	0.1047

Table S58: Validation-selected correction strengths, with counts ordered as $\gamma = 0, 0.25, 0.5, 1$. A zero correction is the exact deployed ridge predictor. A failed arm is recorded separately as a baseline fallback, not as an observed zero-strength validation selection. Selection counts summarize tasks and do not supply an additional inference sample.

Procedure	Synthetic gamma counts	Fallback	Archive gamma counts	Fallback
Fixed, IS KAHM	0 / 0 / 0 / 36	0	1 / 5 / 1 / 5	0
Fixed, IS Affine	0 / 0 / 1 / 35	0	2 / 3 / 2 / 5	0
Fixed, IS Locality	0 / 0 / 1 / 35	0	0 / 3 / 5 / 4	0
Fixed, IS RBF local	0 / 0 / 0 / 36	0	0 / 3 / 5 / 4	0
Fixed, OOF KAHM	0 / 0 / 0 / 36	0	2 / 5 / 2 / 3	0
Fixed, OOF Affine	0 / 0 / 1 / 35	0	3 / 4 / 4 / 1	0
Fixed, OOF Locality	0 / 0 / 1 / 35	0	0 / 7 / 4 / 1	0
Fixed, OOF RBF local	0 / 0 / 0 / 36	0	1 / 4 / 6 / 1	0
Learned, OOF KAHM	0 / 0 / 1 / 35	0	1 / 4 / 6 / 1	0
Learned, OOF Affine	0 / 0 / 1 / 35	0	1 / 4 / 5 / 2	0
Learned, OOF Locality	0 / 0 / 1 / 35	0	1 / 4 / 7 / 0	0
Learned, OOF RBF local	0 / 0 / 0 / 36	0	0 / 5 / 5 / 2	0

Table S59: Descriptive sensitivity to collection composition: each column omits the named collection and recomputes the mean scaled loss over the remaining three collections, divided by the corresponding raw-ridge loss. Values below one favor the procedure. Every omission is shown; none is selected as the primary result. These ratios are not independent replications, confidence intervals, or additional confirmatory tests.

Procedure	Electricity	Bitcoin	Pedestrian	Air quality
Raw ridge	1.000	1.000	1.000	1.000
Raw RBF	1.526	0.941	1.702	1.548
Persistence / seasonal naive	1.344	1.912	1.481	1.227
Joint raw + KAHM	1.127	0.913	1.199	1.130
Fixed, IS KAHM	0.974	0.969	0.977	0.968
Fixed, IS Affine	0.968	0.964	0.973	0.964
Fixed, IS Locality	0.974	0.967	0.978	0.966
Fixed, IS RBF local	0.965	0.959	0.977	0.965
Fixed, OOF KAHM	0.983	0.963	0.994	0.979
Fixed, OOF Affine	0.979	0.957	0.994	0.976
Fixed, OOF Locality	0.981	0.967	0.991	0.973
Fixed, OOF RBF local	0.980	0.976	0.985	0.978
Learned, OOF KAHM	0.979	0.966	0.985	0.977
Learned, OOF Affine	0.972	0.967	0.988	0.972
Learned, OOF Locality	0.975	0.952	0.989	0.971
Learned, OOF RBF local	0.988	0.991	0.990	0.985

SL Direct reconstruction protocol and reproducibility

The protocol in `extensions/direct_reconstruction/PROTOCOL.md` specifies the model, input preparation, optimization, analysis, and exact task roster. Its public checkpoint includes the implementation, numerical dependencies, source metadata, and prepared input arrays and precedes evaluation on D. The exploratory gate-training pilot has a separate design and does not contribute observations to the primary interval family.

Local shape controls and gate normalization. Use the row-stored reference matrix, principal basis, and center from Section S9. Let $b(c)$ be the partition block containing local model c, and set

$$a_c(z) = \mu_c + P_c P_c^\top (z - \mu_c), \quad u_c(z) = P_c^\top (z - \mu_c), \quad n_c(z) = (I_L - P_c P_c^\top)(z - \mu_c).$$

The scalars t_c and e_c are the positive reference-radius and projection-residual scales in Equation (S22). For a horizon-specific shape parameter ξ_b, the affine control uses

$$r_c^{\mathrm{A}}(z; \xi_b) = \mu_c + e^{\xi_b} P_c P_c^\top (z - \mu_c).$$

Both this reconstruction and the KAHM reconstruction determine a folding score s_c through Equation (S21). Their gate logit is $\ell_c(z) = 16 \log[1 - \mathrm{clip}_{[0,1]}\, s_c(z) + 10^{-6}]$. The other two controls use logits

$$\ell_c^{\mathrm{loc}}(z; \xi_b) = -e^{\xi_b} \frac{\|n_c(z)\|_2^2}{e_c} - 0.1 e^{-\xi_b} \frac{\|z - \mu_c\|_2^2}{t_c},$$
$$\ell_c^{\mathrm{RBF}}(z; \xi_b) = -\frac{e^{\xi_b} \|u_c(z)\|_2^2 + e^{-\xi_b} \|n_c(z)\|_2^2}{2 t_c}.$$

For every family, the temperature parameter η_b gives the gate

$$\phi_c(z) = \frac{\exp\{e^{\eta_b} \ell_c(z)\}}{\sum_{j \in \mathcal{C}_b} \exp\{e^{\eta_b} \ell_j(z)\}}, \qquad c \in \mathcal{C}_b.$$

Both parameters lie in $[-\log 4, \log 4]$. The fixed model sets both to zero, the gate-trained model fixes $\xi_b = 0$, and the shape-trained model fits both. The principal coordinates u_c themselves remain fixed in all three procedures.

The optimizer is L-BFGS-B with numerical forward differences, at most 40 iterations, at most 20 line-search steps per iteration, function tolerance 10^{-10}, and gradient tolerance 10^{-6}. Its 256-evaluation ceiling counts actual objective calls, including calls used for numerical differentiation. A zero parameter vector initializes every optimization, and the best finite evaluated parameter vector supplies the fitted state.

Exploratory gate-training pilot. The pilot used three independent seeds per law, separately implemented generators, and at most 160 objective evaluations. Gate-temperature training reduced mean KAHM MSE relative to its fixed version by 0.53% on nonlinear AR, 30.21% on Van der Pol, and 11.29% on Duffing. The best directly trained local control was more accurate on every law; raw RBF was also substantially stronger on Van der Pol and Duffing. The reproduction materials contain all 99 method–task prediction records and their optimization outcomes. These nine exploratory tasks are excluded from the direct-reconstruction interval family.

Chronology and seasonal prediction. Synthetic seeds are 84200–84211 for each law, with 72-observation episodes and thirteen forecast origins per episode. Implementation checks use seed 932001, outside the evaluation roster. Whole-episode inner validation selects the raw baseline in A. For natural data, inner A training targets must have matured by its first validation query. Every outer target lies inside its segment. A history can include observations from the preceding segment because these have already been observed at the forecast origin. No future missing-value imputation or label-based series replacement is used.

Dominick has history 52, offsets $(1, 2, 4)$ and seasonal period 52. Rideshare uses only `price_mean` series, history 24, offsets $(1, 3, 6)$ and period 24. An A-only availability check reduced the proposed history from 48 to 24 before the public checkpoint and any forecast evaluation; the seasonal period and eligibility threshold were unchanged. Wikipedia has history 28, offsets $(1, 3, 7)$ and period 7. Sunspot uses the last 4,096 observations independent of their values, history 28, offsets $(1, 3, 7)$ and period one. Each of the first three collections contributes three hash-selected eligible series; sunspot contributes its single series. Every source record, archive checksum, selected identifier, original origin index, split boundary and training-only scale is included in the reproduction records. Natural row caps are 512/128/128/128. The minimum A eligibility is 96 complete windows with a positive finite scale. A later unavailable segment remains a recorded failure. The three selected rideshare series have no complete B windows, identified during preparation after A-only selection; all three are retained as planned feasibility failures. This precludes a complete four-collection accuracy aggregate.

Synthetic task MSE averages squared errors over the three offsets and all D origins. Natural task MSE is divided by an A-only scale: the mean of finite squared differences at the collection's seasonal period. If this mean is unavailable, nonfinite, or nonpositive, the corresponding first-difference mean is used. A task with no positive finite scale is ineligible. The scale remains fixed throughout selection and evaluation. All source-level means weight the selected series equally.

For a primary synthetic comparison, let d_r be comparator MSE minus reconstruction-trained KAHM MSE in repetition r, for $r = 1, \ldots, 12$. With sample mean $\bar{d}$ and sample standard deviation s_d, the 95% Bonferroni-adjusted interval is

$$\bar{d} \pm t_{11,1-0.05/(2\cdot 18)} \frac{s_d}{\sqrt{12}},$$

where $t_{\nu,q}$ denotes the q quantile of Student's t distribution with ν degrees of freedom. Under independent normal paired differences, these intervals jointly cover all eighteen population mean differences with probability at least 95%; otherwise the coverage uses a normal approximation. The statistical unit is the complete repetition, not an individual forecast origin.

Seasonal naive predicts each offset from the most recent observed value with the appropriate seasonal phase. For period p this is $y_{t+h-p\lceil h/p \rceil}$, using only indices at or before t. Period one therefore gives persistence. All declared histories include the required seasonal lag. The selected source records are `https://doi.org/10.5281/zenodo.4654802`, `https://doi.org/10.5281/zenodo.5122114`, `https://doi.org/10.5281/zenodo.4656080`, and `https://doi.org/10.5281/zenodo.4654773`; the source manifest retains attribution and archive-license metadata.

Baseline and correction selection. Raw RBF and boosting retain the fitted ridge state for a recorded horizon-wise D fallback; a D failure cannot change their C forecast or the C-selected policy. Its baseline cost and serialized bytes are included. Raw ridge uses penalties $10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 100$, selected inside A and then refitted on A. The raw RBF reference retains raw standardized history together with at most 64 Nyström landmarks; bandwidth factors $0.5, 1, 2$ multiply the median positive landmark squared-distance scale, and B selects among them and the same seven penalties. Histogram gradient boosting fits 100 iterations with L2 penalty one, learning rate 0.05 or 0.1, and maximum leaves 7 or 15; B selects the grid jointly across horizons without external early stopping, using a separate fitted tree ensemble per target. All local corrections use the same A residual targets, sum-loss penalties $0.01, 0.1, 1$, A-standardized features and unpenalized intercept. C chooses their mixing weights separately by horizon with the zero-correction tie-breaking rule. A budget exit or nonfinite trial is logged; the best finite incumbent is retained.

The affine gain and locality/RBF anisotropy are deliberately different functional interventions from the KAHM bandwidth. Their common parameter counts, supervision and evaluation ceilings support a matched experiment at that level; they do not prove equal approximation power or equal computation. Likewise, a gate-only arm has three parameters per horizon and a reconstruction/shape arm has six, so their comparison asks whether the additional intervention helps within the declared budget.

Computation and state. Numerical-library thread counts are fixed to one. Warm measured nodes cover the baseline, shared bank, KAHM-specific factors, all paid geometry trials, C selection, and D transformations and predictions. The implementation eagerly prepares fixed gate queries and charges that work even when C selects zero correction or a different bandwidth; the accounting is conservative relative to a possible lazy implementation. Each standalone procedure sums its required dependencies once. Policy costs sum the union of measured candidate dependencies, including their eager query work, not only the selected winner. This component sum is not a separately timed minimal policy execution. Selected fitted states, their serialized byte sizes and serialization time are retained. Downloads, input preparation, imports and evidence artifact writing lie outside the model pipeline. Absolute times and ratios describe this implementation and host, not intrinsic algorithmic constants. A whole-process high-water RSS is not a per-method memory estimate.

The evaluation compares fixed computational search budgets rather than equal measured wall times. It does not evaluate delayed-label updates or drifting streams; those questions are addressed separately in Section S10.

Computational verification. The verification program checks archived inputs and source files, reconstructs windows from retained raw episodes or series, recomputes C/D losses, checks all method and candidate rosters, replays strict selection, checks exact zero correction, reconstructs cost dependencies and replays serialized selected predictions. The program passes 543,017 consistency checks, including reconstruction of 144 generated episode batches and replay of 731 selected states on C and D. A separate calculation recomputes scalar losses, A-only natural scales, and all 18 intervals without importing the model or its analyzer. Neither check independently refits every optimization trial or supplies an external experimental replication.

Bitwise regeneration also depends on the numerical environment. The recorded execution uses NumPy 2.3.5 on an AVX512-capable x86-64 host. A post-execution diagnostic reproduces all four episode batches of one declared Duffing task exactly on that host, but disabling AVX512 dispatch changes some values at floating-point roundoff scale. Package versions and single-thread execution alone therefore do not guarantee identical regenerated bytes on other processors. The archived input arrays therefore provide the exact evaluated inputs. Regeneration checks compare their bytes exactly, whereas selected-state prediction replay uses an absolute and relative tolerance of 10^{-10}. A separate raw-RBF state probe crosses the declared 10^{-10} absolute and relative replay tolerances under altered dispatch, while changing its C/D mean squared errors by less than 10^{-14} relatively. Verification succeeds on the original host and on one compatible hosted environment. The altered-dispatch failures identify a limitation of strict numerical replay across processor configurations.

SL.1 Complete comparisons and resource records

Table S60: The complete 18-comparison family for reconstruction-trained KAHM. With $\overline{L}$ denoting mean MSE across twelve repetitions, gain is $100(\overline{L}_{\text{control}} - \overline{L}_{\text{KAHM}})/\overline{L}_{\text{control}}$. Intervals are absolute paired MSE differences, not percentage intervals; their positive direction favors KAHM.

Law	Control	Gain (%)	Difference interval	Outcome
AR	Raw RBF	2.87	[6.65e-06, 0.000278]	positive
AR	Gate KAHM	0.09	[-3.86e-05, 4.71e-05]	unresolved
AR	Reconstruction affine	-0.11	[-0.0001, 8.97e-05]	unresolved
AR	Reconstruction locality	1.11	[-3.46e-05, 0.000142]	unresolved
AR	Reconstruction RBF-local	-0.18	[-9.5e-05, 7.77e-05]	unresolved
AR	Selected non-KAHM	-0.88	[-0.000147, 6.26e-05]	unresolved
VDP	Raw RBF	-31.51	[-0.0042, 0.000112]	unresolved
VDP	Gate KAHM	41.43	[0.00435, 0.00771]	positive
VDP	Reconstruction affine	-12.99	[-0.00181, -0.000155]	negative
VDP	Reconstruction locality	8.26	[-0.00121, 0.00275]	unresolved
VDP	Reconstruction RBF-local	-9.33	[-0.00206, 0.000604]	unresolved
VDP	Selected non-KAHM	-251.52	[-0.0079, -0.0043]	negative
Duffing	Raw RBF	-22.04	[-0.00146, -0.00048]	negative
Duffing	Gate KAHM	5.51	[-5.75e-05, 0.000682]	unresolved
Duffing	Reconstruction affine	-13.74	[-0.000992, -0.000303]	negative
Duffing	Reconstruction locality	-21.83	[-0.0016, -0.000317]	negative
Duffing	Reconstruction RBF-local	-6.73	[-0.00109, 0.00041]	unresolved
Duffing	Selected non-KAHM	-29.32	[-0.00159, -0.000841]	negative

Table S61: Collection-specific scaled MSE for every procedure. Each available collection weights its selected series equally. All three A-selected rideshare series have no complete B windows, so their entries and the planned four-collection aggregate are undefined.

Procedure	Dominick	Rideshare	Sunspot	Wikipedia
Ridge	0.3574	undefined	0.34016	0.45932
Raw RBF	0.34874	undefined	0.22123	0.51905
Boosted trees	0.27423	undefined	0.28535	0.49702
Seasonal naive	0.42346	undefined	0.25627	0.70067
Fixed KAHM	0.37401	undefined	0.34016	0.46804
Fixed affine	0.34612	undefined	0.32203	0.47187
Fixed locality	0.33466	undefined	0.33485	0.45928
Fixed RBF-local	0.35577	undefined	0.34016	0.46267
Gate KAHM	0.38055	undefined	0.34016	0.46787
Gate affine	0.35066	undefined	0.32307	0.49483
Gate locality	0.39325	undefined	0.29447	0.45928
Gate RBF-local	0.34289	undefined	0.29803	0.46245
Reconstruction KAHM	0.31635	undefined	0.34016	0.4593
Reconstruction affine	0.35629	undefined	0.28051	0.50837
Reconstruction locality	0.38436	undefined	0.28051	0.45962
Reconstruction RBF-local	0.33356	undefined	0.28772	0.47022
Selected non-KAHM	0.35628	undefined	0.28051	0.51063

Table S62: Measured complete procedure dependencies and selected model state. Means weight the 36 synthetic tasks equally or the seven available natural tasks equally. Natural loss tables instead weight their three available collections equally. Times include selected-state packing and serialization, and policy search costs count shared dependencies once. State bytes are not peak resident memory.

Procedure	Synthetic time (s)	Natural subset time (s)	Synthetic state (KiB)	Natural subset state (KiB)
Ridge	0.002035	0.00173	0.7686	1.795
Raw RBF	0.01715	0.01298	42.26	57.88
Boosted trees	0.6534	0.655	386.7	383.3
Seasonal naive	8.219e-05	5.728e-05	0.07031	0.07031
Fixed KAHM	0.07743	0.0502	174.5	124.8
Fixed affine	0.04852	0.03264	10.01	16.4
Fixed locality	0.0425	0.02986	10.01	13.88
Fixed RBF-local	0.04199	0.02957	10.04	10.94
Gate KAHM	0.9139	0.734	174.4	132.9
Gate affine	0.8478	0.6936	10.04	16.5
Gate locality	0.8422	0.5606	10.01	14.22
Gate RBF-local	0.8415	0.6556	10.08	13.22
Reconstruction KAHM	10.73	7.929	174.5	104.2
Reconstruction affine	6.973	5.345	10.07	18.61
Reconstruction locality	2.722	2.133	10.01	16.33
Reconstruction RBF-local	2.766	2.184	10.04	17.74
Selected non-KAHM	15.57	12.16	155.9	17.49

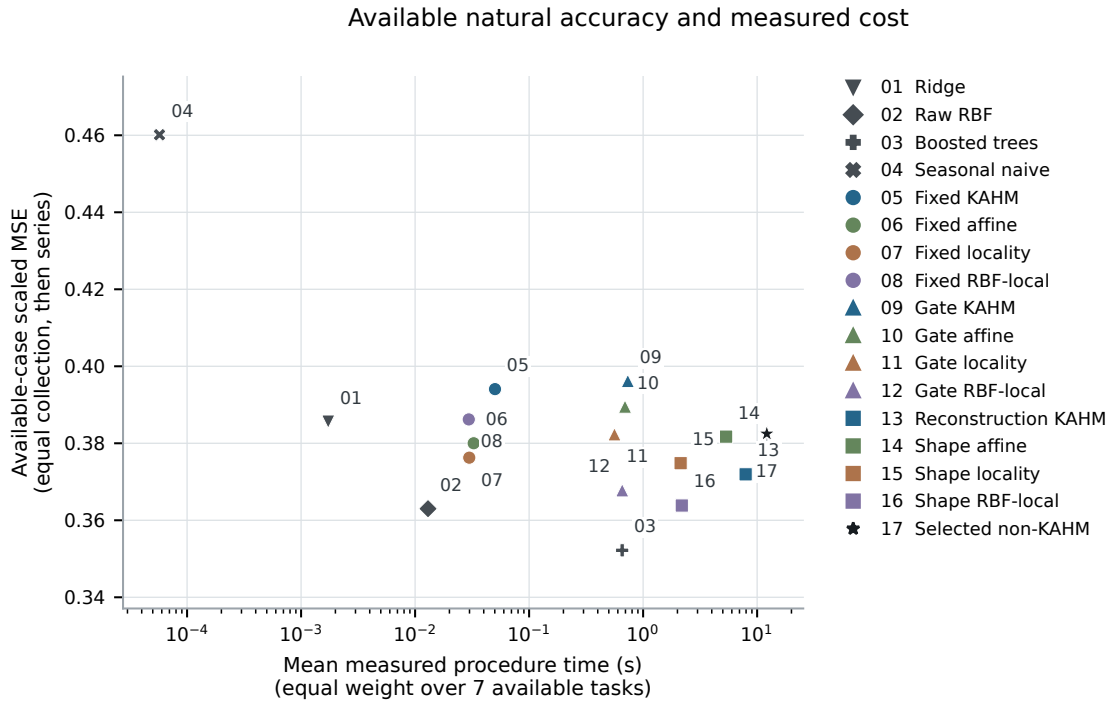


Figure S1: Available-case natural accuracy and implemented cost for all 17 procedures. Accuracy averages equally across three collections; time averages equally across seven feasible task suites, with a logarithmic horizontal scale. The missing rideshare collection is not replaced. This descriptive plot does not establish population uncertainty or an application-specific acceptable-cost threshold.

SM Online adaptation: protocol and resource measurements

Inputs and temporal roles. The eight synthetic streams contain 2,048 retained observations each, after 256 regime-A burn-in observations. Four seeds, 20261101–20261104, pair stationary and recurring schedules by innovations. Histories and horizons are respectively 12 and $(1, 4, 12)$. The retained Dominick series T7830, T25851 and T108533 each contain 393 weekly observations, with histories 52 and horizons $(1, 2, 4)$. The Sunspot input is the finite final 4,096-point crop of the previously selected daily T1 series, starting at original raw index 69,828; histories are 28 and horizons $(1, 3, 7)$. No interpolation is used. The source archives and version identifiers are documented with the reproduction materials (Godahewa et al., 2021).

At observation time t, the evaluator scores previously issued forecasts whose targets mature at t, releases the newly available labels, updates the model, then issues forecasts for $t + h$. Initial fitting uses common joint-horizon complete rows, and calibration weights are selected before processing the first evaluation observation using only earlier targets. Matured earlier evaluation labels may update the model; the update rule and selected weights remain fixed. The model therefore uses labels from earlier evaluation times once they become available, while the update algorithm remains fixed. Scale factors use the fitting prefix only. Losses average origins within horizon, horizons within series, then series within collection; Dominick and Sunspot receive equal weight. The four seeds define independent simulation replications; forecasts from overlapping windows within a stream are dependent. No broad transfer intervals are claimed from the two selected natural collections.

Reference policy and calibration scope. The geometric bank contains four local models of rank at most two. Every 16 observations, an adaptive arm considers one local model in round-robin order, retains half its initial references as anchors, and replaces at most four other references with observed histories assigned to that local model. The affected geometry and all residual heads are rebuilt coherently from their bounded replay. No bandwidth, penalty, history, reference-policy or refresh search is performed. The kernel comparator uses an initial-prefix median-distance bandwidth, regularizer 0.1 and FIFO supports, with recorded inverse-residual checks and rebuilds. Calibration selects only the four possible horizon-specific correction weights stated in Section S10. For the kernel comparator, the correction is its kernel forecast minus the common raw ridge forecast. The same calibration rule therefore selects a ridge–KRR mixture for each horizon: weights one and zero recover KRR and ridge, respectively. All reported method losses include this selection.

Prespecified accuracy and service criteria. The decision to expand the evaluation requires one capacity setting to satisfy all accuracy and service criteria. On recurring streams, adaptive KAHM must reduce mean loss by at least 3% against each of adaptive affine, adaptive RBF-local and kernel ridge, with wins on at least three of four paired seeds against each, and by at least 1% against both fixed KAHM and raw ridge. On the natural aggregate, it must improve each of the three main comparators by at least 1%; at least one collection must show improvements of at least 3% against all three comparators, and neither collection may be more than 5% worse than any. KAHM must also meet the per-stream 99th-percentile service-time screen, with no unexpected exception or cap violation in any method. These thresholds were fixed before the original run and were also used for the repeated evaluation after the numerical amendment, designated A1. Neither capacity setting meets the recurring-stream accuracy criteria. The protocol therefore classifies both settings as insufficient for expanding the experiment. Every criterion is evaluated within a single capacity setting.

Resource scope. Workers execute serially in fresh processes with one numerical thread and the same CPU affinity. Initialization and calibration are measured separately. The service interval covers history/pair assembly, all matured horizon updates, feature/head rebuilds, kernel inverse audits and prediction; scoring, serialization and state inspection are excluded. Buffer-state samples and whole-worker peak RSS have the distinct scopes stated in Table S63. The 20 ms allowance is a host-specific engineering screen, not an application-derived service-level requirement. Different support sharing across horizons prevents interpreting the tier labels as equal memory or equal computation.

Solver amendment and computational verification. The original implementation accumulated correction cross-products; large rows entering and leaving a finite window produced cancellation in the reproduced

Table S63: Measured service and state by method/tier. P99 denotes the 99th percentile and is first computed per stream, then maximized across the 12 streams. Other columns also report maxima; they may occur on different streams. Owned NumPy buffers exclude Python objects and temporary work arrays; worker resident set size (RSS) includes interpreter, libraries and evidence storage. Neither is a precise incremental deployment footprint.

Method	Tier	P99 (ms)	Max (ms)	NumPy (KiB)	RSS (MiB)
Raw ridge	Small	0.55	1.77	300.34	150.48
Raw ridge	Large	1.26	4.54	465.72	150.48
Fixed KAHM	Small	4.98	29.01	466.29	150.48
Fixed KAHM	Large	8.61	27.24	685.38	150.48
Adaptive KAHM	Small	9.89	49.10	518.95	150.48
Adaptive KAHM	Large	11.75	23.60	824.52	150.48
Adaptive affine	Small	9.56	24.75	509.83	150.48
Adaptive affine	Large	7.00	13.44	791.39	150.48
Adaptive RBF-local	Small	5.05	12.05	509.83	150.48
Adaptive RBF-local	Large	4.58	8.00	791.39	150.48
Sliding-window KRR	Small	3.15	3.95	477.34	150.48
Sliding-window KRR	Large	3.42	13.49	1011.72	150.48

failing state. The amended implementation (A1) recomputes residuals against the current baseline from the same bounded replay, centers to eliminate the free intercept, equilibrates both data and penalty columns, and solves the augmented ridge objective with pivoted QR. It changes arithmetic, not the specified objective or advancement rule. The original 144 runs remain in `extensions/adaptive_development/archives/initial_attempt.zip`. Because the amended runs repeat the same inputs, they do not increase the number of independent experimental replications. The verification program passes 6,713,837 computational consistency checks, covering chronology, coverage, calibration selection, aggregation, decision gates, resources, shared baselines and matched reference changes. It imports neither the forecasting model nor its analyzer, but remains an internal computational check rather than independent scientific replication.

The reproduction materials contain both versions of the protocol and source, the numerical diagnostic states, input hashes, predictions, update traces, and summaries. The manuscript builder `extensions/tmlr_integration/build_adaptive_assets.py` reconstructs the displayed collection means and resource maxima from saved task records, checks them against the analysis and decision files, and records input/output hashes. It performs no model fitting. Both execution checkpoints were recorded locally. The study was not publicly preregistered.